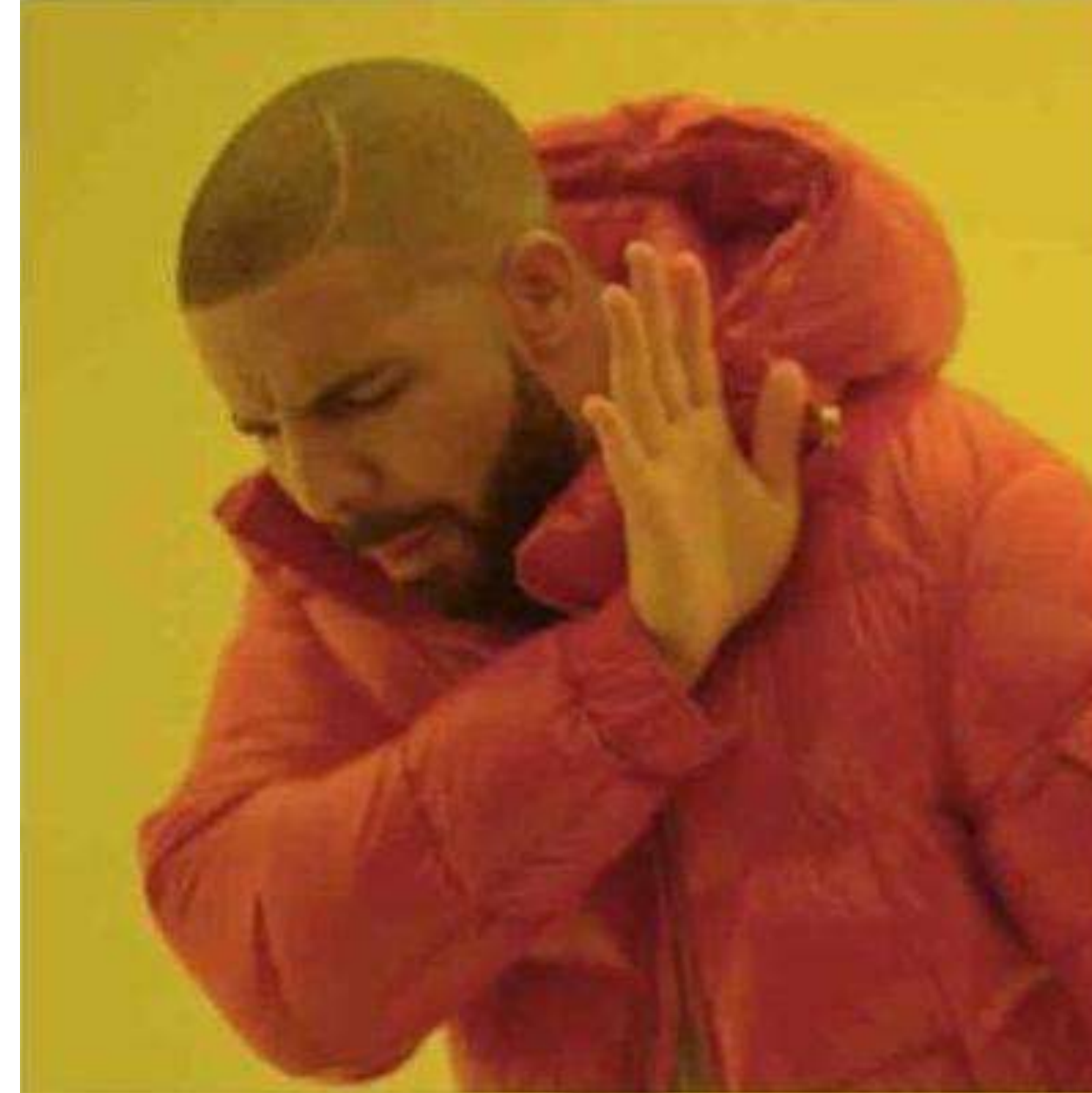
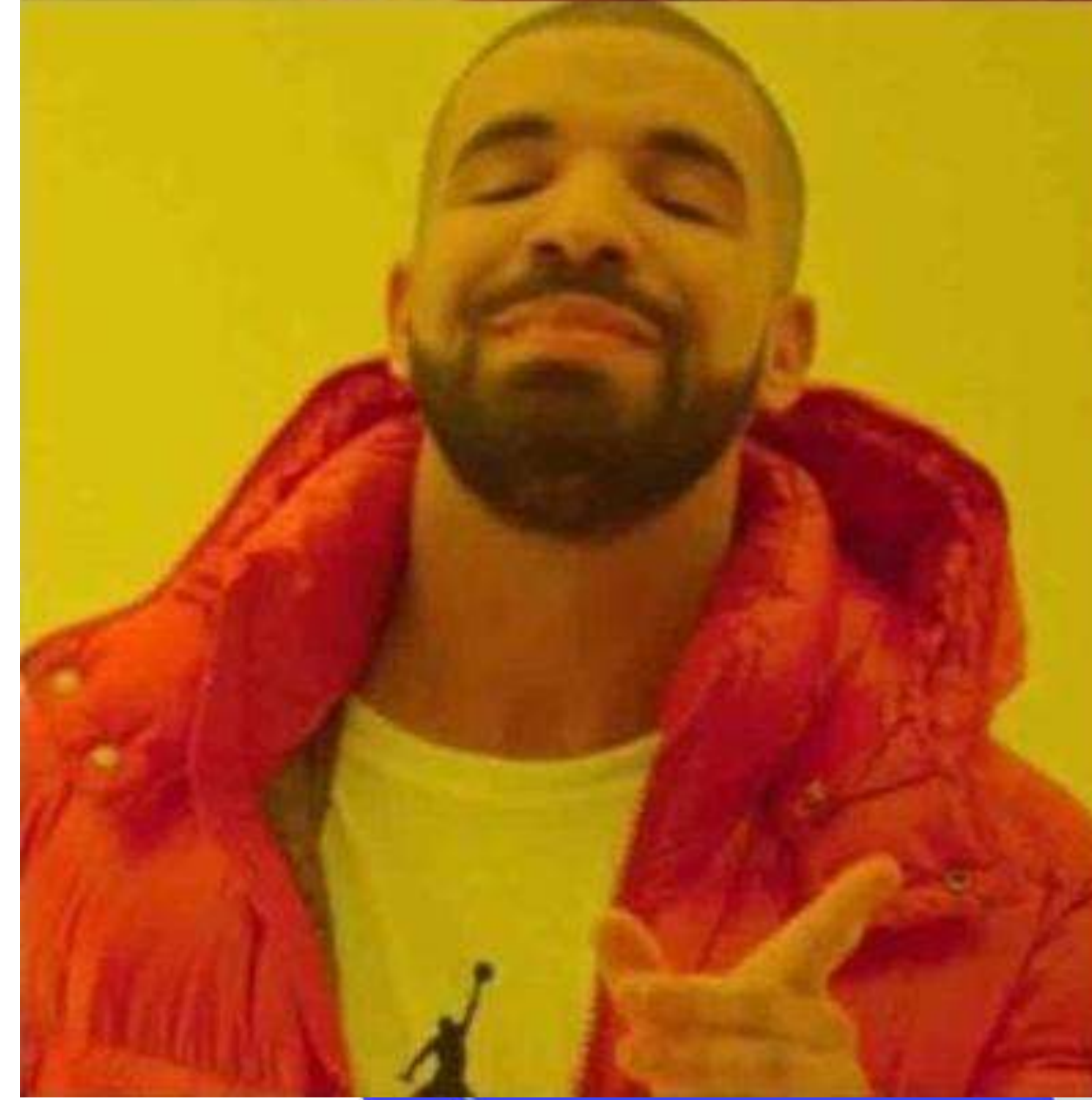


Autoregressive Visual Generation



Diffusion



Autoregressive

Common opinion in the community



Diffusion



AR models

Why AR visual models?

Making use of pretrained LLMs: even text-text pretrain is useful for visual generation

Why AR visual models?

Making use of pretrained LLMs: even text-text pretrain is useful for visual generation

Not AR vs Diffusion, but **AR + Diffusion**



e.g.

Long video generation: diffusion for spatial dimensions and AR for temporal

Outline

01

Classical visual AR modeling

02

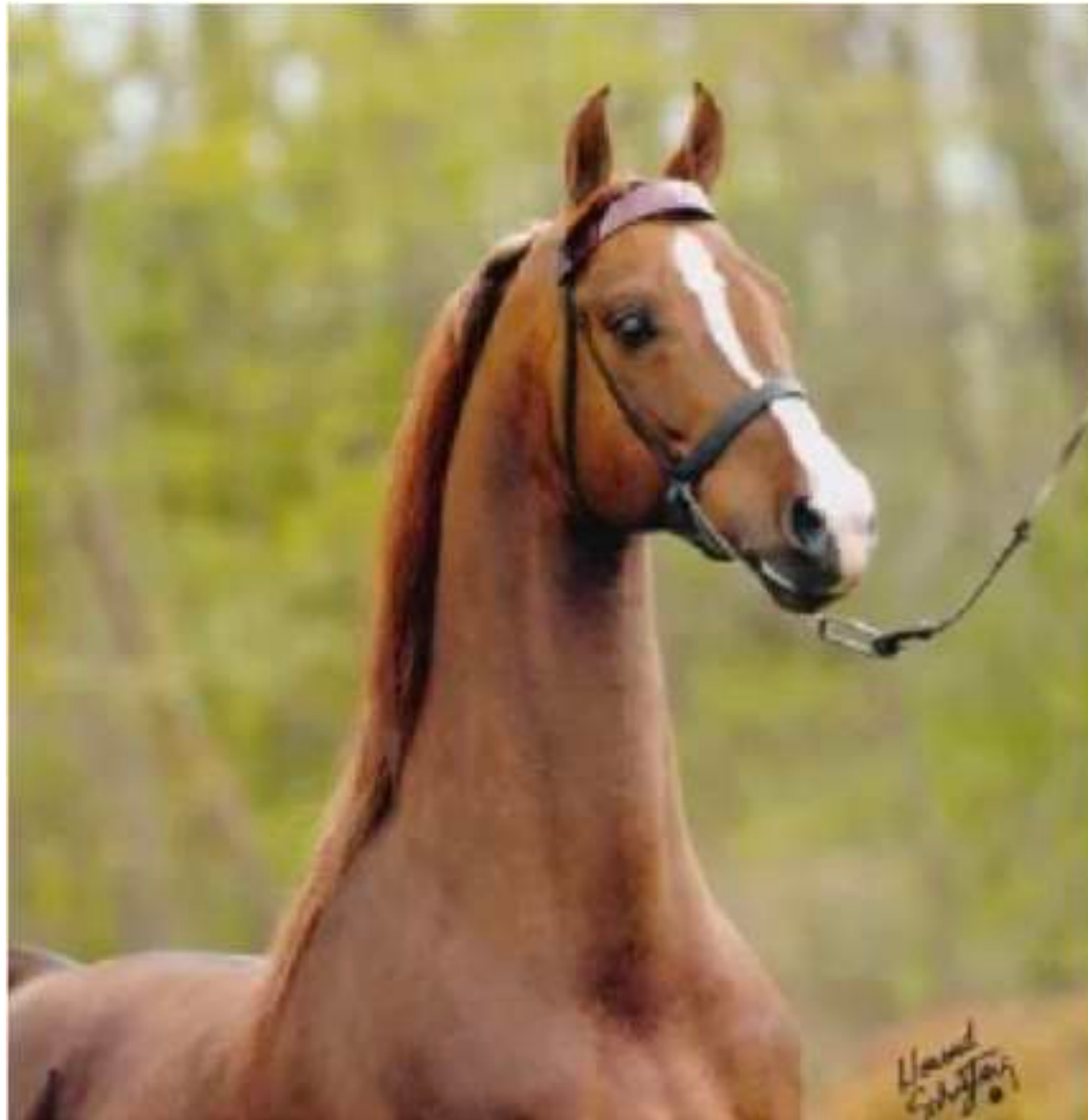
Next-scale prediction modeling

03

Continuous AR modeling

Image tokenization

Similar to LLMs, we first need to represent **images as a sequence of discrete tokens**



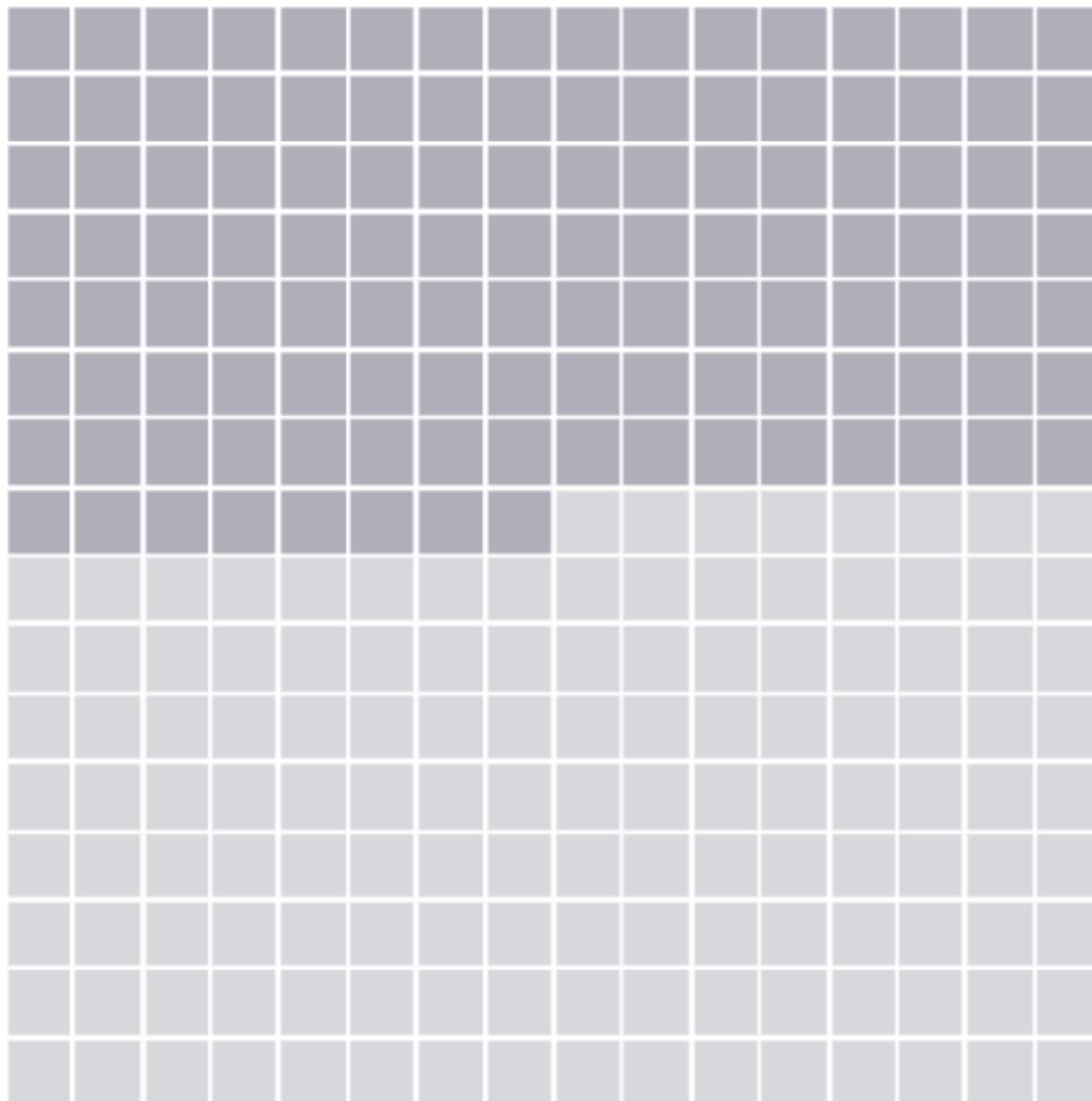
Discrete tokens

Naive image tokenization

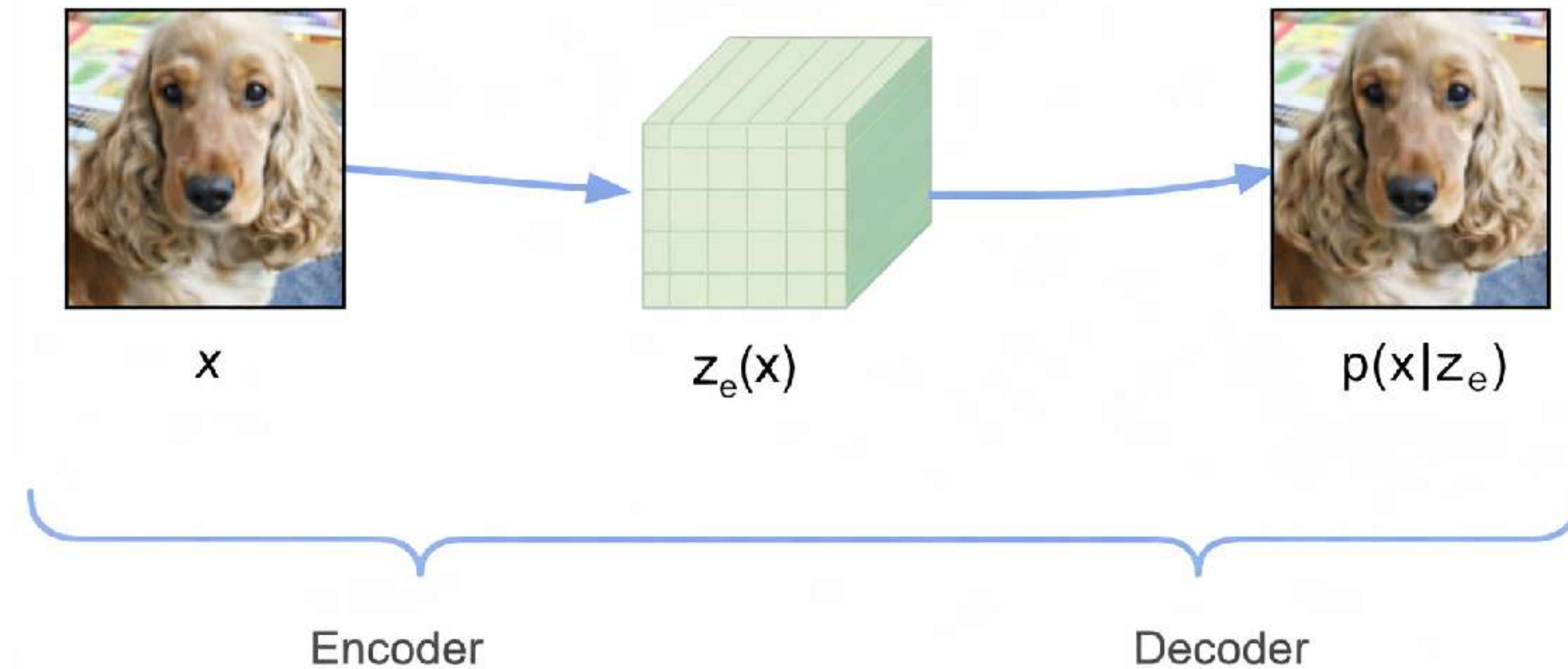
Pixels: each pixels is a discrete value in $[0, 255]$

Extremely expensive even
for low-resolution images

Raster-order image generation

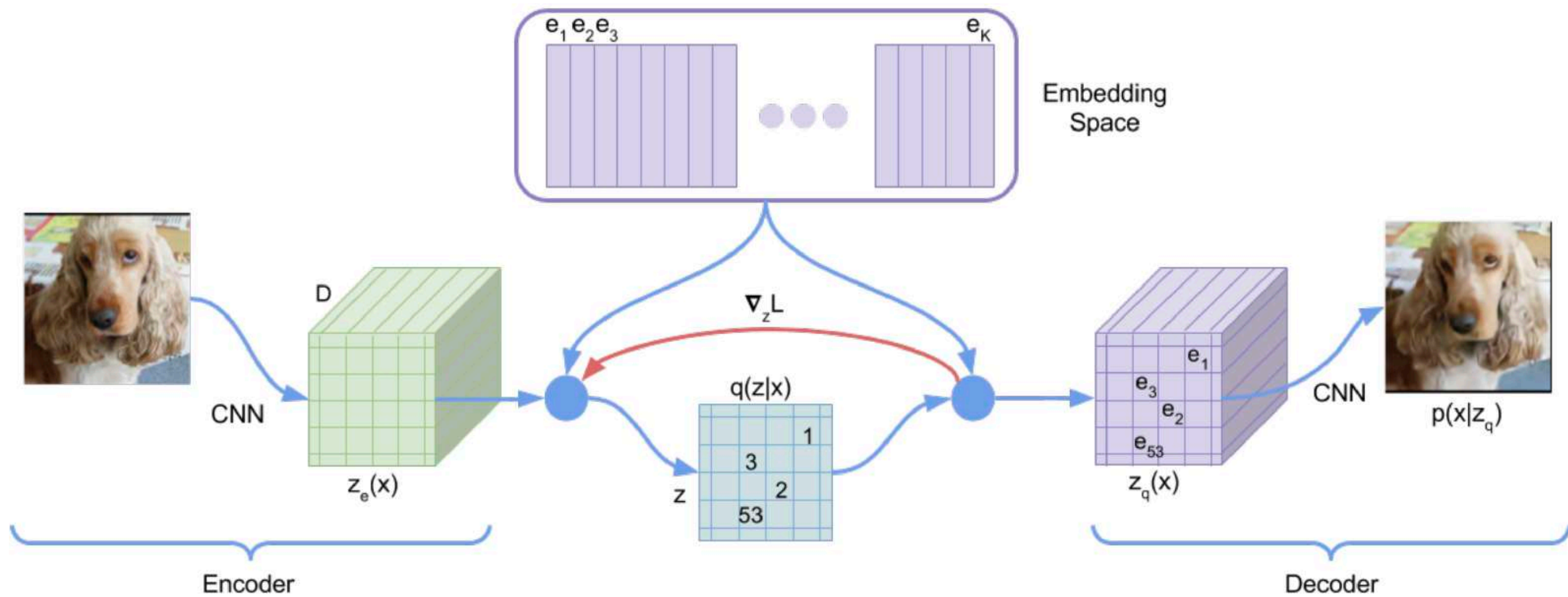


Continuous Autoencoders

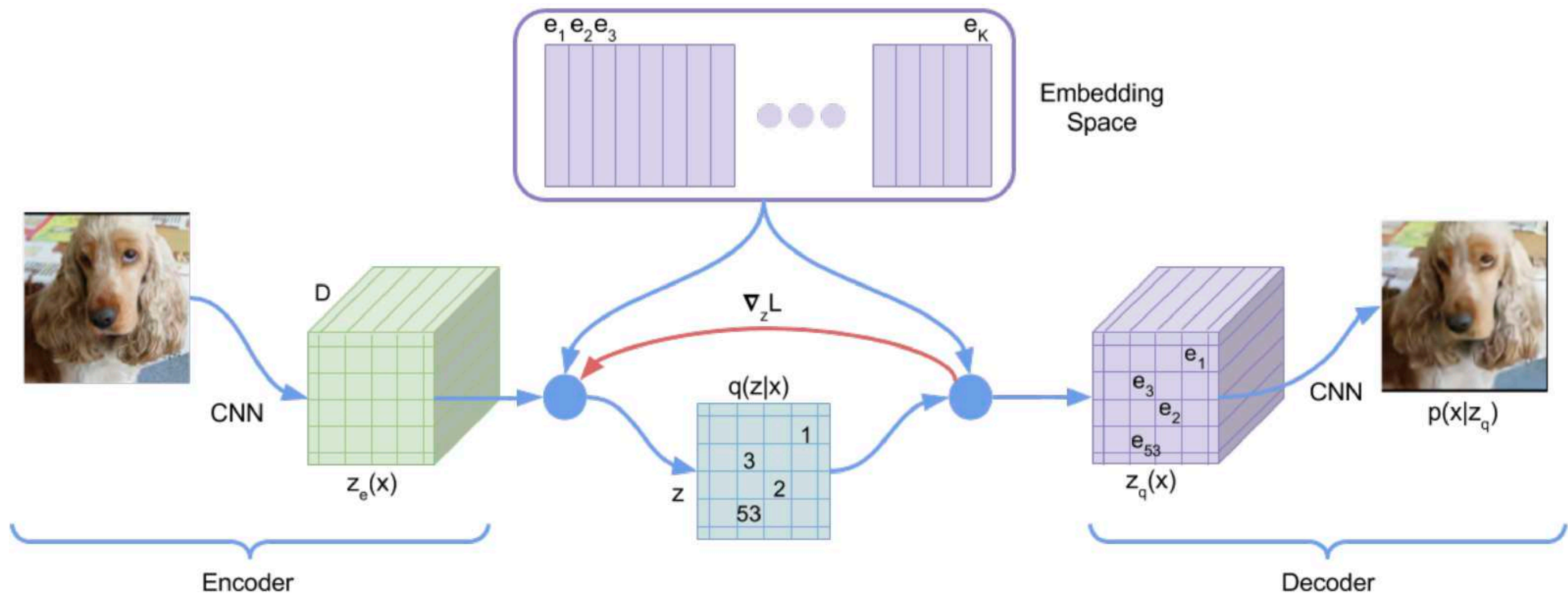


Reconstruction loss: $L = -\log p(\mathbf{x} \mid \mathbf{z}_e(x))$, e.g., MSE.

Vector Quantization Autoencoders



Vector Quantization Autoencoders



VQ-VAE loss: $L = -\log p(\mathbf{x} | \mathbf{z}_q(x)) + \left\| \text{sg}[\mathbf{z}_e(x)] - e \right\|_2^2 + \beta \left\| \mathbf{z}_e(x) - \text{sg}[e] \right\|_2^2$

VQ-VAE follow-ups

VQ-GAN — VQ-VAE but add GAN and LPIPS losses for better reconstruction

ViT-VQ-GAN

— Convolutional AE → ViT-AE (Not critical)

Address bad codebook utilization:

- Normalize latents and codebook vectors
- Reduce dimensionality of codebook vectors

In practice: convolutional VQ-GAN + ViT-VQ-GAN tricks

Other discrete VAEs

REINFORCE / Log-derivative trick — universal way to pass gradients through sampling

Bad convergence even with variance reduction tricks

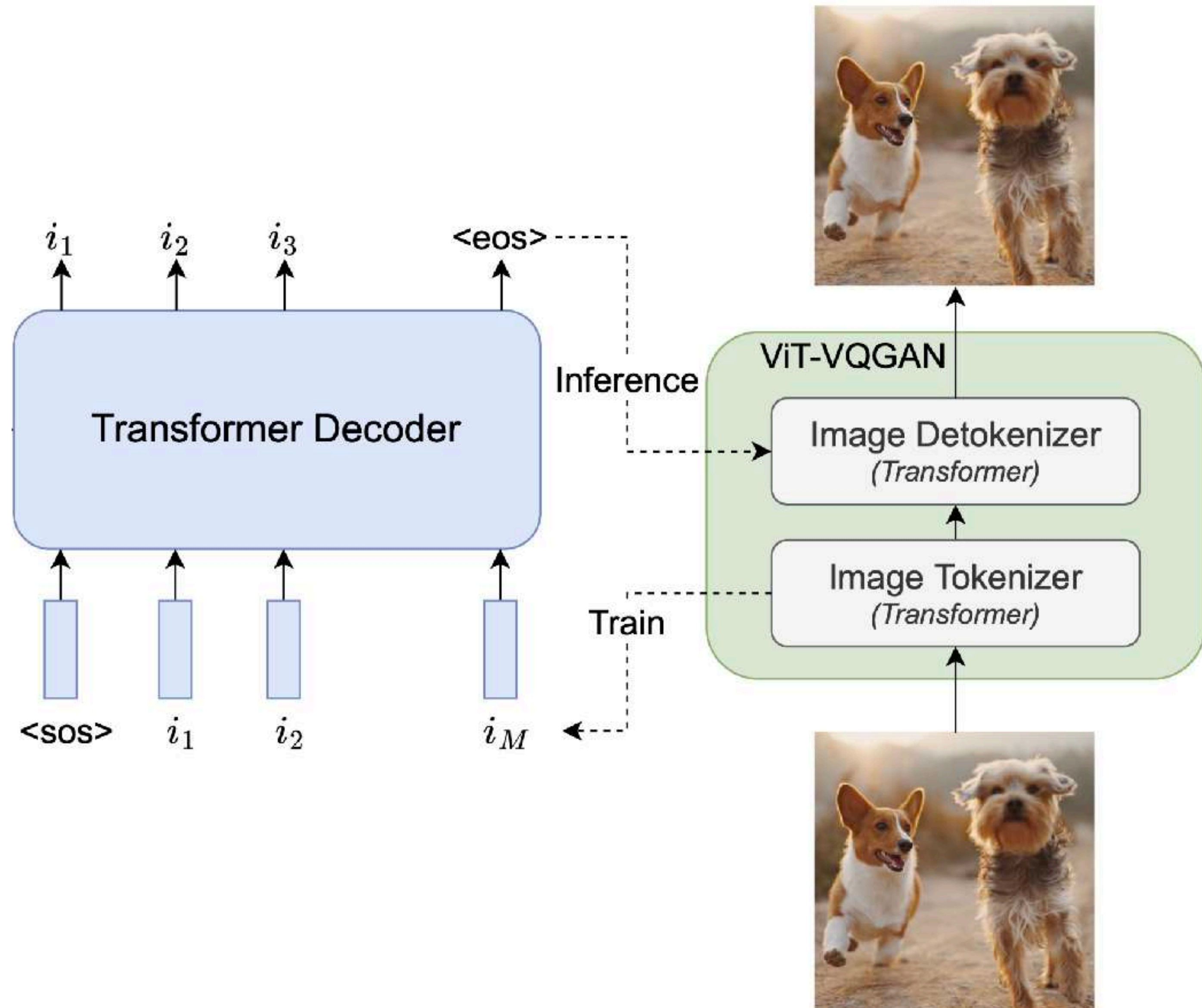
Gumbel-Softmax distribution — relaxing categorical distribution

— Getting continuous samples and pass gradients using reparameterization trick

— *Straight-Through Gumbel estimator*: decoder takes categorical (hard) samples but the gradients are passed via the relaxed samples

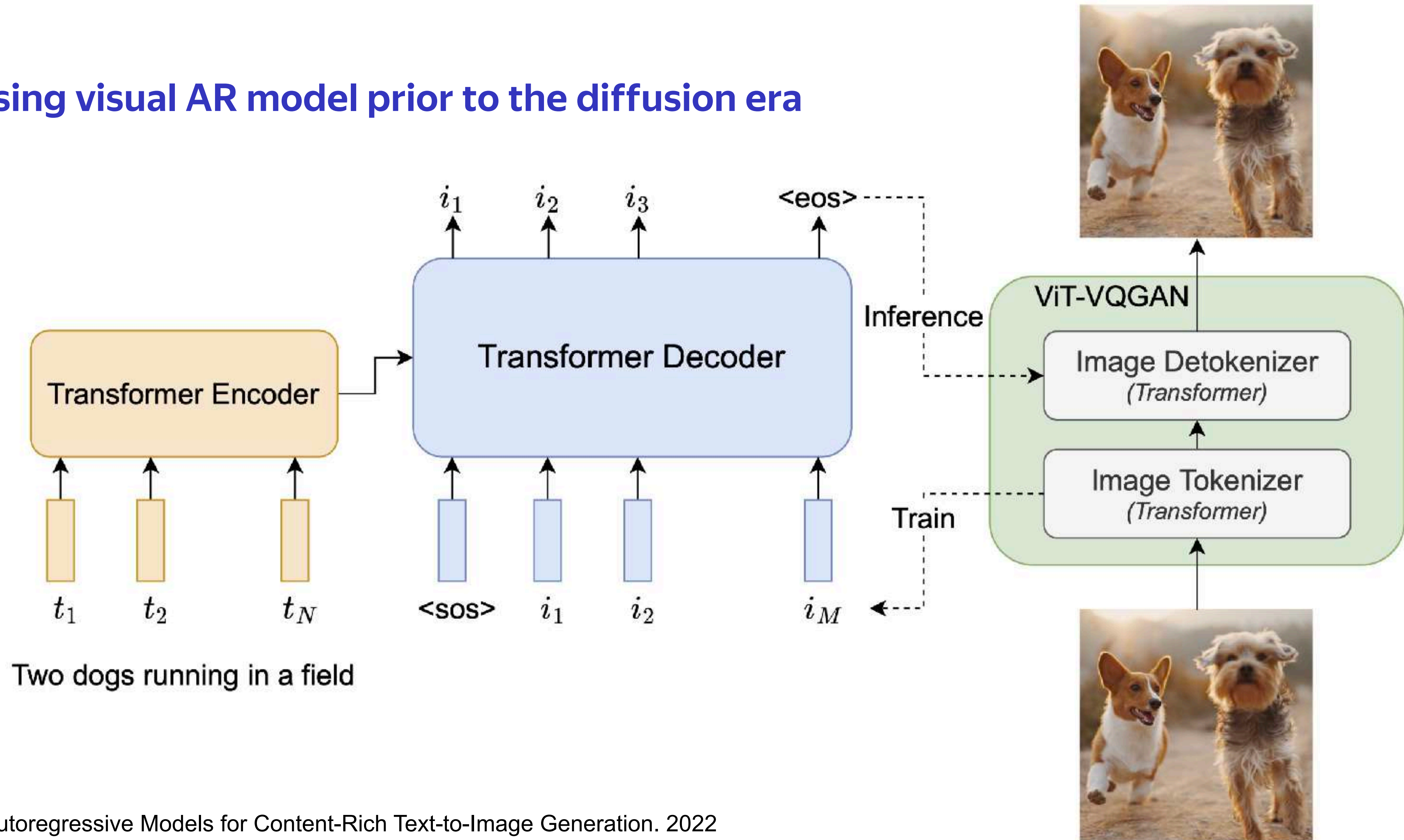
Gumbel-Softmax can provide comparable results to VQ-like methods, if properly trained

Image AR generation



Text-to-image AR generation | Parti

Last promising visual AR model prior to the diffusion era





A FEW

~~MOMENTS~~ LATER

YEARS

LlamaGen

Using more recent LLM architectures

RMSNorm pre-normalization

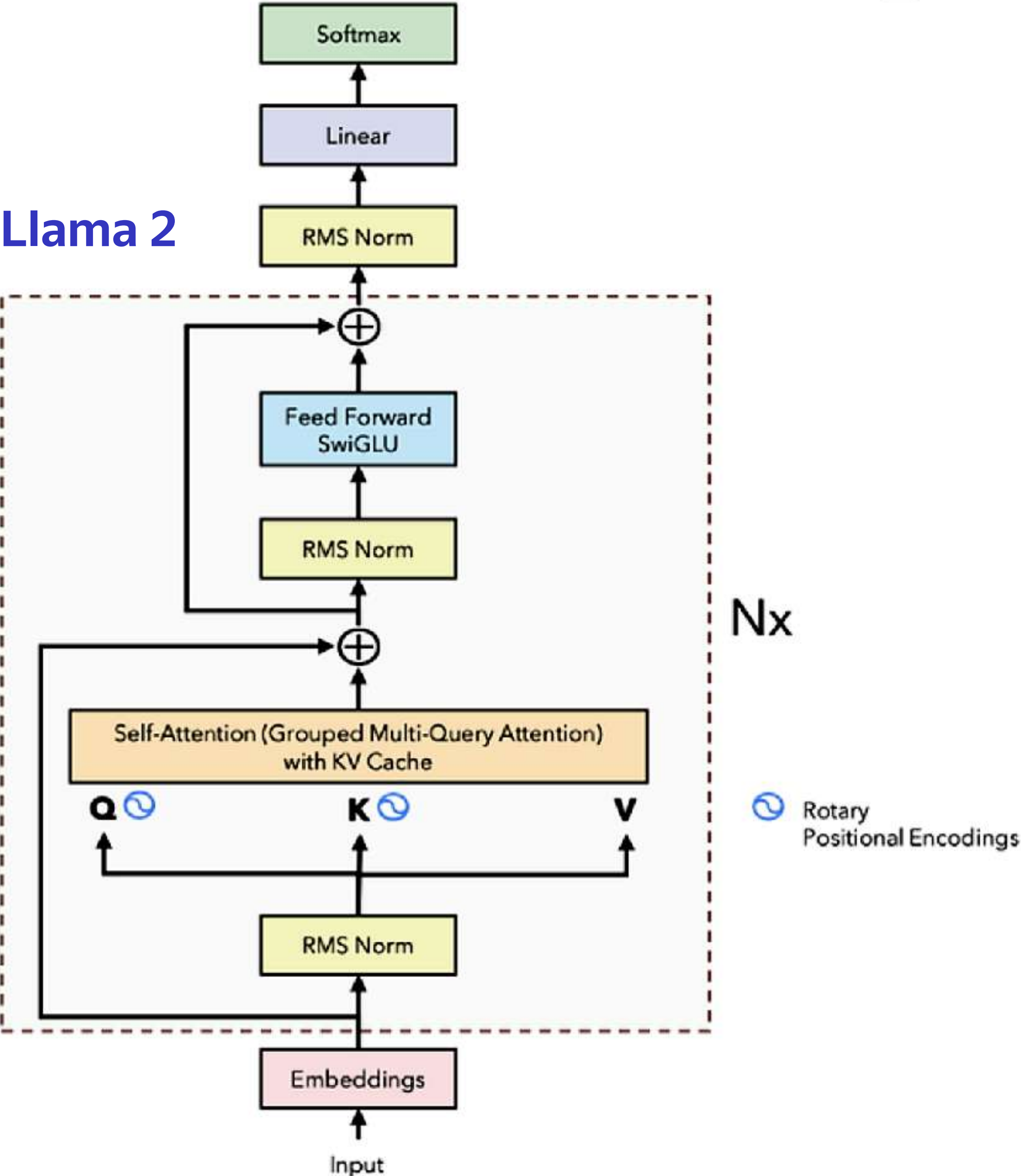
2D RoPE

In-context conditioning

SwiGLU

Moving closer to diffusion but no...

Llama 2



Sampling with discrete models

All typical sampling techniques from LLMs are valid here

Top-k and Top-p (nucleus) sampling

Temperature in softmax

Beam search *but I didn't see it in practice*

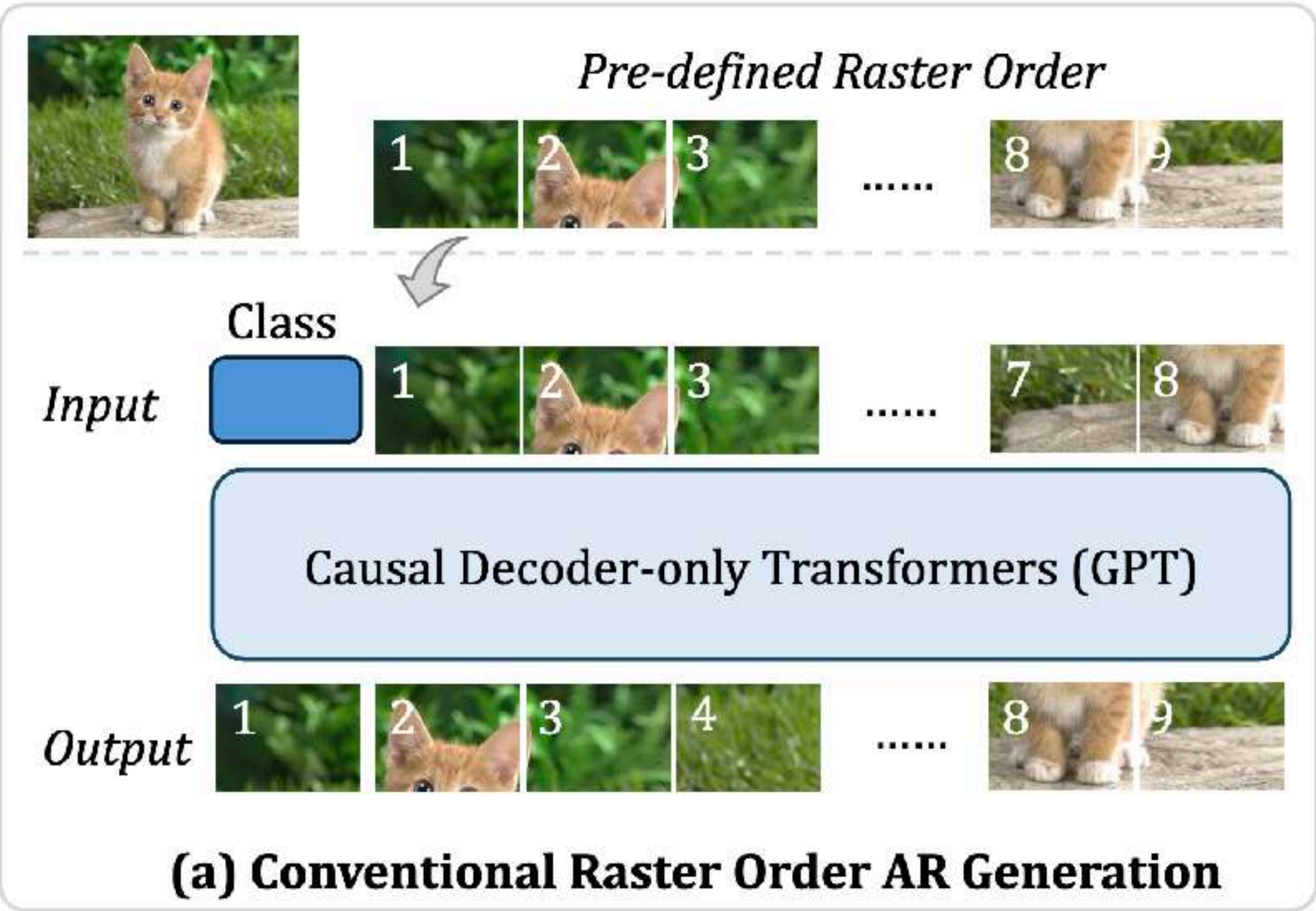
There are more image specific techniques

CFG: $\text{logits}_{cf\text{g}} = \text{logits}_{uncond} + \gamma \cdot (\text{logits}_{cond} - \text{logits}_{uncond})$

Gumbel-softmax averaging *quite specific thing, never mind*

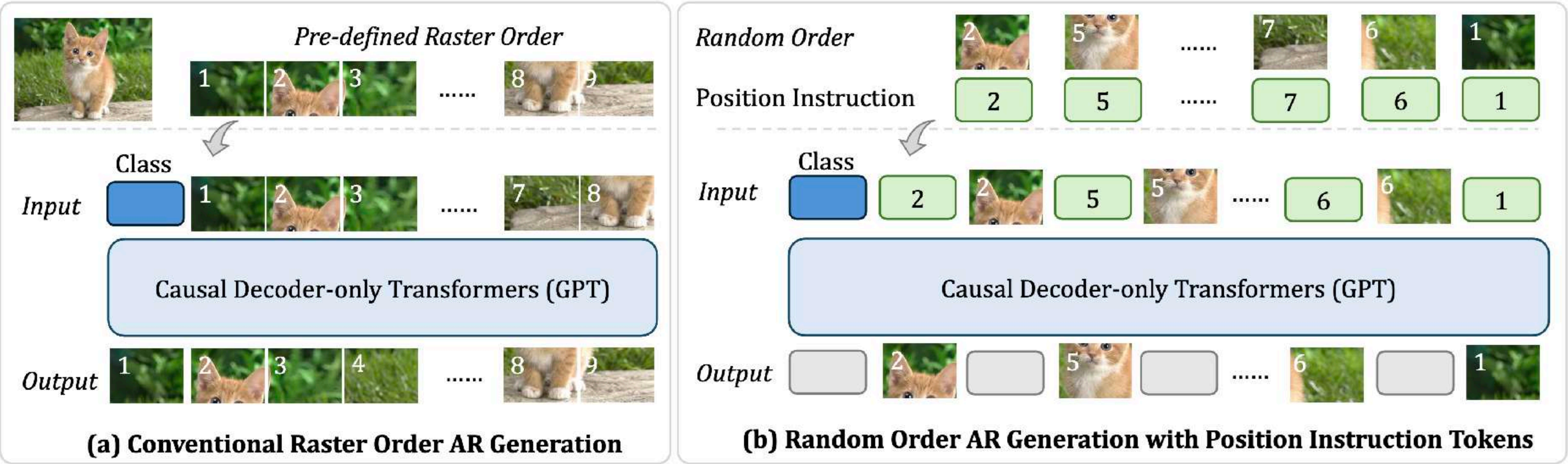
Raster-order AR models

LlamaGen performs raster-order AR generation



Any-order AR models

RandAR enables AR generation in any orders by introducing **Position Instruction Tokens**

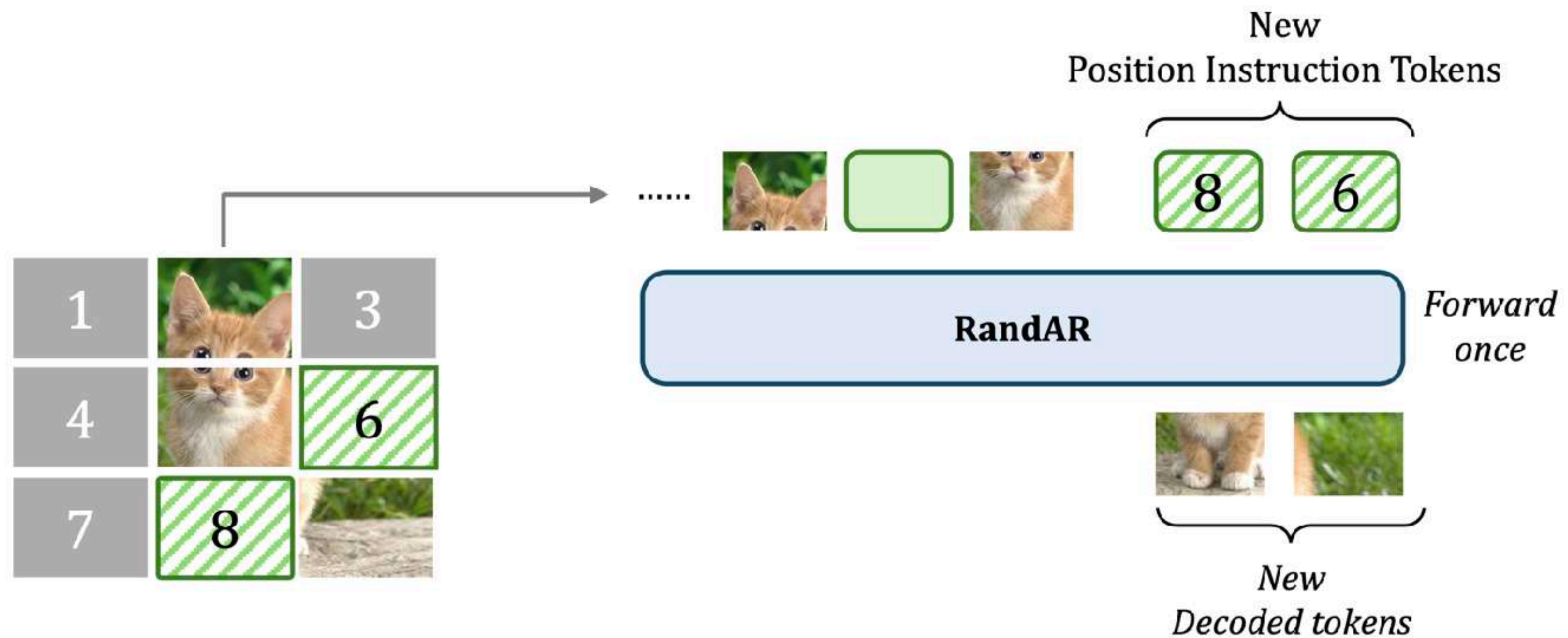


Any-order AR models

Does any order improve image quality? No, but...

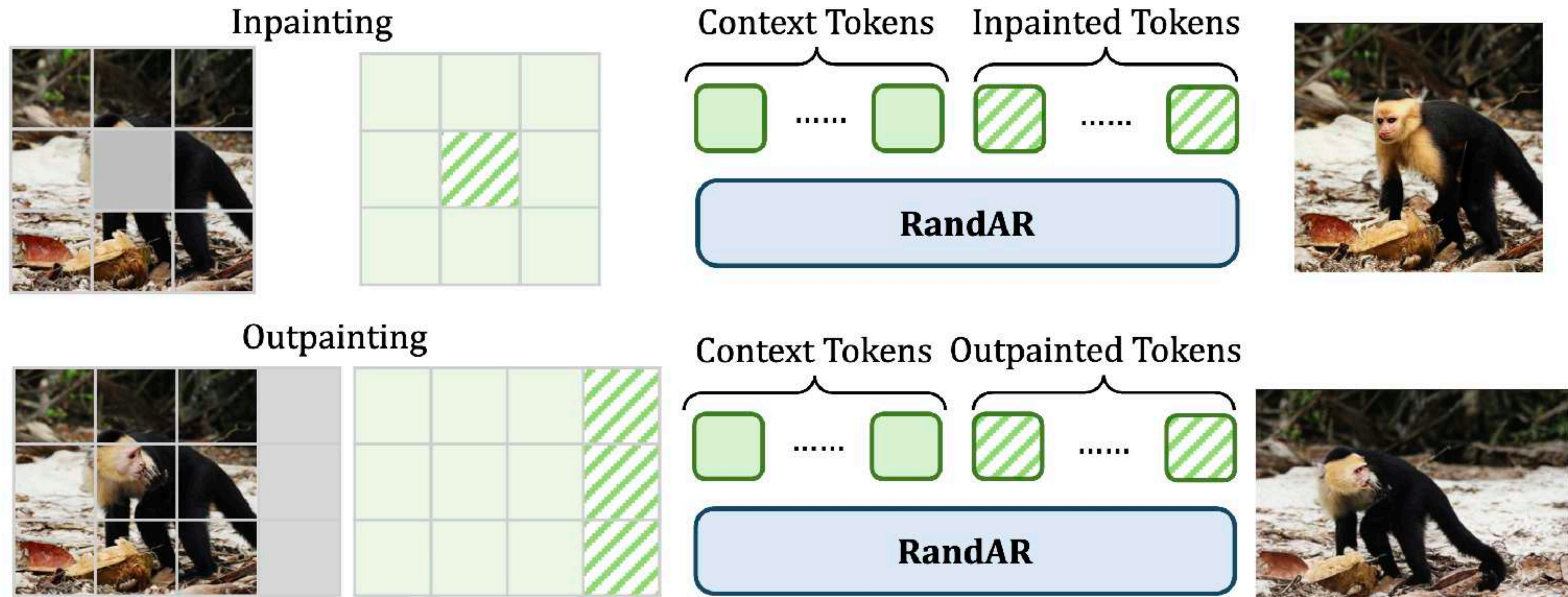
Any-order AR models

Does any order improve image quality? No, but **parallel decoding**



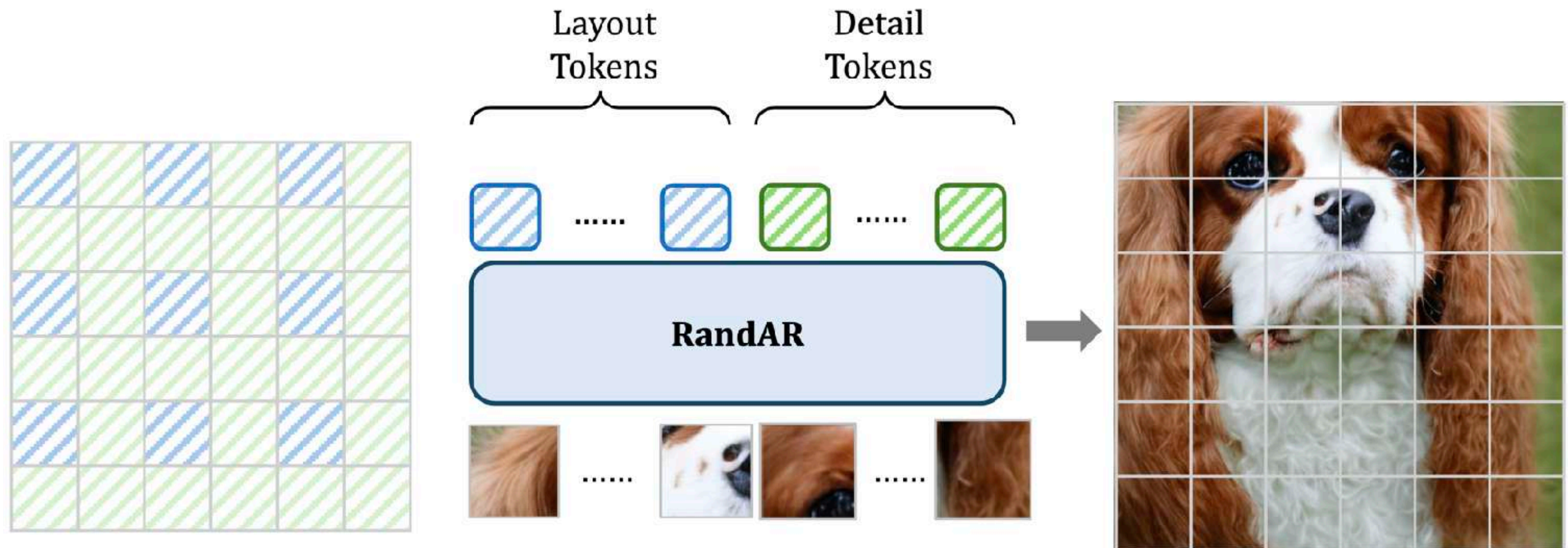
Any-order AR models

Does any order improve image quality? No, but **inpainting & outpainting**



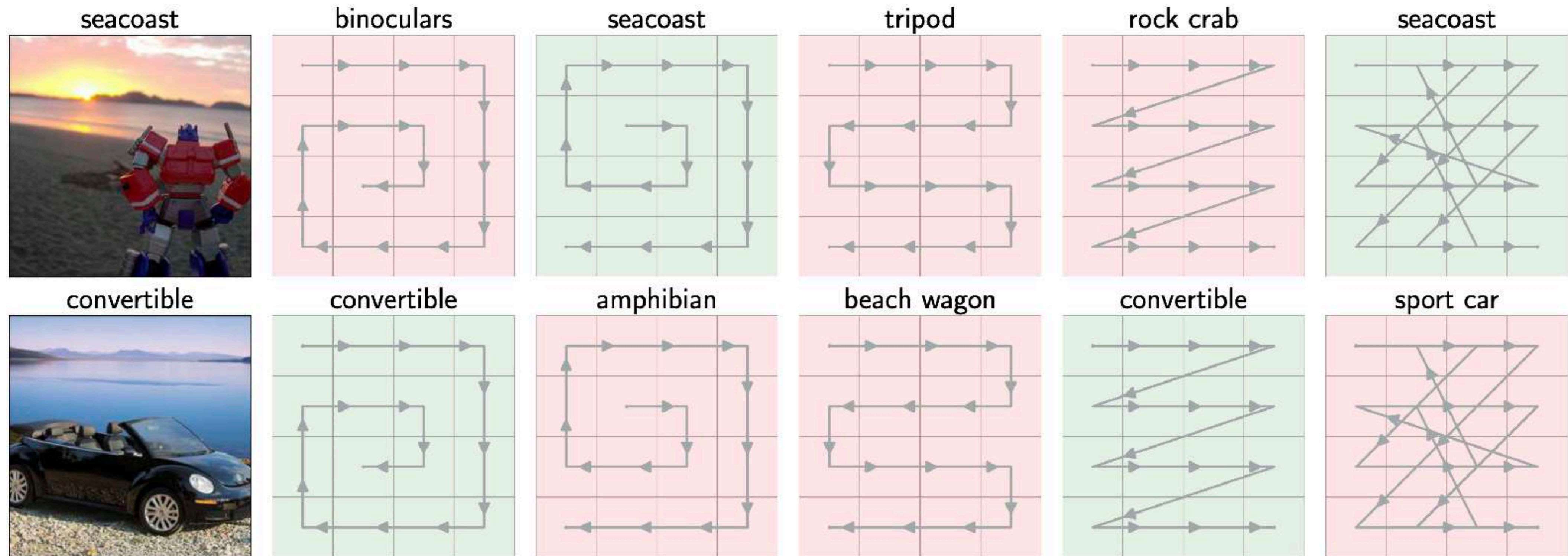
Any-order AR models

Does any order improve image quality? No, but **zero-shot resolution extrapolation**



Any-order AR models

Does any order improve image quality? No, but **better AR-based image classification**

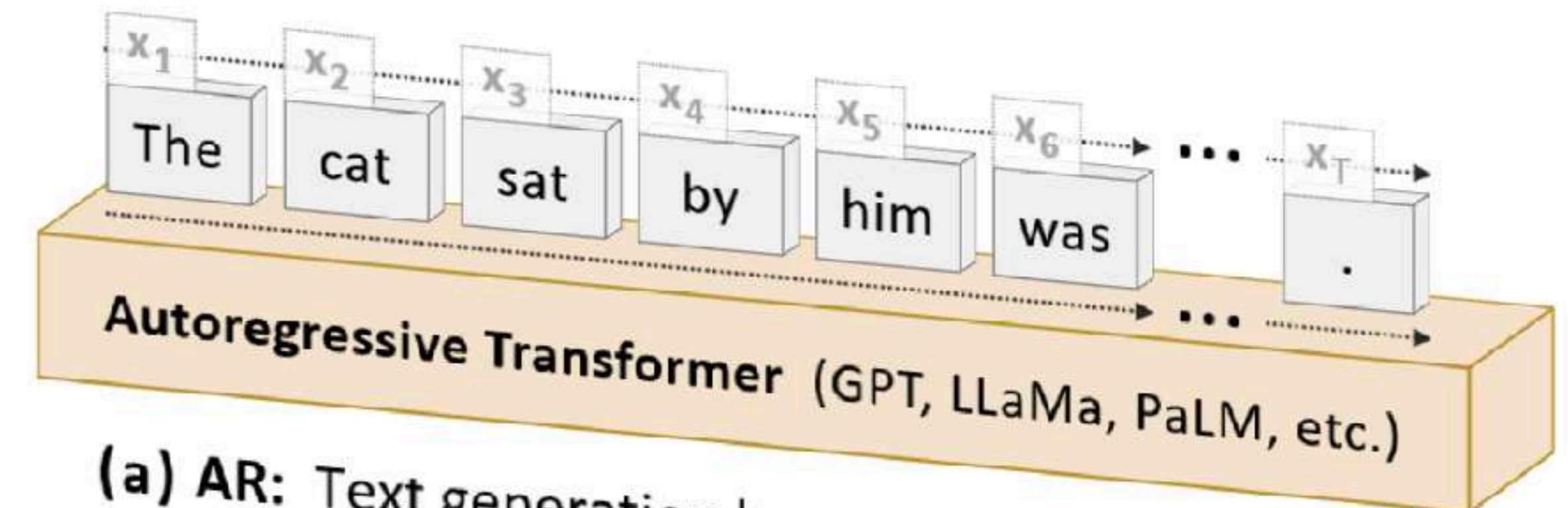


Why GPT-like AR is bad for visual domains?

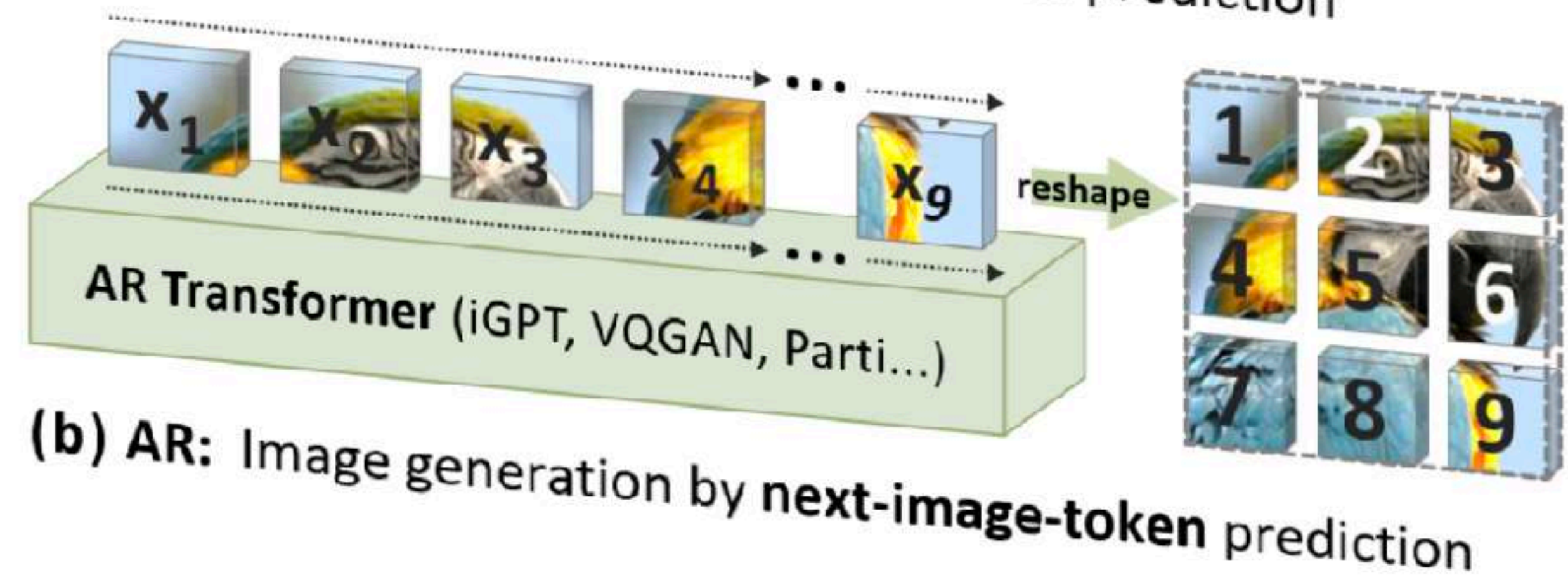
Next-token prediction is **unnatural for visual data**

Discrete representations are worse for both image reconstruction and generators

Low inference speed even in latent space



(a) AR: Text generation by **next-token** prediction



(b) AR: Image generation by **next-image-token** prediction

Checkpoint

GPT-like AR

Unnatural inductive bias

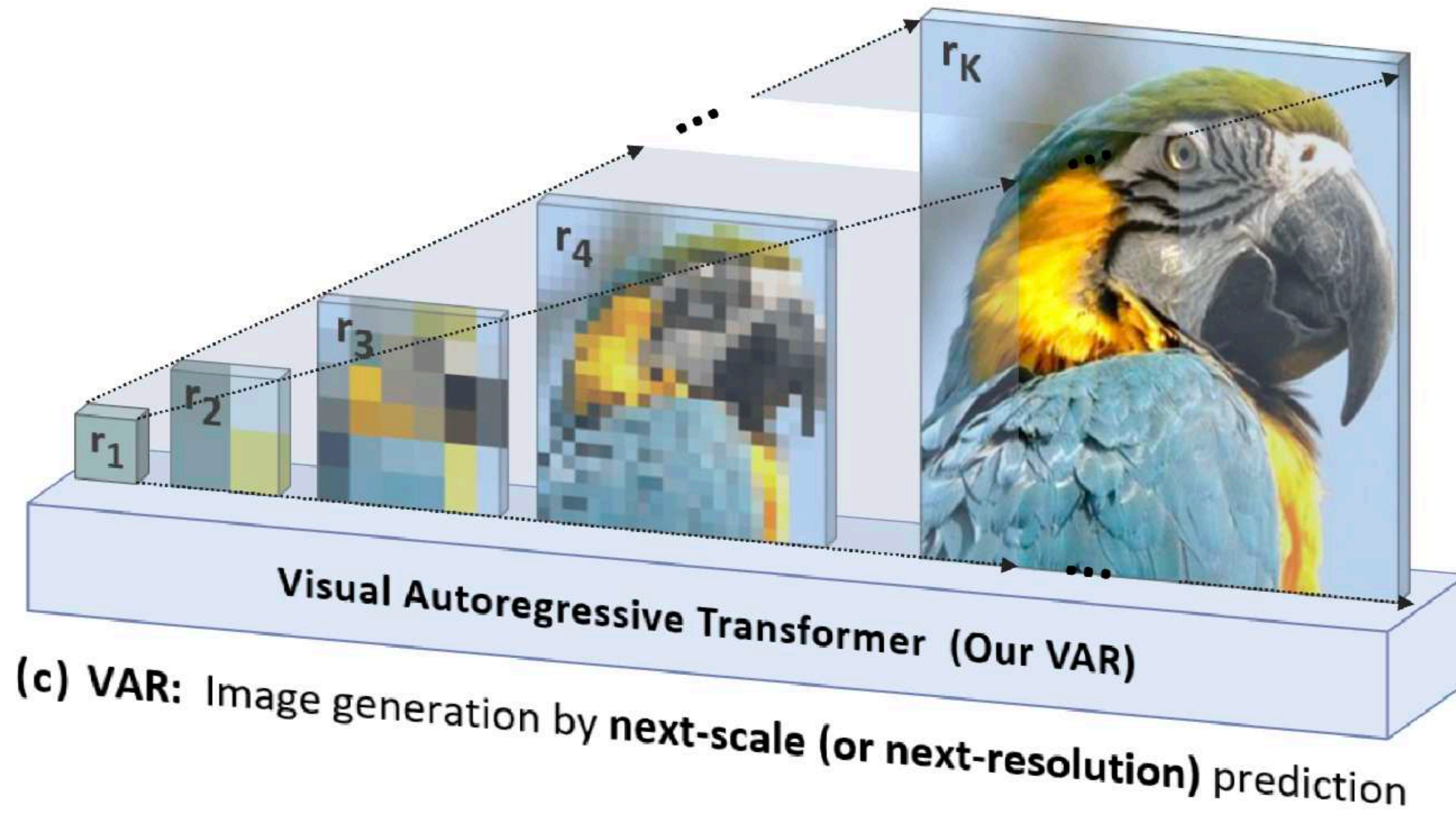
Discrete

Slow inference

Next-scale prediction AR models

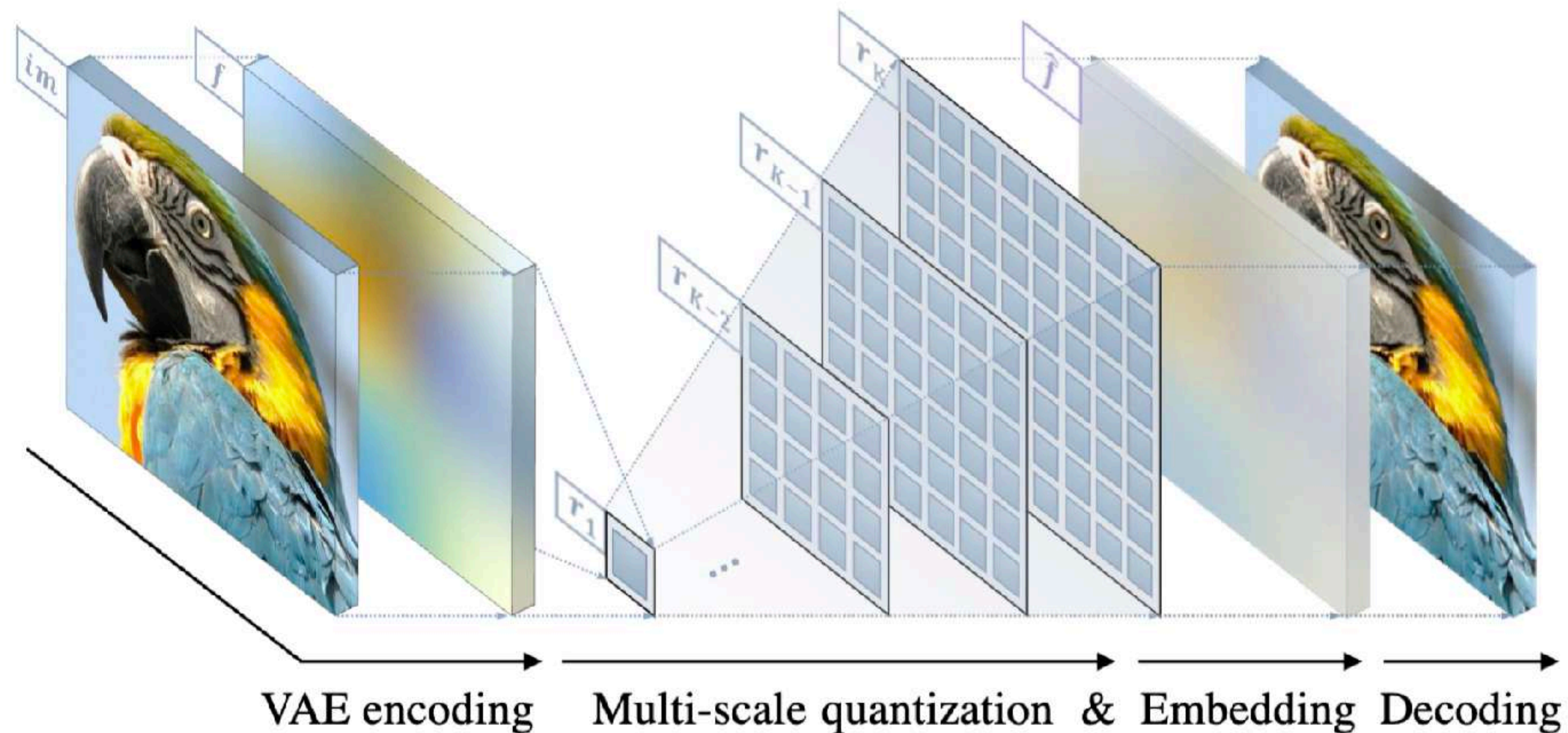


Scale-wise Visual AR Generation

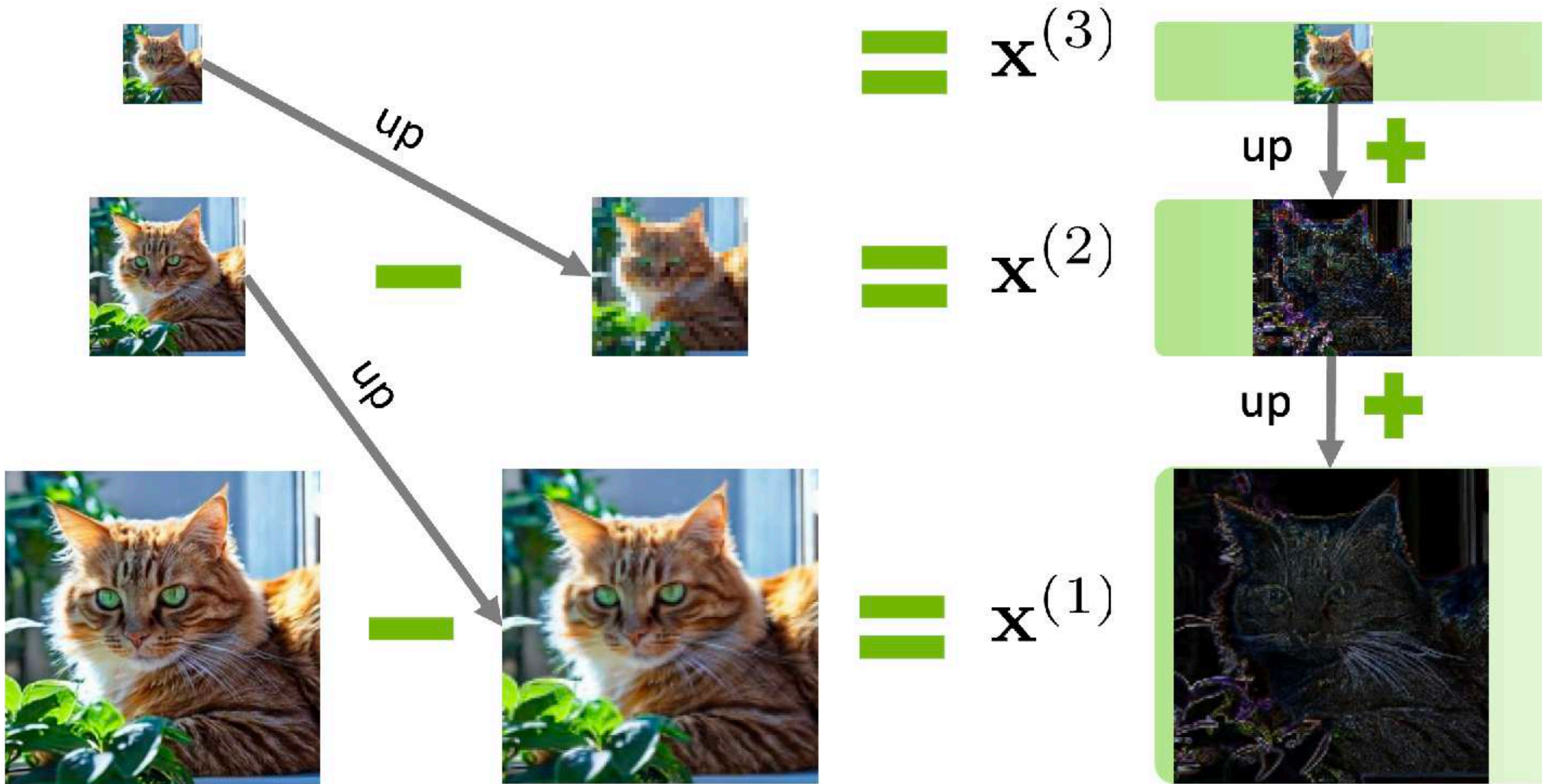


Hierarchical multi-scale VQ-VAE

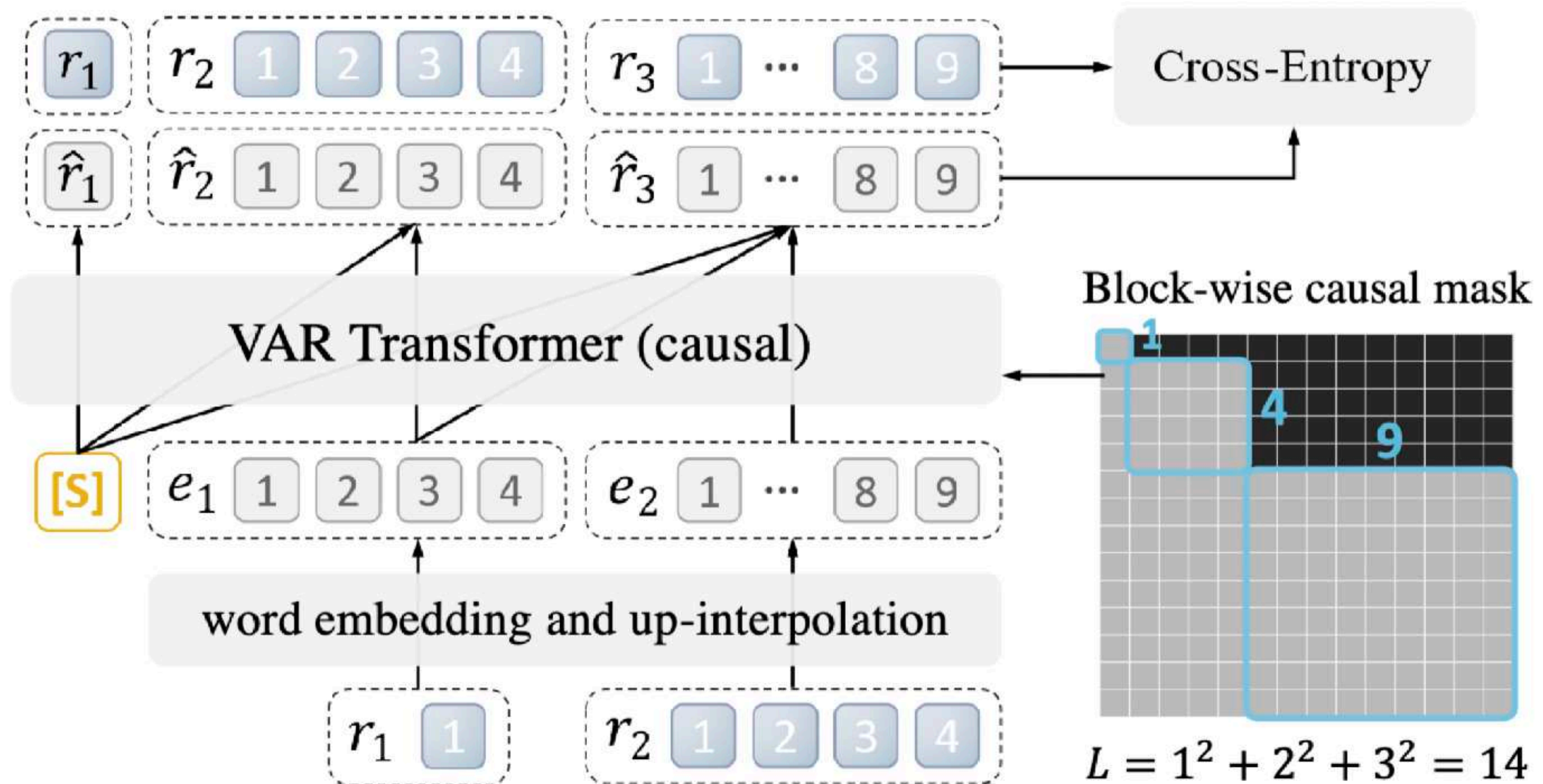
Residual Quantization: $\hat{\mathbf{f}} = up(\mathbf{r}_1, 8\times) + up(\mathbf{r}_2, 4\times) + up(\mathbf{r}_3, 2\times) + \mathbf{r}_4$



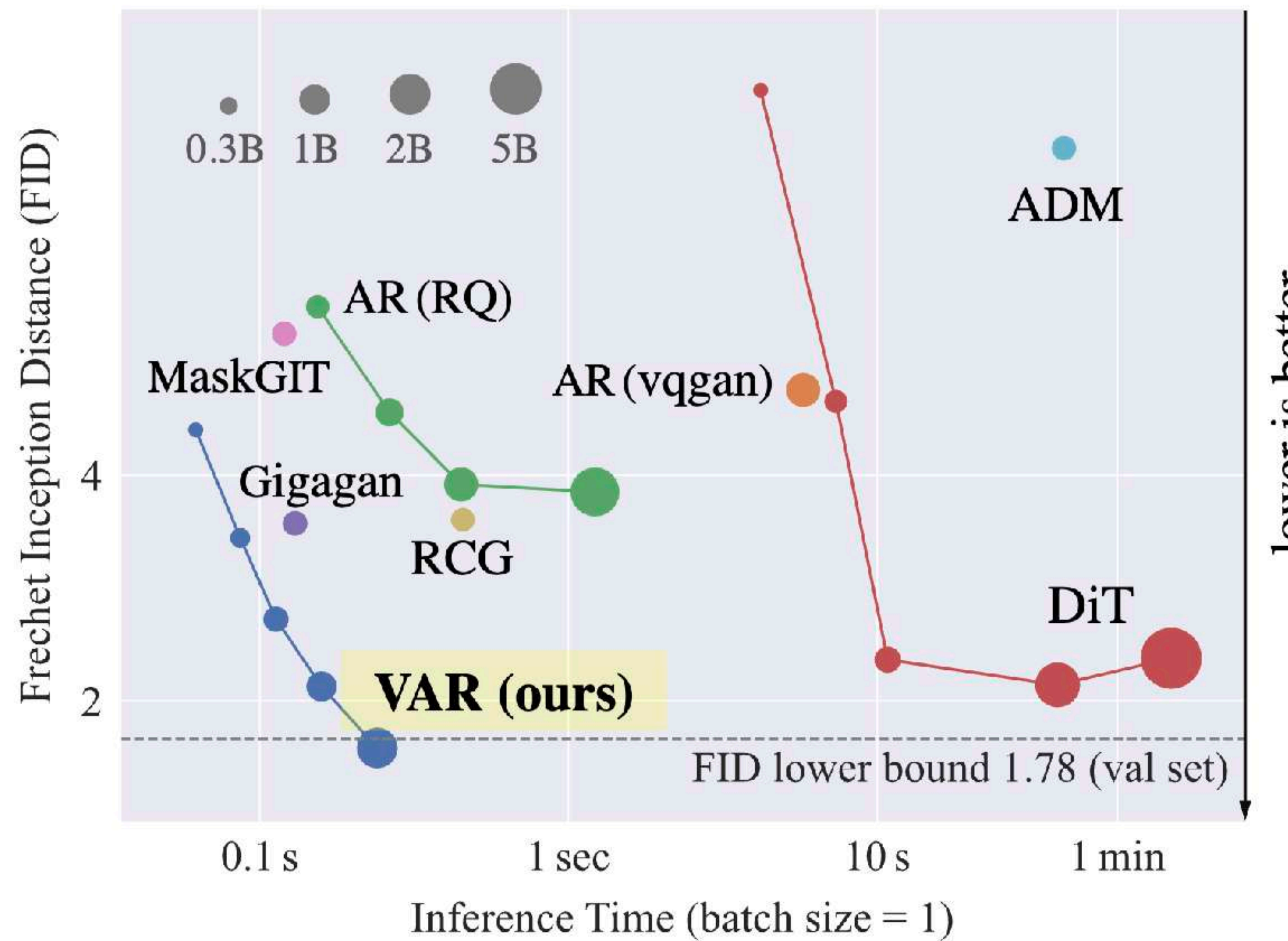
Laplacian decomposition



VAR training

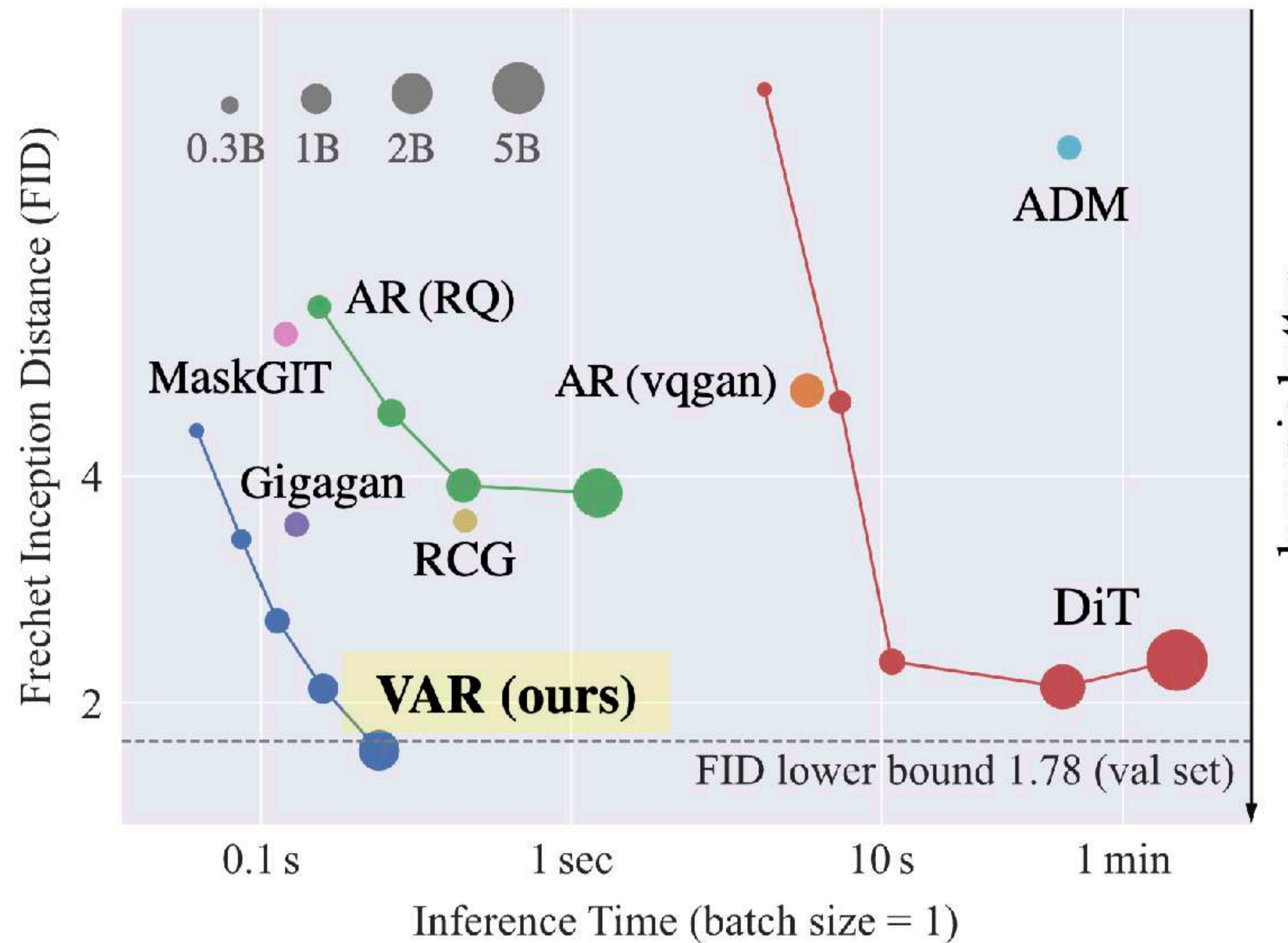


VAR results



VAR results

What about text-to-image generation?



SWITTI: Designing Scale-Wise Transformers for Text-to-Image Synthesis

Anton Voronov^{1,2,3}

Denis Kuznedelev^{1,4}

Mikhail Khoroshikh⁶

Valentin Khrulkov^{1,5}

Dmitry Baranchuk¹

¹Yandex Research

²HSE University

³MIPT

⁴Skoltech

⁵AIRI

⁶ITMO

<https://yandex-research.github.io/switti>



Figure 1. SWITTI produces high quality and aesthetic 1024×1024 image samples in around 0.5 seconds.

Designing and scaling T2I VAR

Goal: design and train first T2I VAR

Project directions

T2I VAR Architecture

*CLIP text conditioning
via CA and AdaLN*

Non-causal attention

*“Sandwich” normalization
for stability*

Multi-scale RoPE

Large-scale training

2.5B model

100M dataset

320 A100 GPUs

256 → 512 → 1024 stages

SFT post-training

Analysis

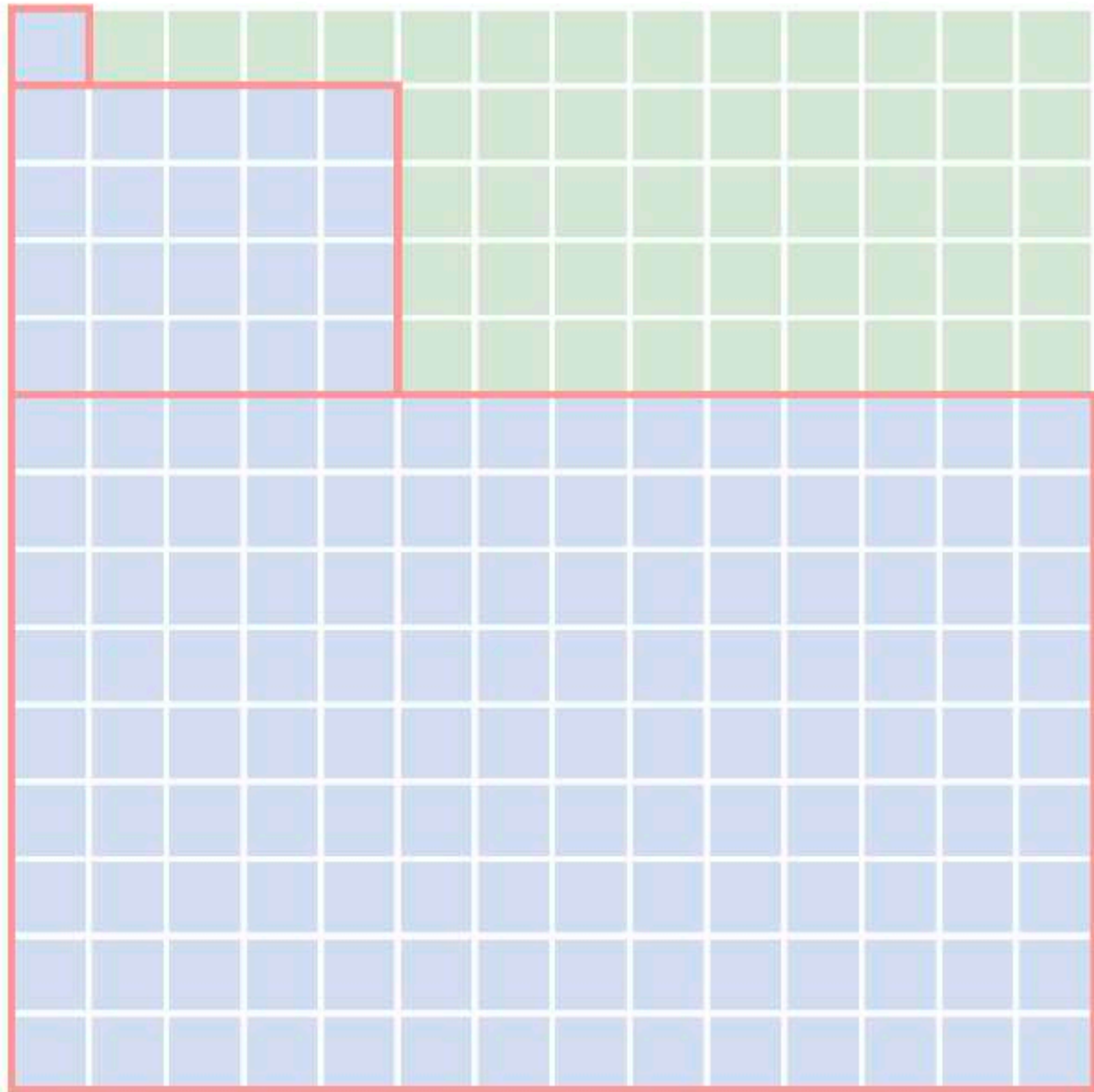
Activation norms

Attention behavior

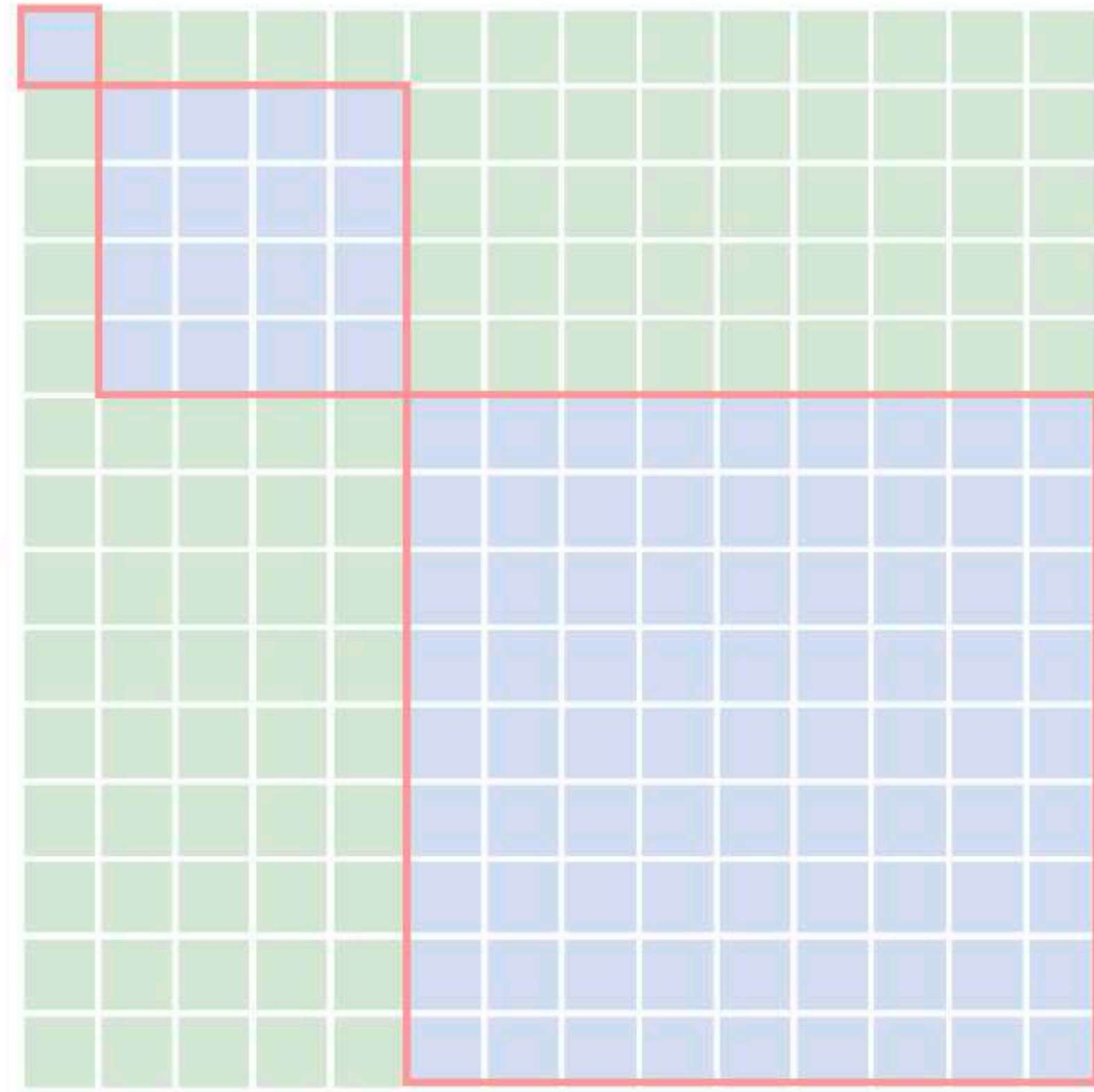
Role of text-conditioning

Switti is no longer **explicitly** autoregressive

Block-wise causal mask (VAR)



Block-wise non-causal mask (Switti)



Switti is **implicitly** autoregressive

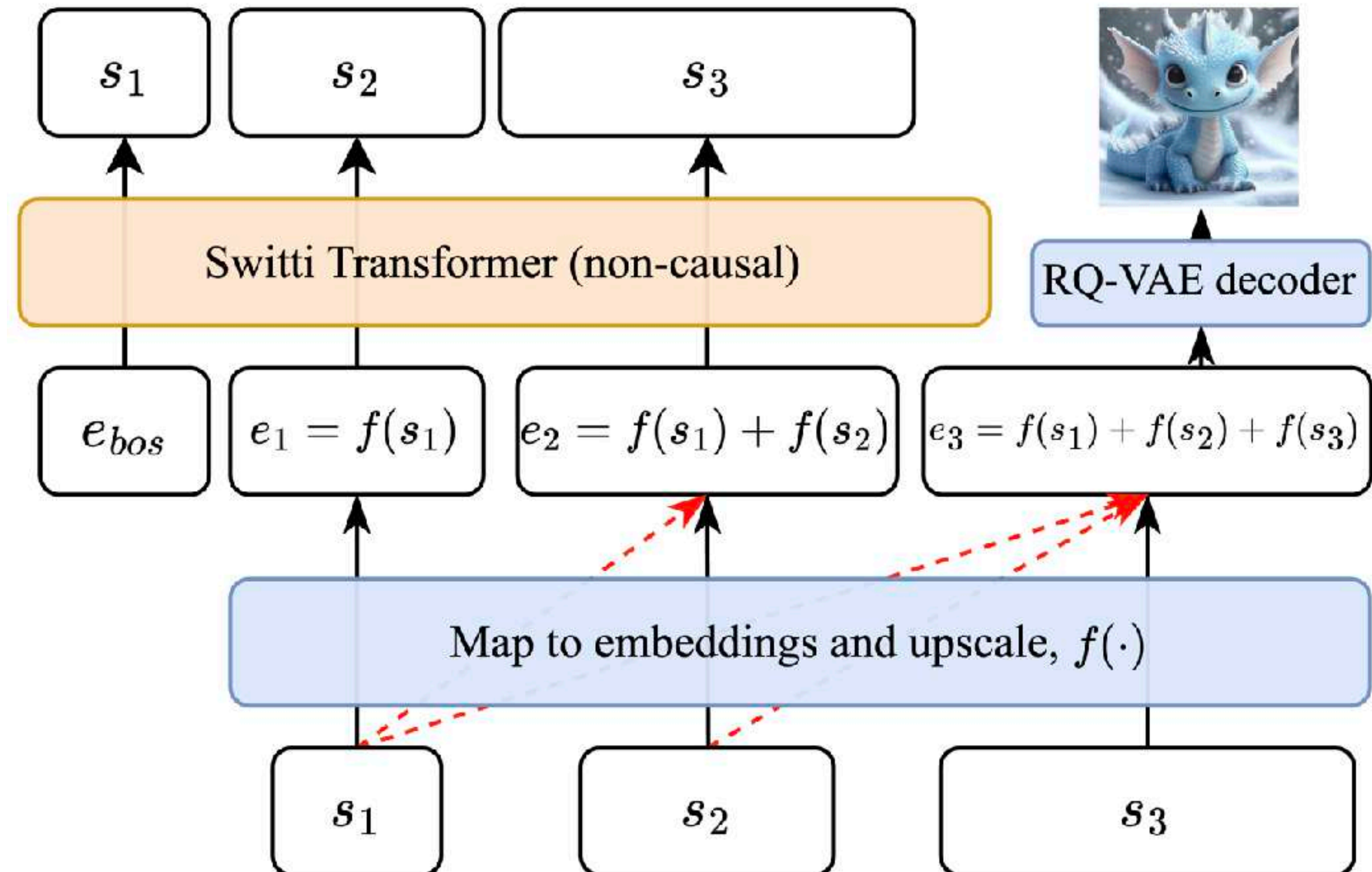
Flexible training

- No need to process all scales at once

More efficient inference

- No KV-cache (saves 2.4GB/image)
- Cheaper attention calculation (~21%)

Slightly higher quality



Effect of text conditioning

Weak text reliance
at last scales



Turn off CFG



~32% speedup

Less defects



VAR Summary

Highly natural visual inductive bias

Much faster than DMs by design

Still requires discrete tokenizers

Promising results but still needs much work to achieve top-tier DM performance



Checkpoint

GPT-like AR

Unnatural inductive bias

Discrete

Slow inference

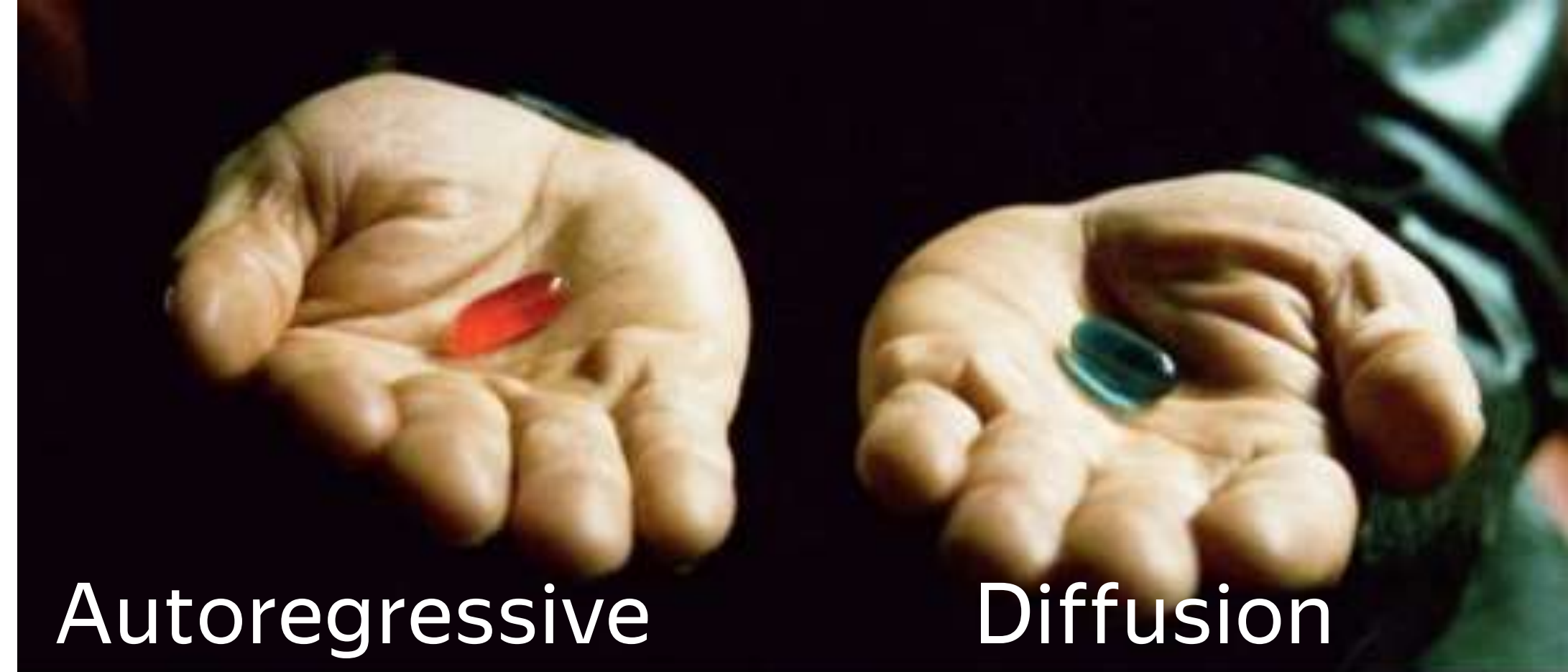
Next-scale prediction

Natural inductive bias

Discrete

Fast inference

Continuous AR models



Why discrete tokenizers?

Discrete AR models

For AR generation, we need to define $p(\mathbf{x} \mid \mathbf{z})$

$\mathbf{x}$ — discrete VQ token

$\mathbf{z}$ — predicted *continuous* token presentation prior to quantization

$p(\mathbf{x} \mid \mathbf{z})$ — categorical distribution: $\text{softmax}(W\mathbf{z}/\tau)$, where W — codebook, τ — temperature

Loss: cross-entropy

Sampling: Gumbel-Max or inverse transform sampling

What about continuous tokenizers?

Continuous AR models

For AR generation, we need to define $p(\mathbf{x} | \mathbf{z})$

$\mathbf{x}$ — *continuous* token, e.g., a latent patch 2×2

$\mathbf{z}$ — predicted *continuous* patch representation with the AR model

$p(\mathbf{x} | \mathbf{z})$ — ?

Loss: ?

Sampling: ?

What about continuous tokenizers?

Continuous AR models

For AR generation, we need to define $p(\mathbf{x} | \mathbf{z})$

$\mathbf{x}$ — *continuous* token, e.g., a latent patch 2×2

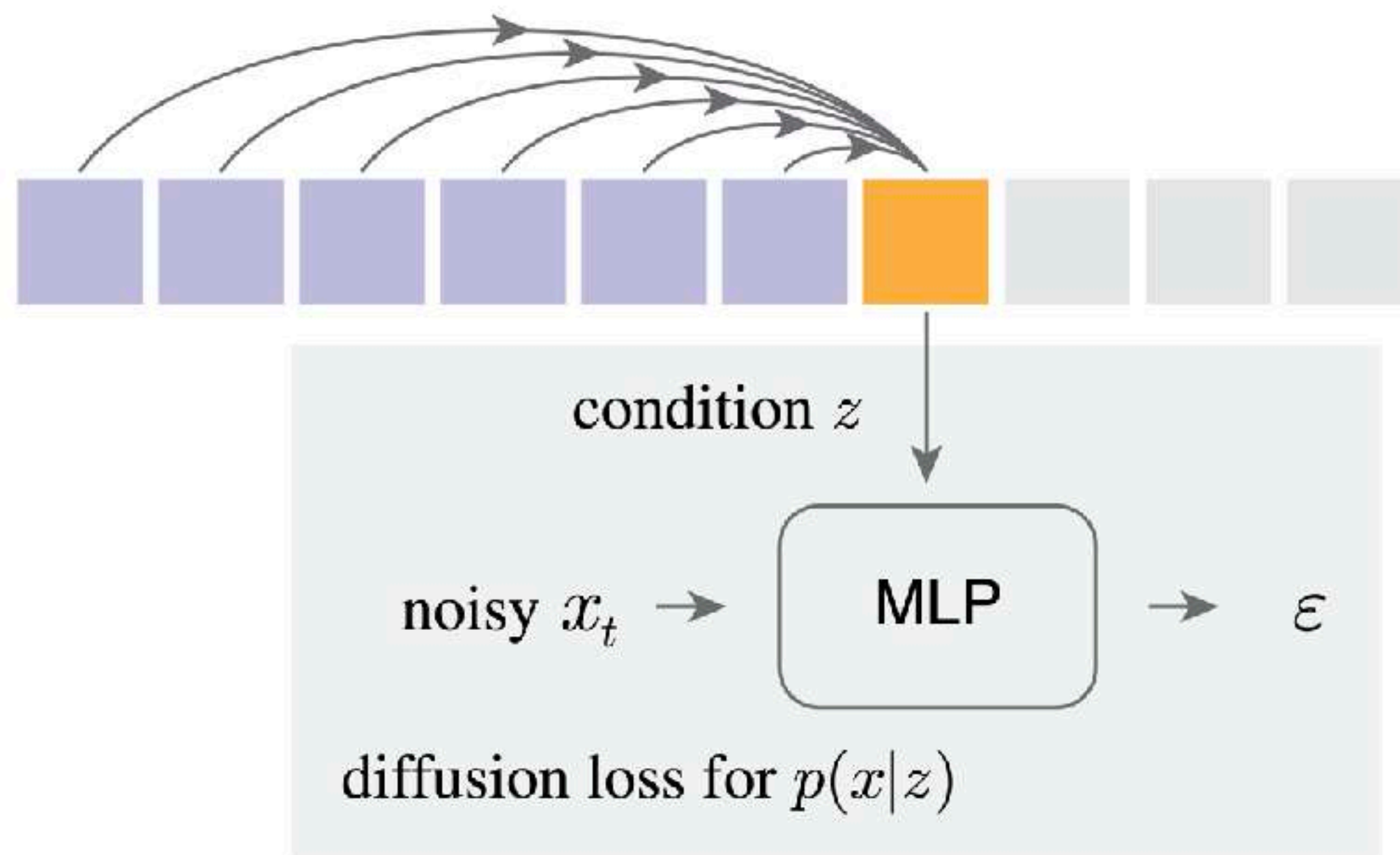
$\mathbf{z}$ — predicted *continuous* patch representation with the AR model

$p(\mathbf{x} | \mathbf{z})$ — **diffusion**

Loss: **diffusion**

Sampling: **diffusion**

Diffusion loss



Diffusion loss

$$\mathcal{L}(z, x) = \mathbb{E}_{\epsilon, t} \left[\|\epsilon - \epsilon_{\theta}(x_t | t, z)\|^2 \right]$$

Diffusion loss plays a role of perceptual or adversarial losses

Getting back to prediction strategies

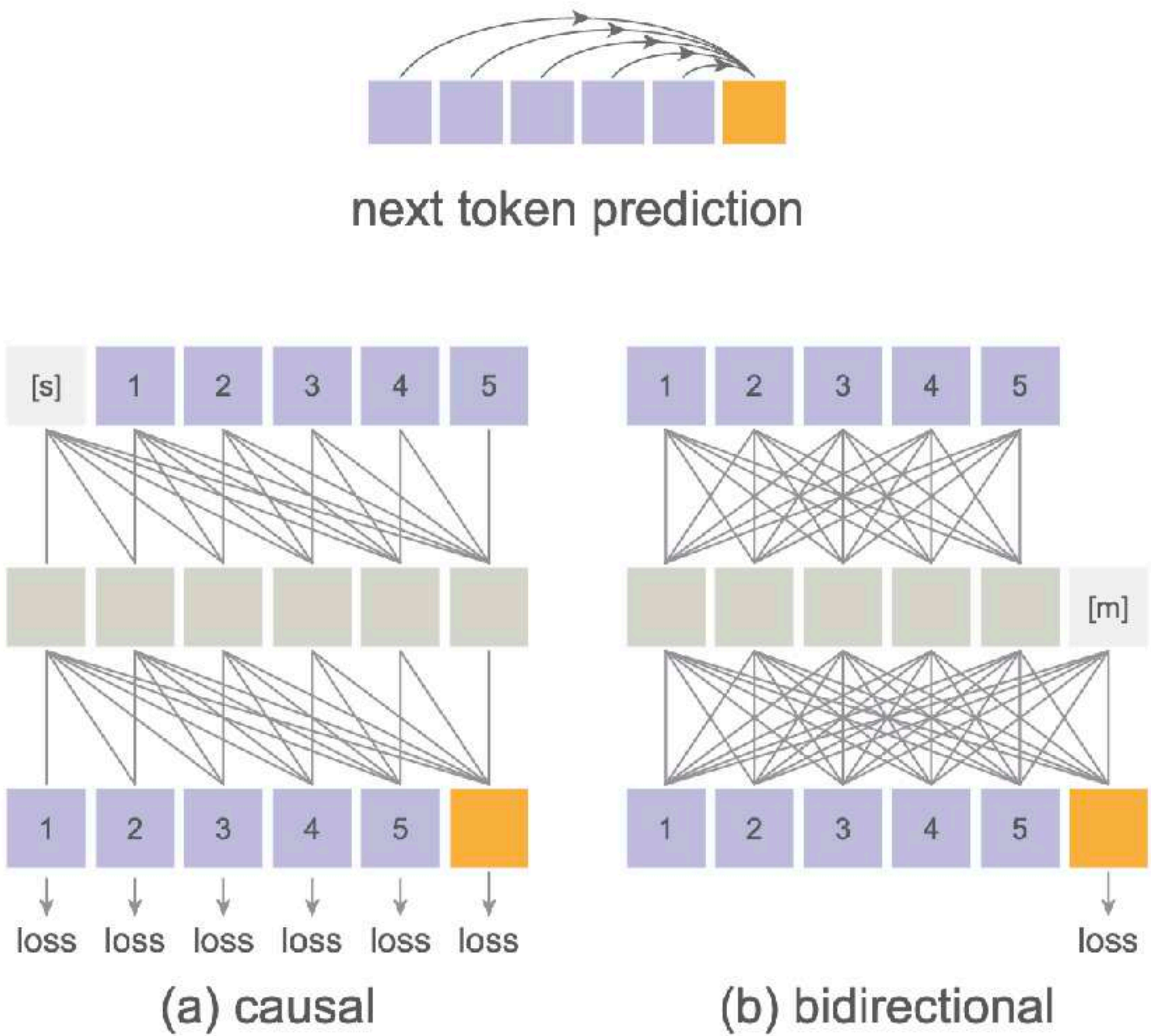
Next-token prediction can be done with
bidirectional transformers

Bidirectional models

Allow parallel decoding
(multiple predictions at each step)

Higher quality due to more connections

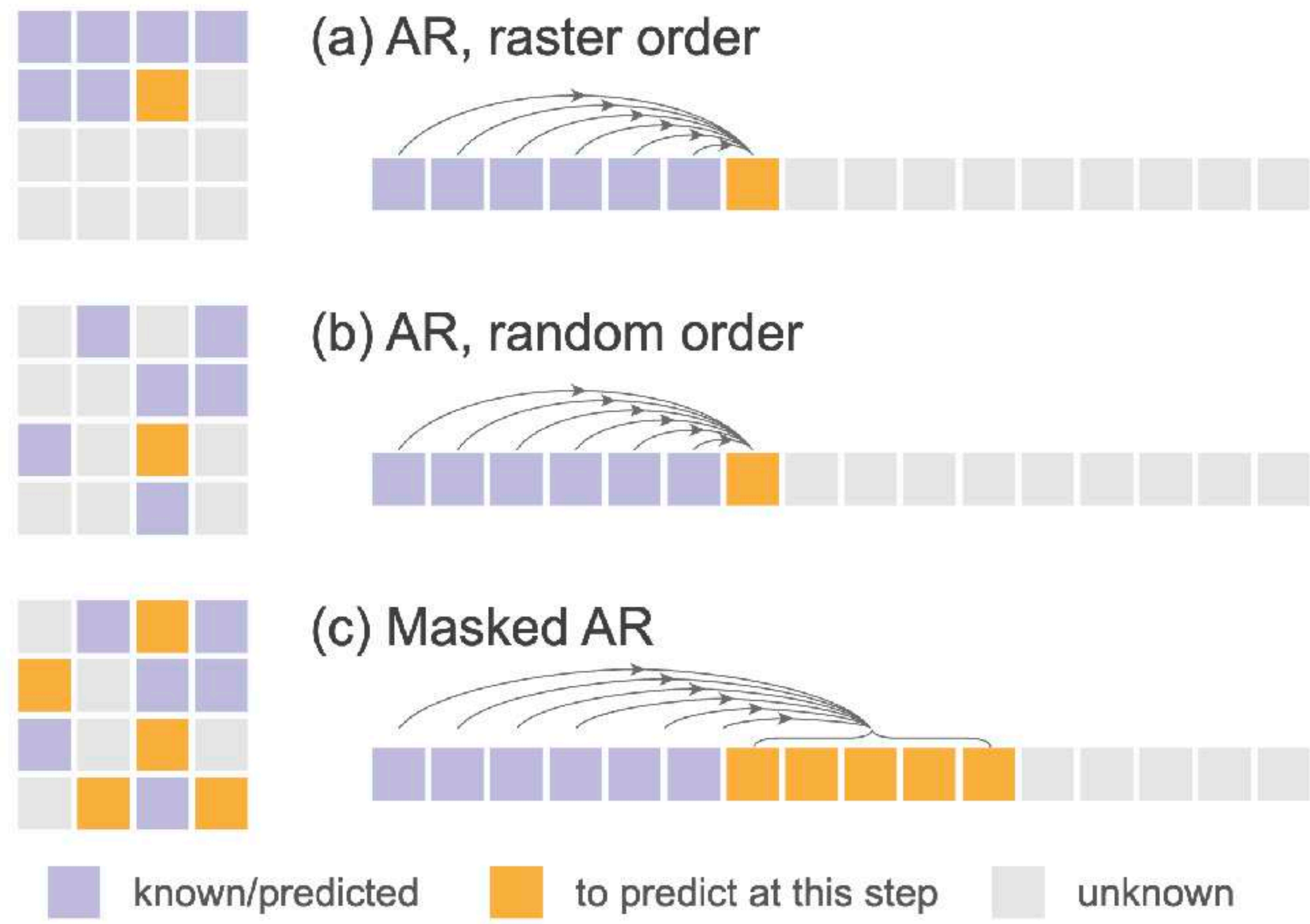
KV-cache is unavailable



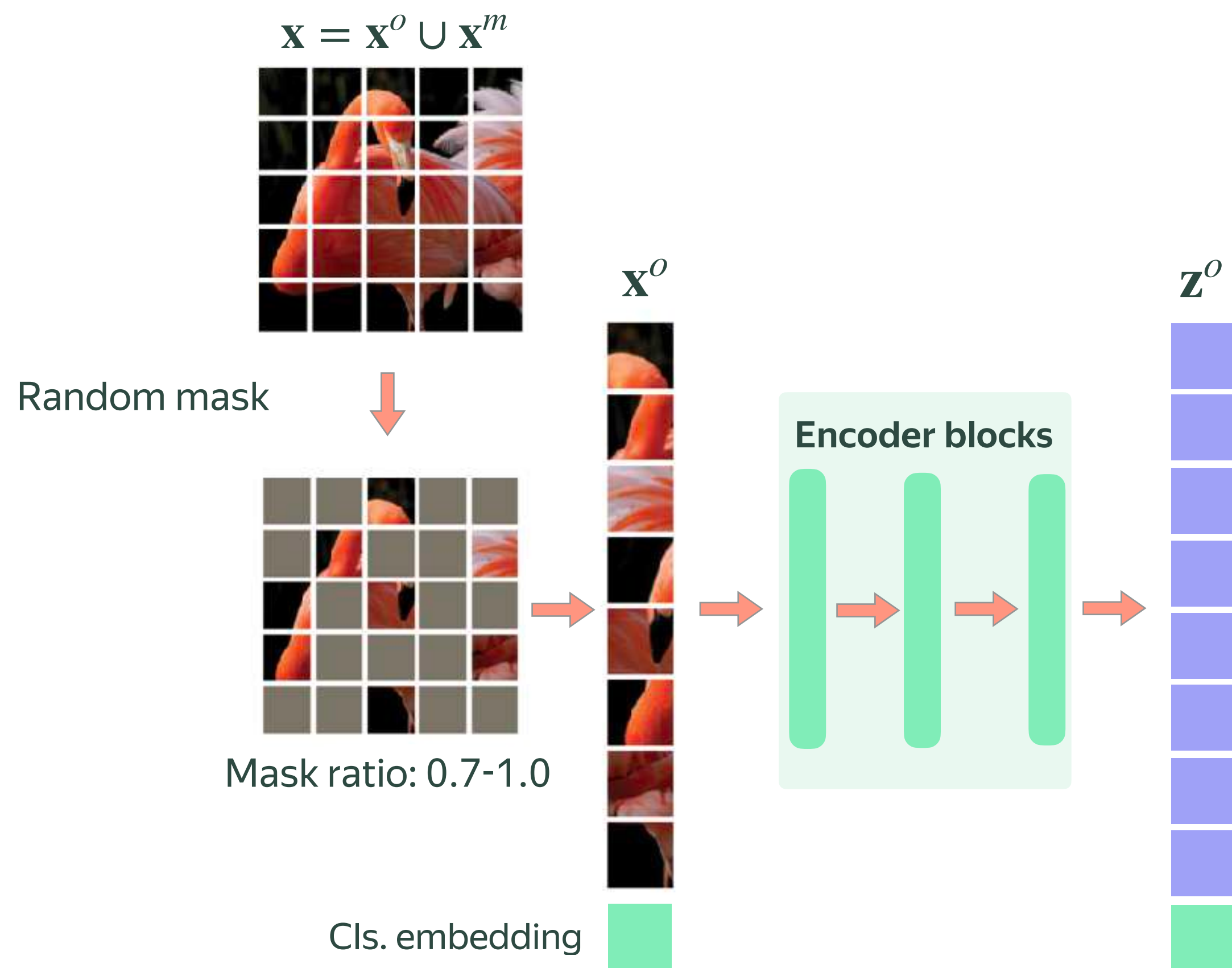
Masked autoregressive models

Masked AR is a form of the next-set prediction strategy

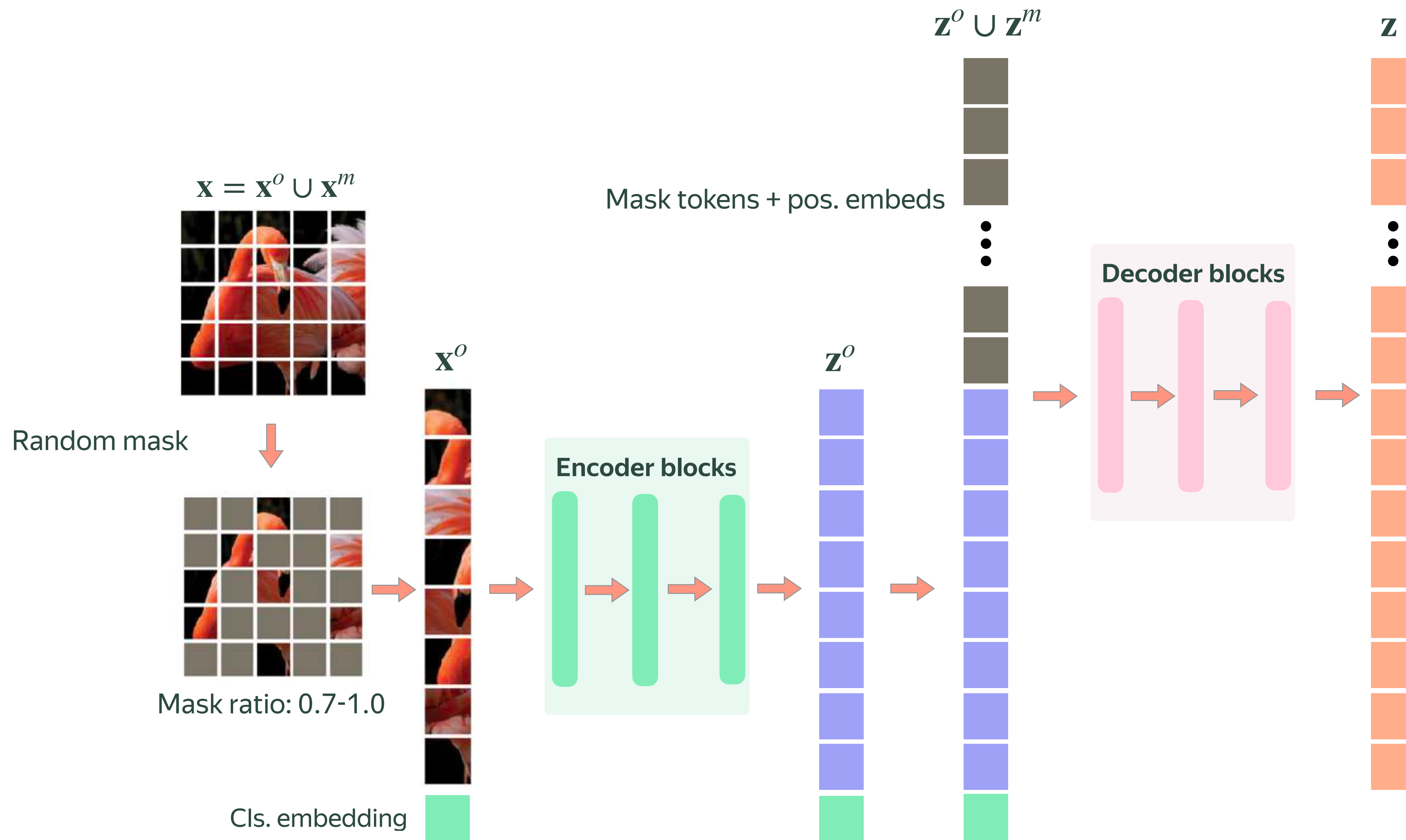
In fact, only predicted tokens need a bidirectional mask



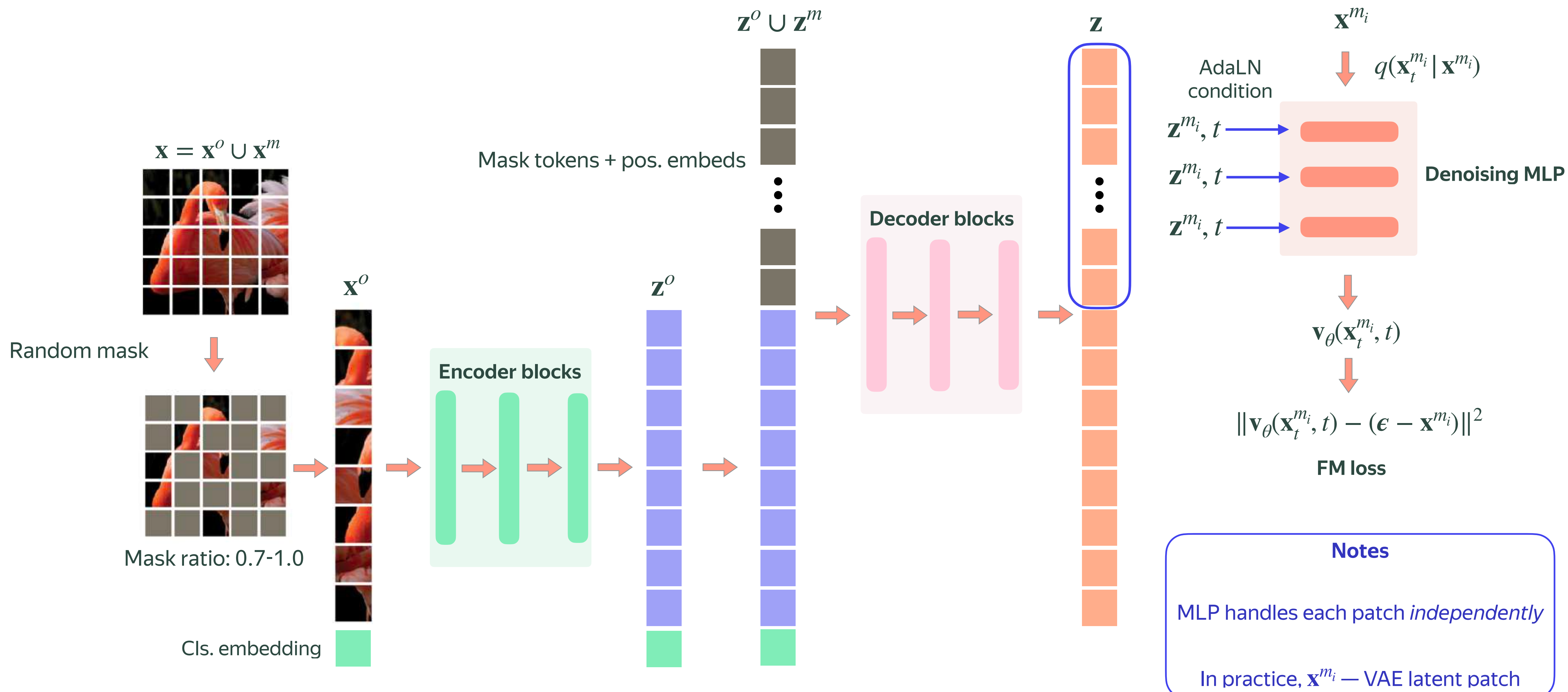
MAR training



MAR training



MAR training



Denoising MLP

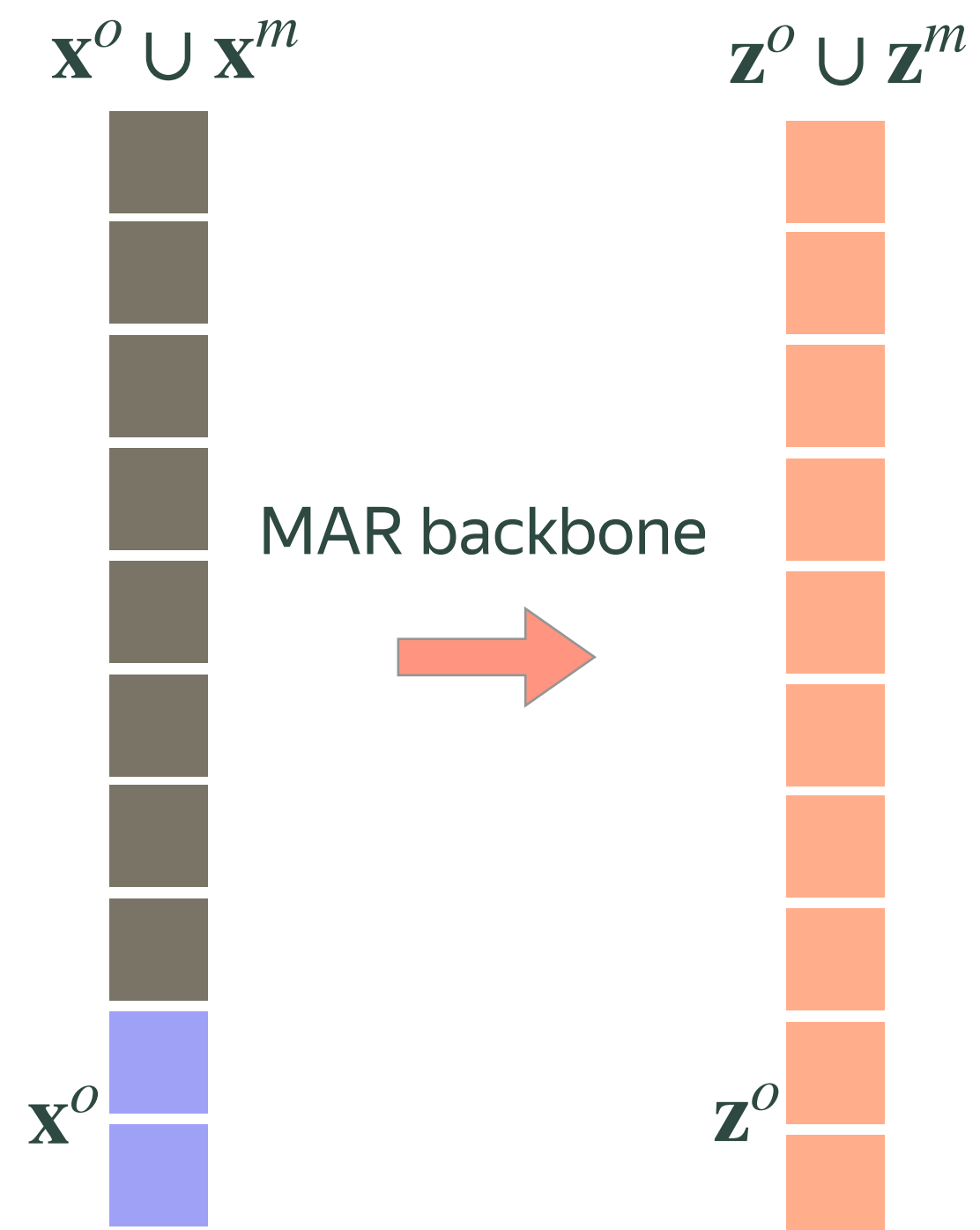
Small compared to the AR backbone

50-100 steps is sufficient and takes minor compute overhead at inference (~10%)

CFG is important, similar to the discrete case

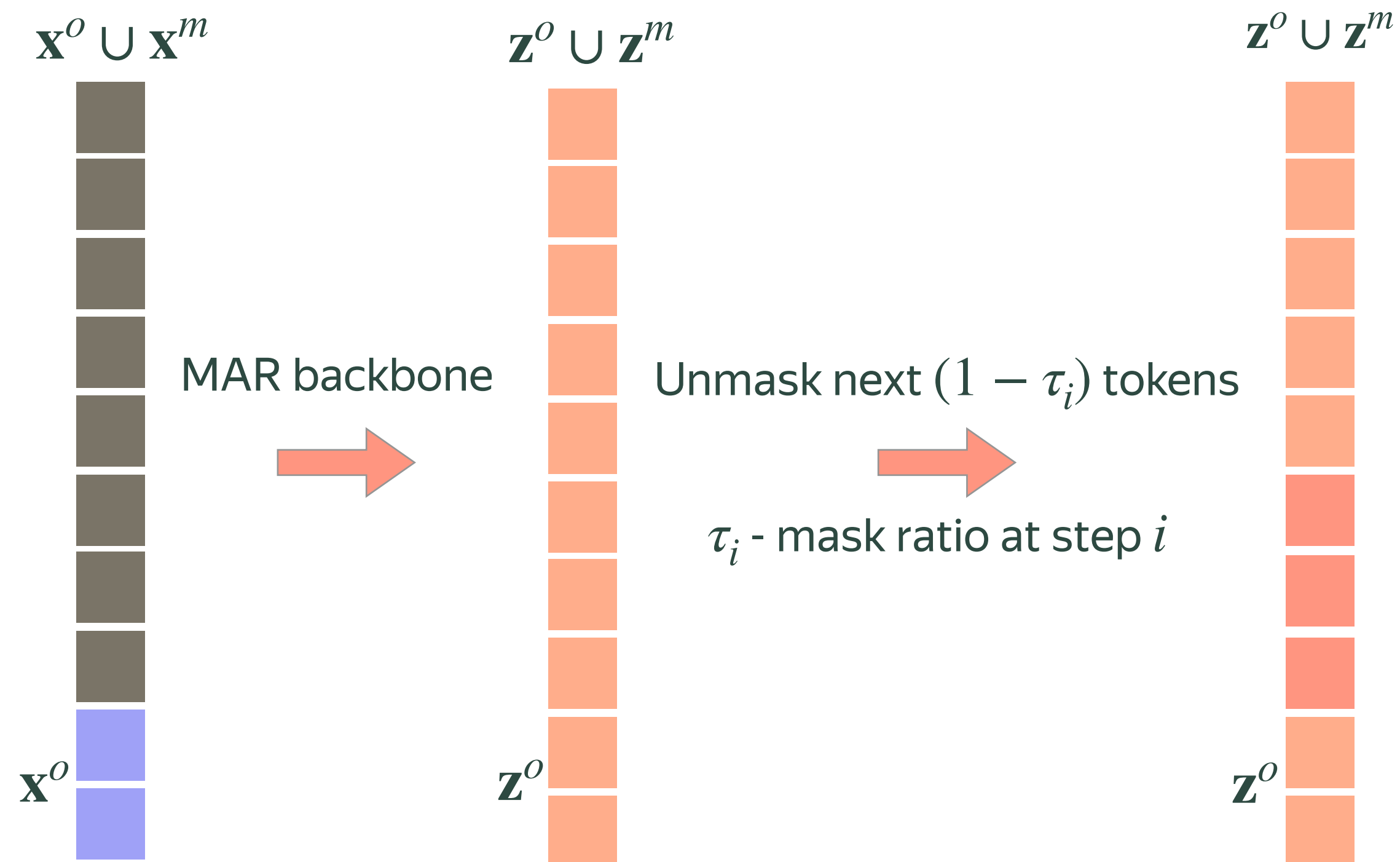
Sampling step

Following the random token order, sampled at the first step



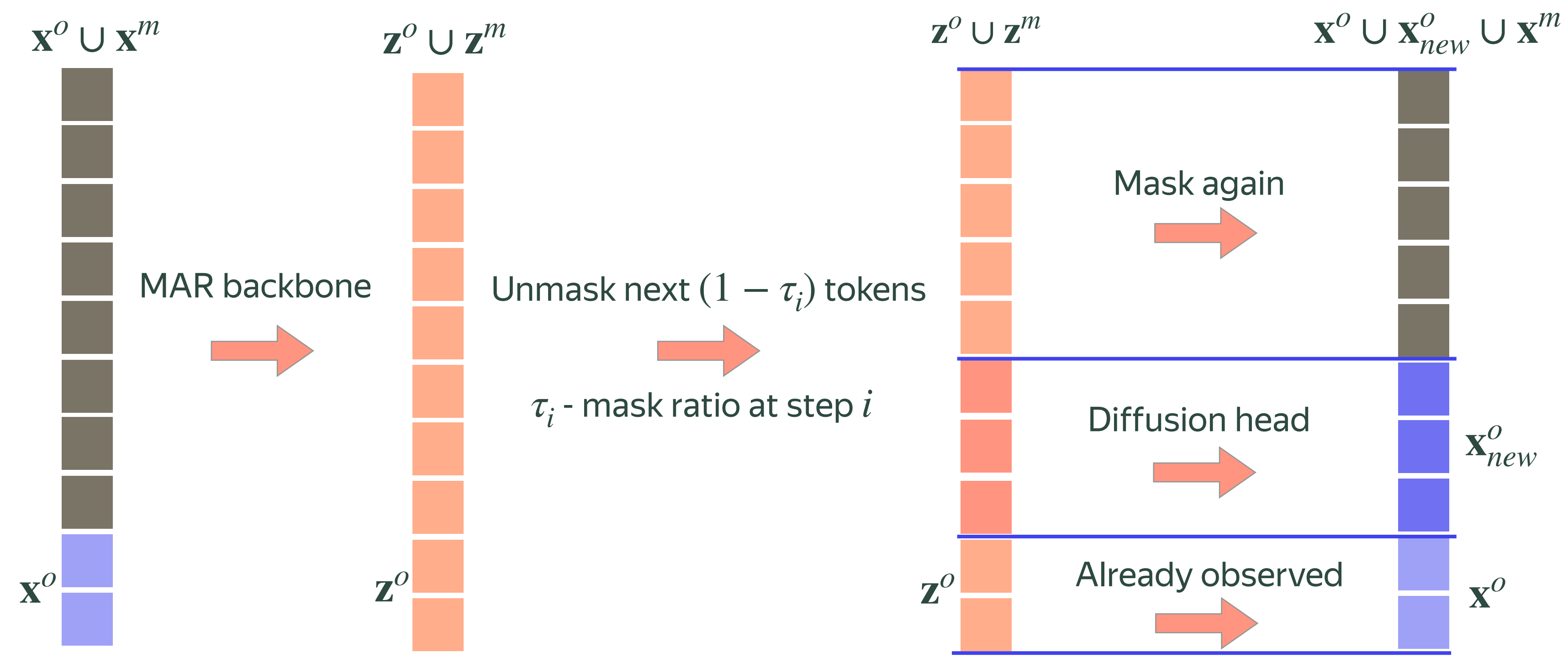
Sampling step

Following the random token order, sampled at the first step



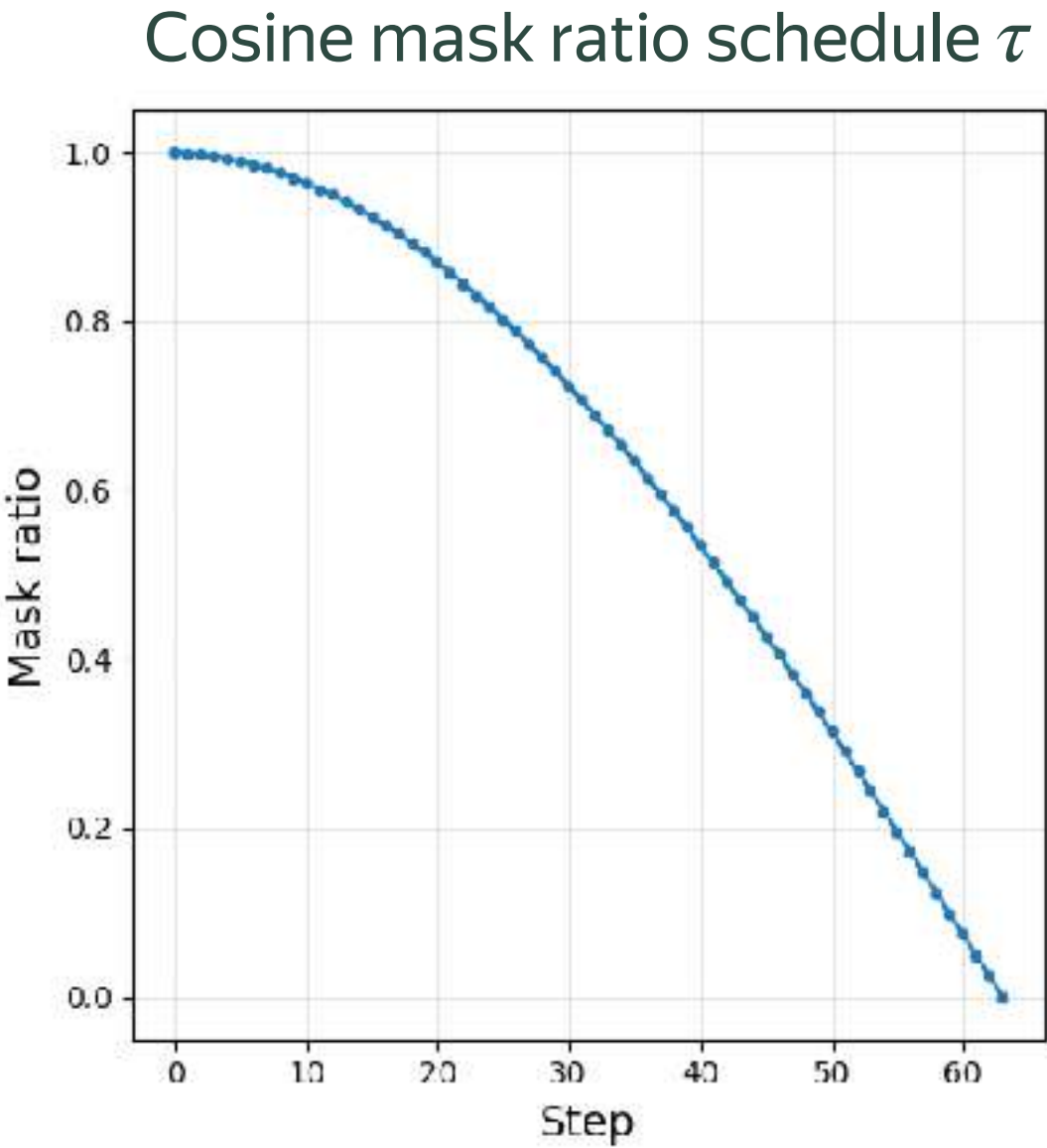
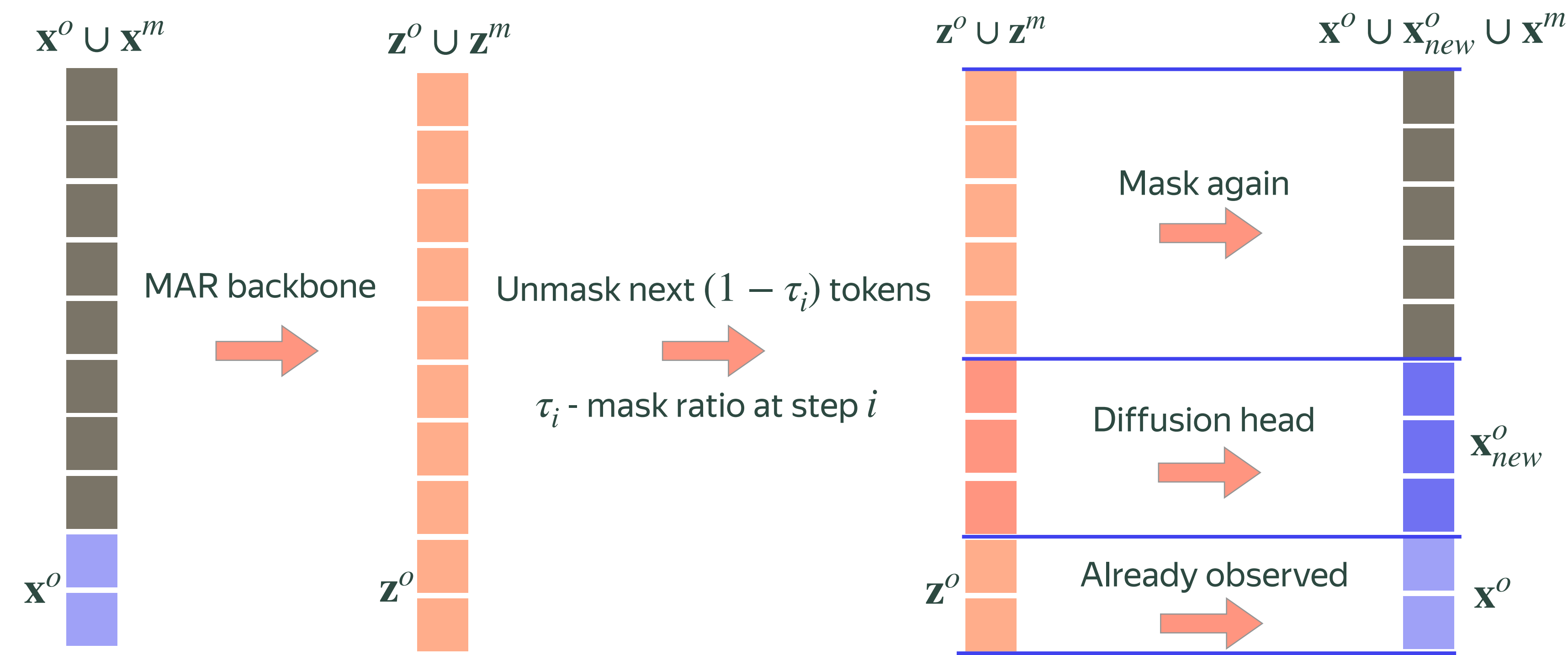
Sampling step

Following the random token order, sampled at the first step



Sampling step

Following the random token order, sampled at the first step



Ablation

Bidirectional > causal

Continuous > discrete: VAE > VQ-VAE and diffusion loss > cross-entropy

Text-to-image MAR | Fluid

FLUID: SCALING AUTOREGRESSIVE TEXT-TO-IMAGE GENERATIVE MODELS WITH CONTINUOUS TOKENS

Lijie Fan^{1,*} Tianhong Li^{2,†} Siyang Qin^{1,†} Yuanzhen Li¹ Chen Sun¹
Michael Rubinstein¹ Deqing Sun¹ Kaiming He² Yonglong Tian^{1,*}

¹Google DeepMind ²MIT *equal contribution, project lead [†]equal contribution



Multimodal MAR | UniFluid

Unified Autoregressive Visual Generation and Understanding with Continuous Tokens

Lijie Fan^{1,*} Luming Tang^{1,*} Siyang Qin^{1,*} Tianhong Li² Xuan Yang¹ Siyuan Qiao¹
Andreas Steiner¹ Chen Sun¹ Yuanzhen Li¹ Tao Zhu¹ Michael Rubinstein¹ Michalis Raptis¹
Deqing Sun^{1,†} Radu Soricut^{1,†}
¹Google DeepMind ²MIT ^{*,†} equal contribution



Checkpoint

GPT-like AR

Unnatural inductive bias

Discrete

Slow inference

Next-scale prediction

Natural inductive bias

Discrete

Fast inference

Continuous-token prediction

Unnatural inductive bias

Continuous

Slow inference

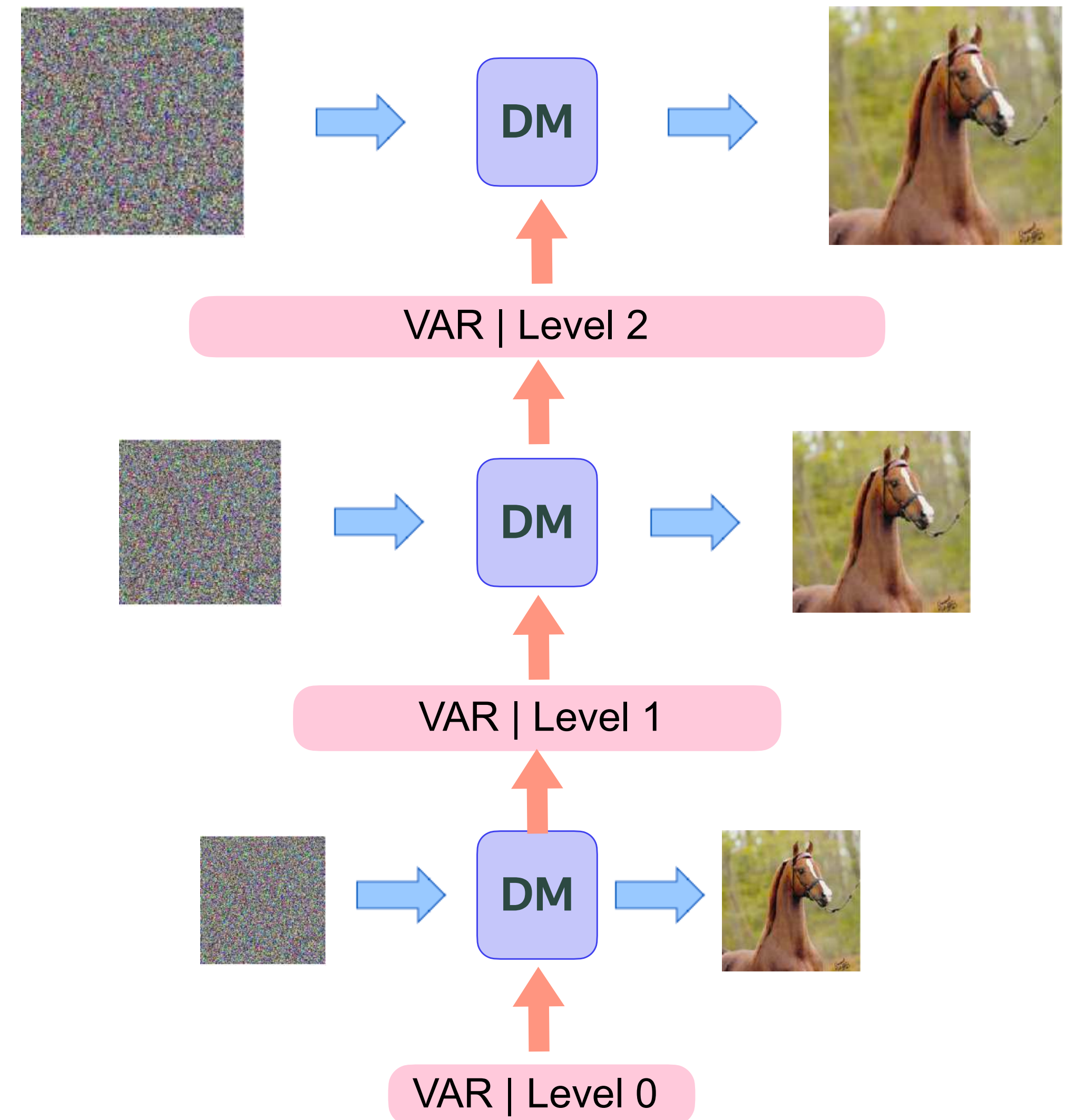
What about ...



MAR + VAR

Reasonable but not quite elegant...
Looks like a cascaded DM

Do we know any other scale-wise and
continuous AR paradigms?



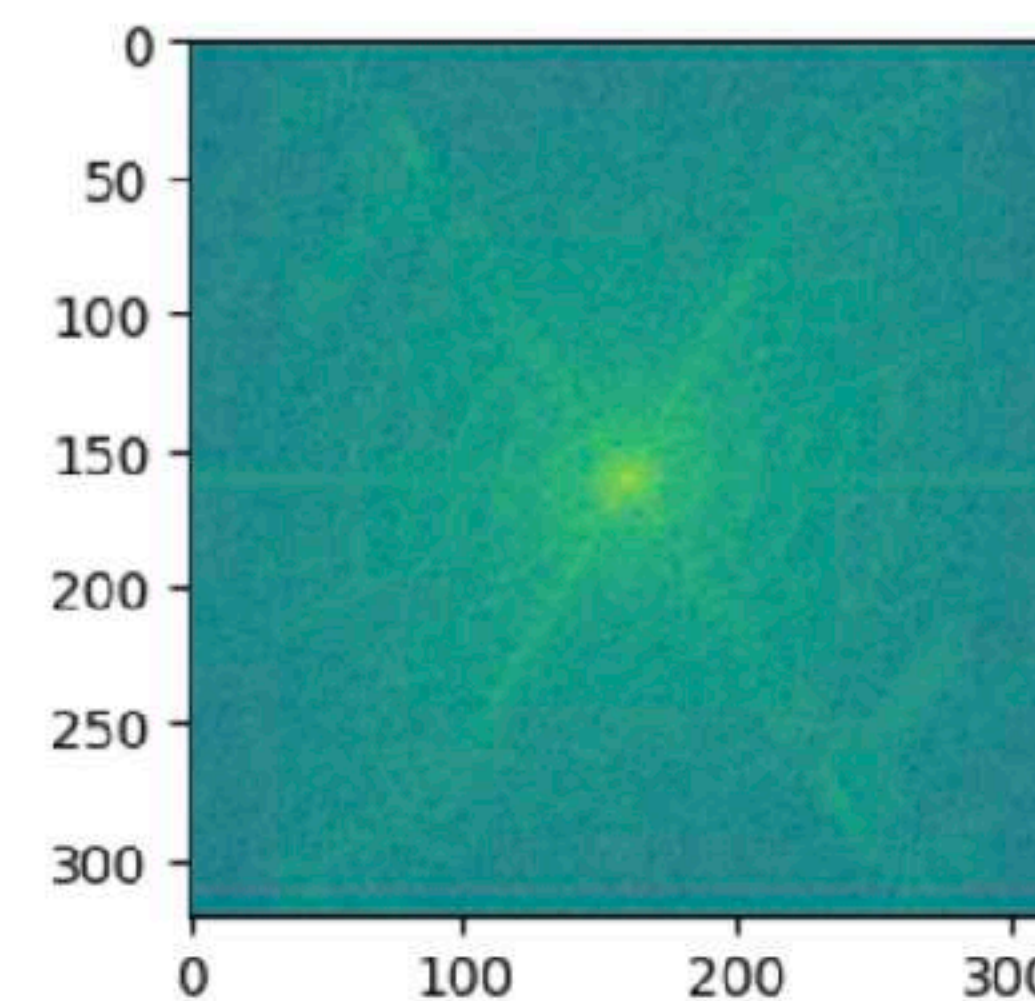
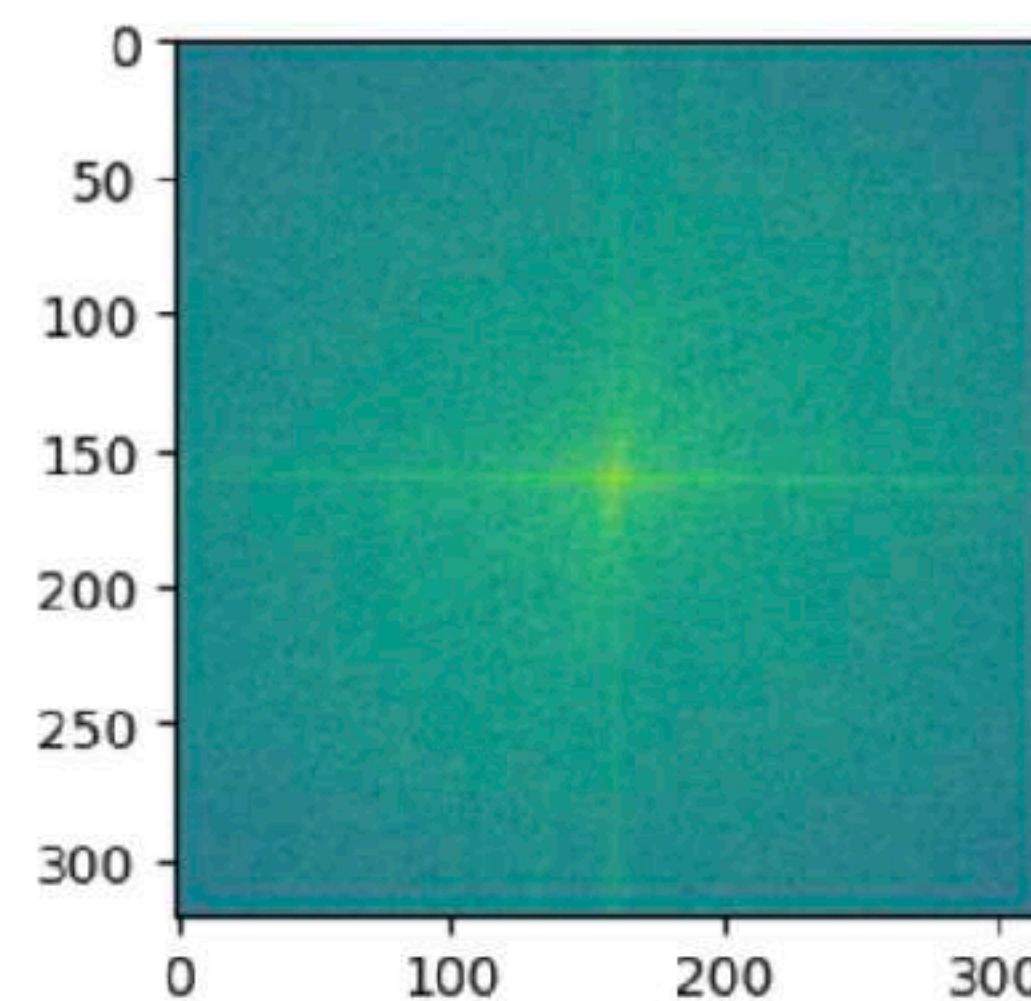
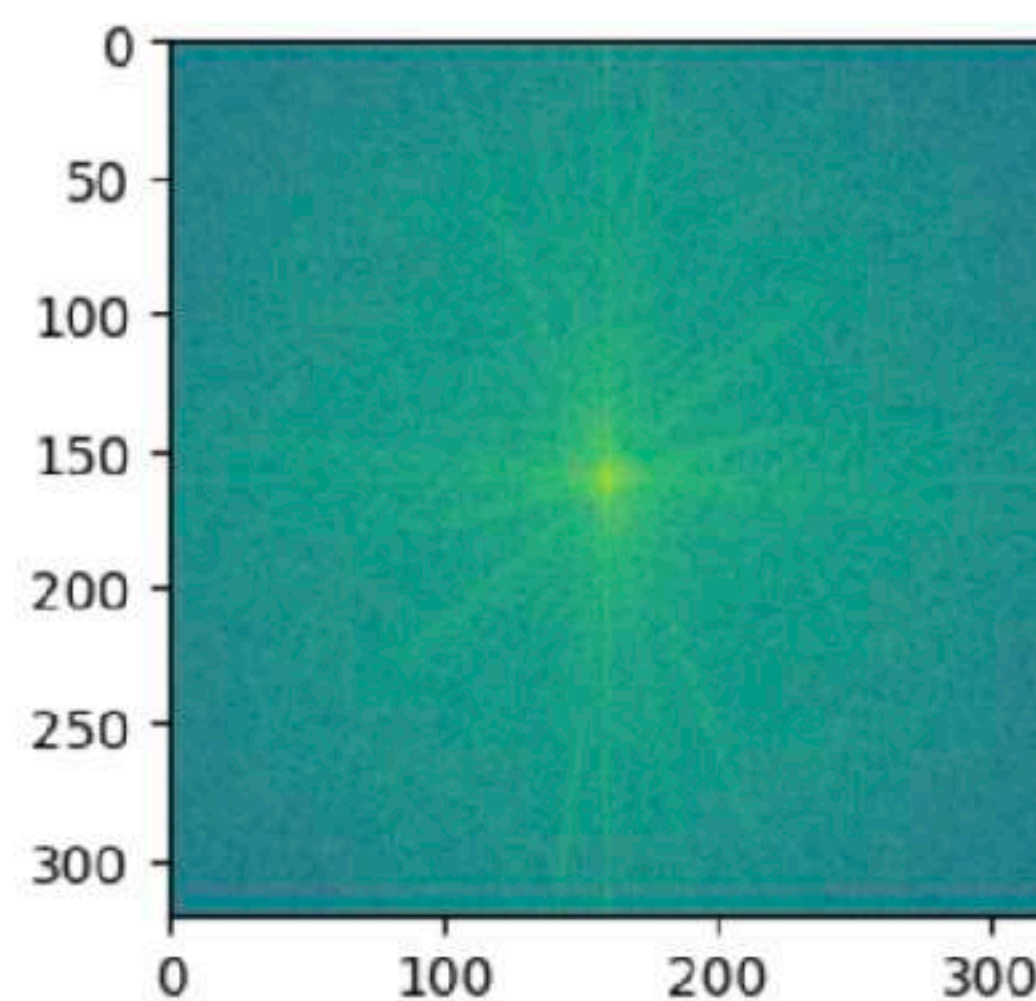
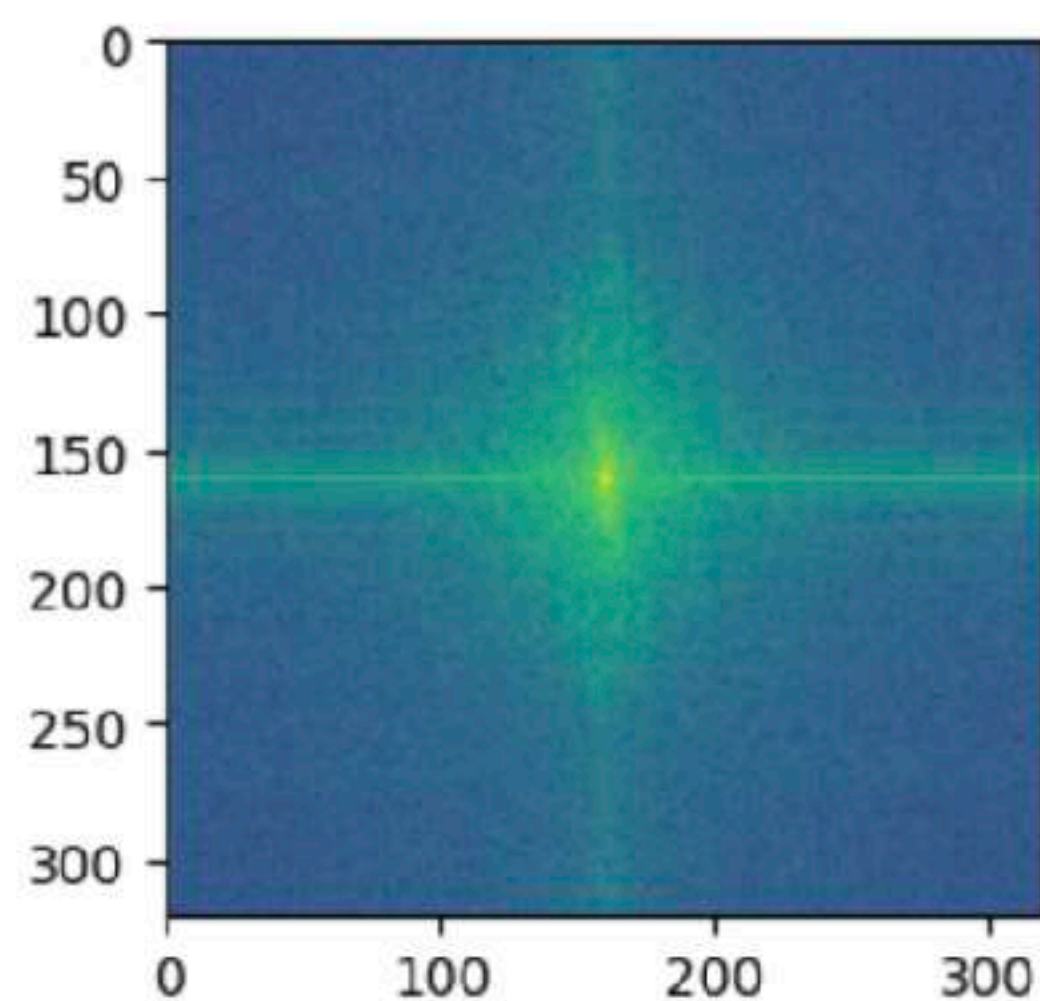
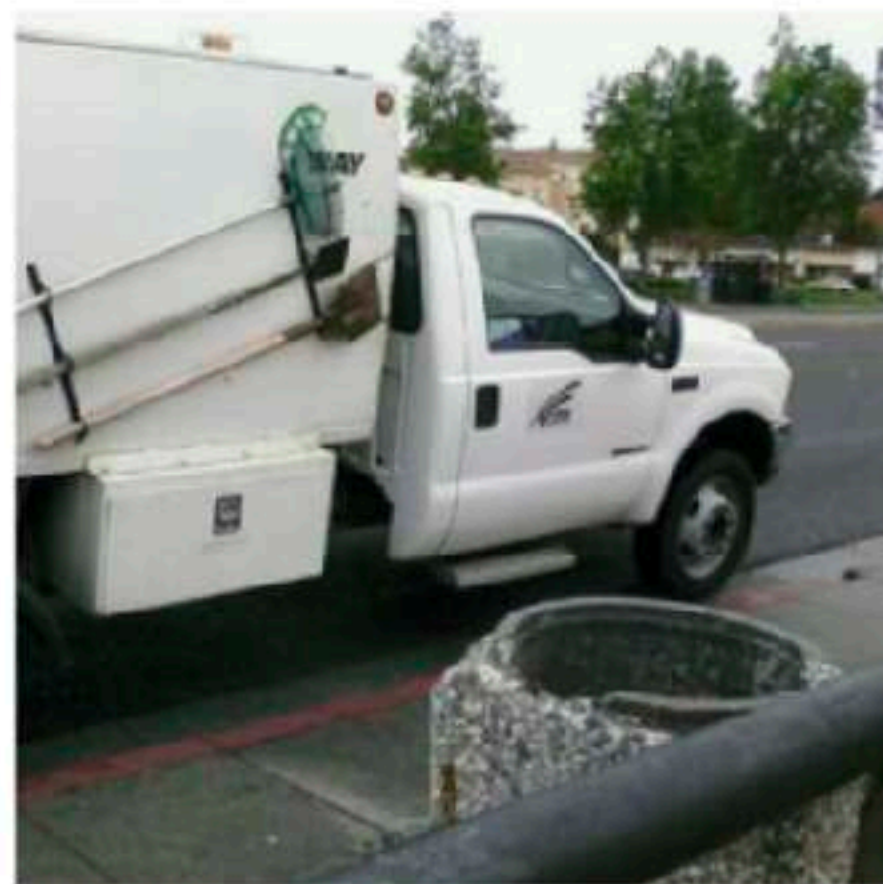
Diffusion models

Implicit continuous AR models

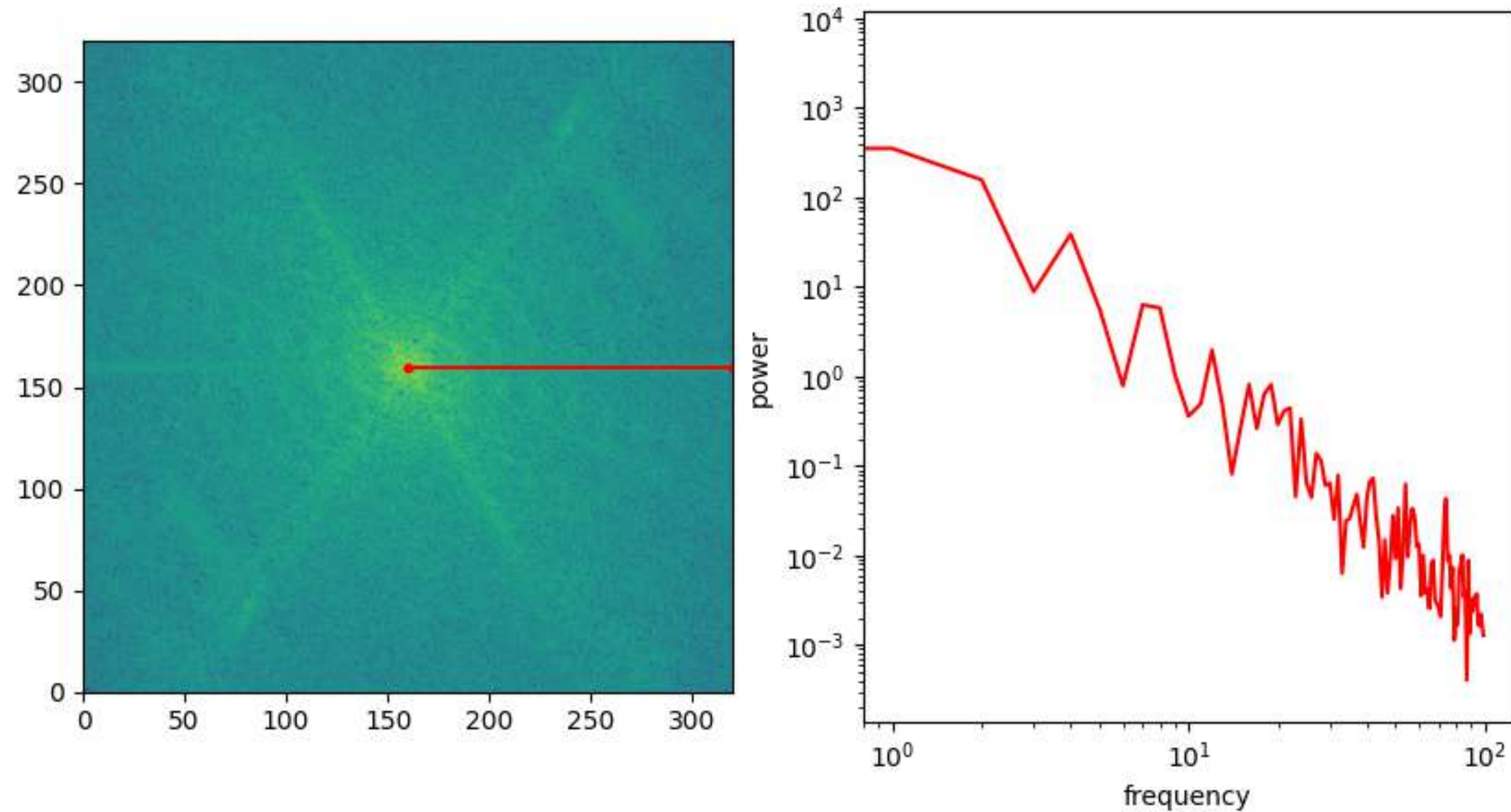
Implicit next-scale prediction



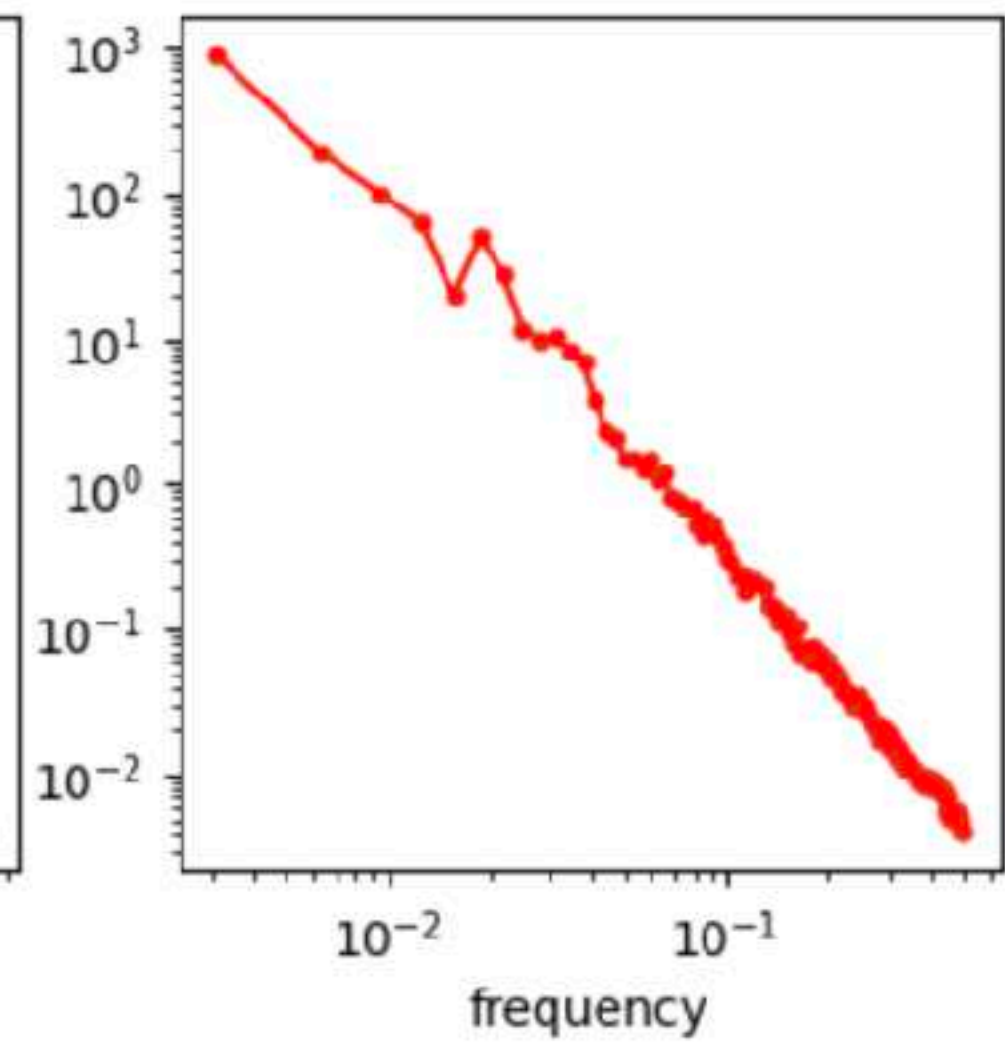
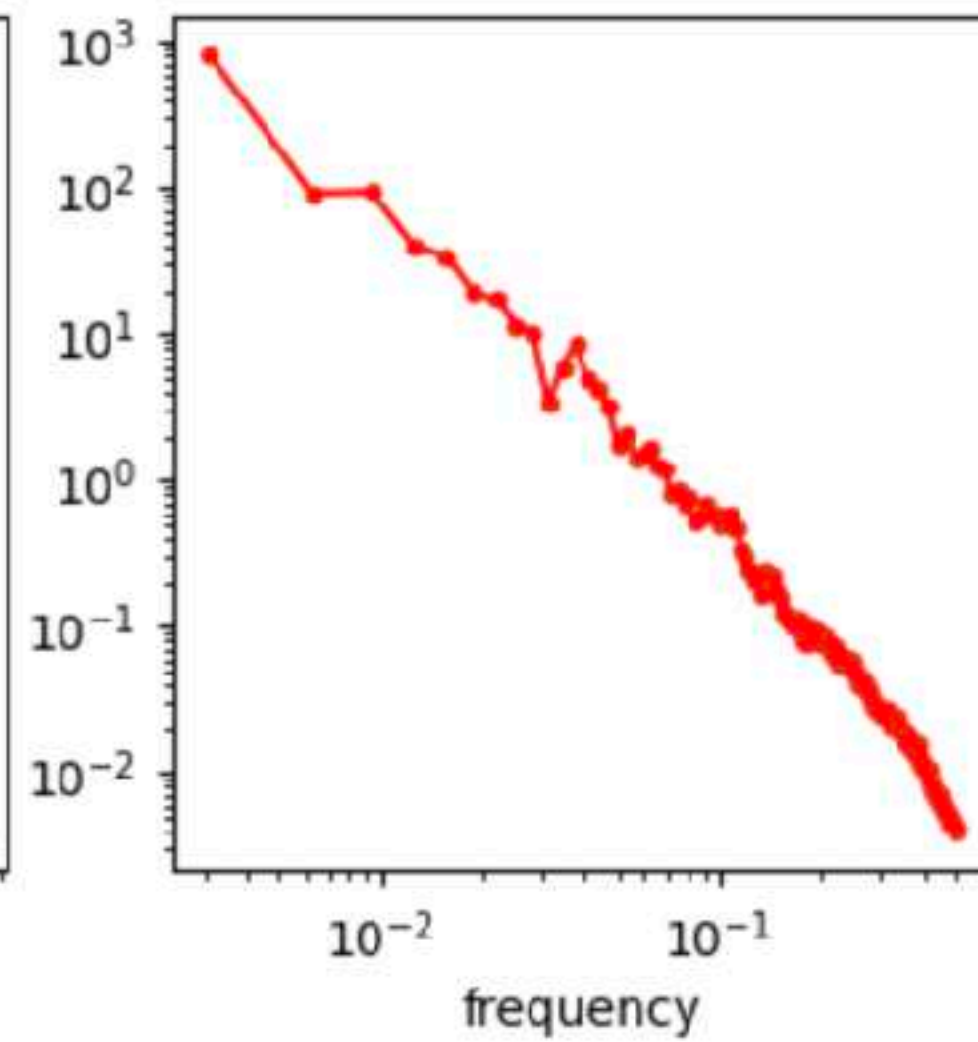
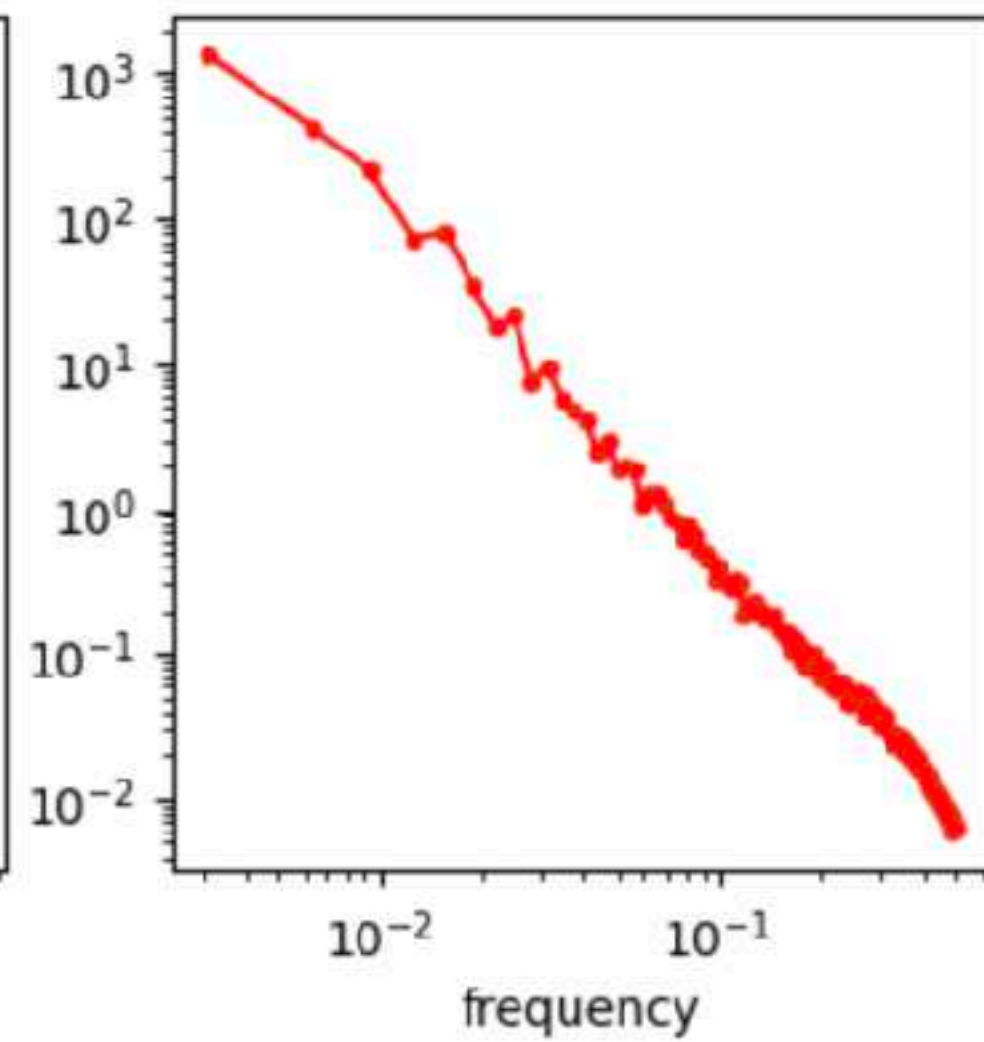
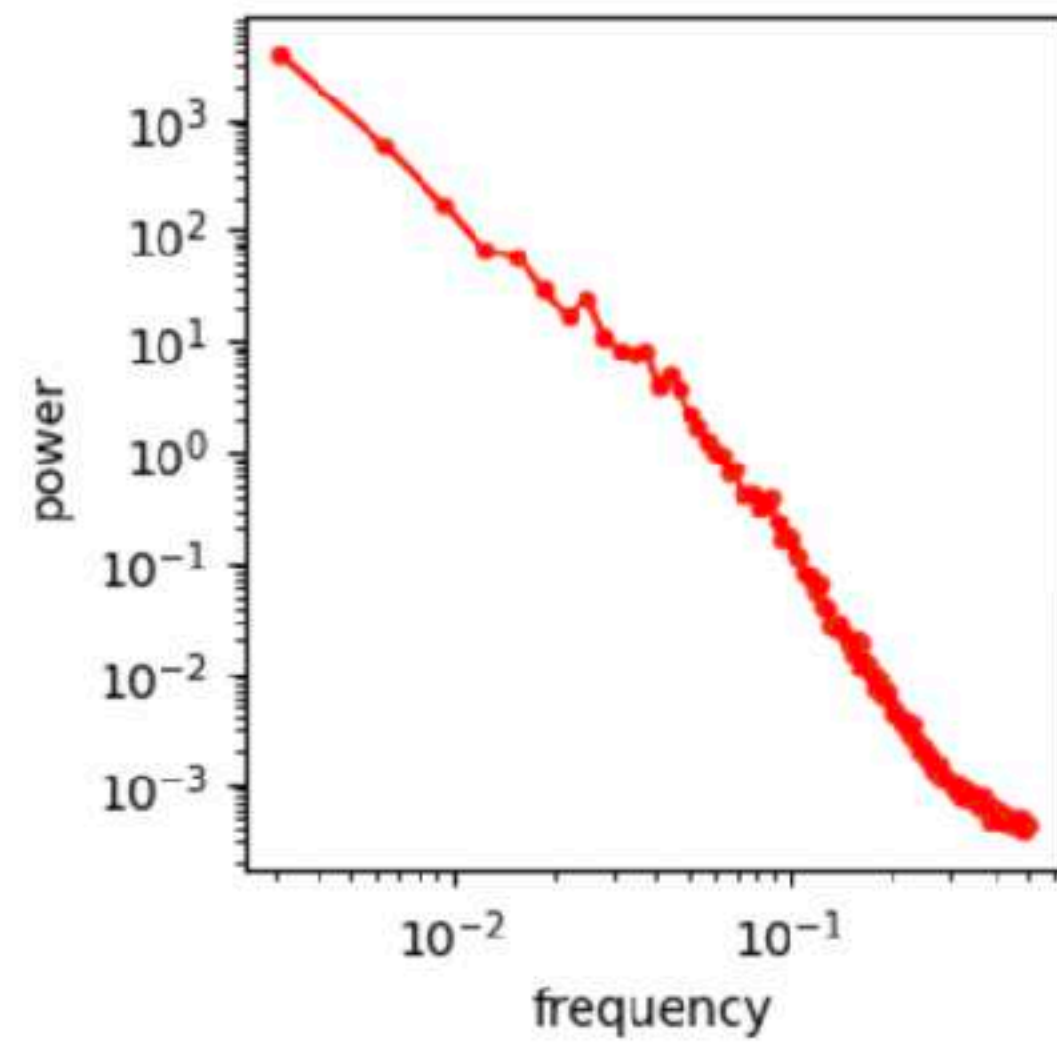
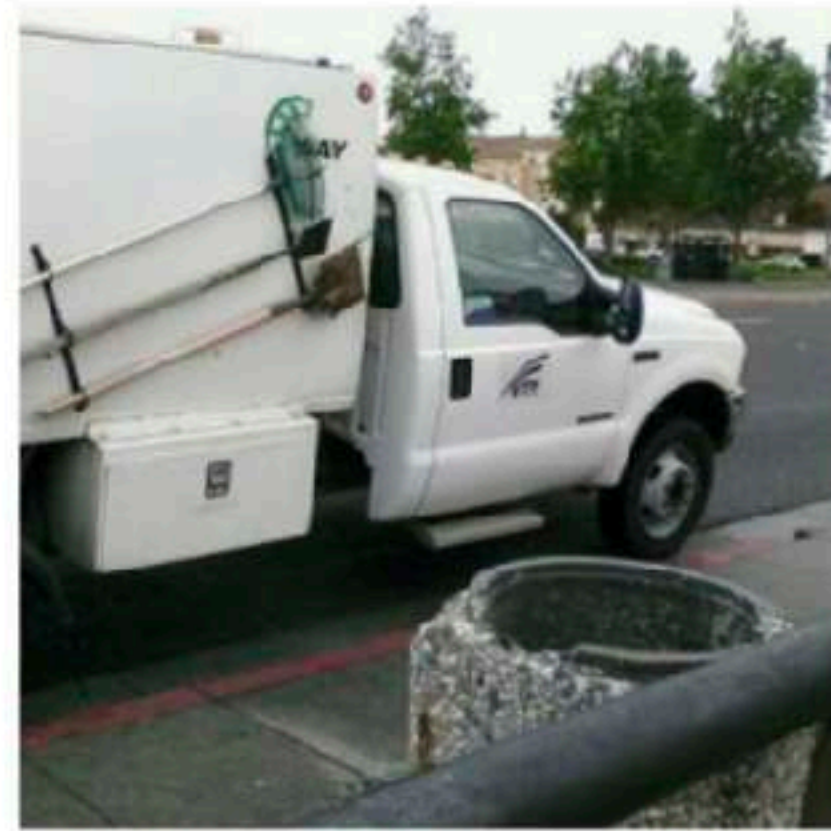
Magnitude spectra of natural images



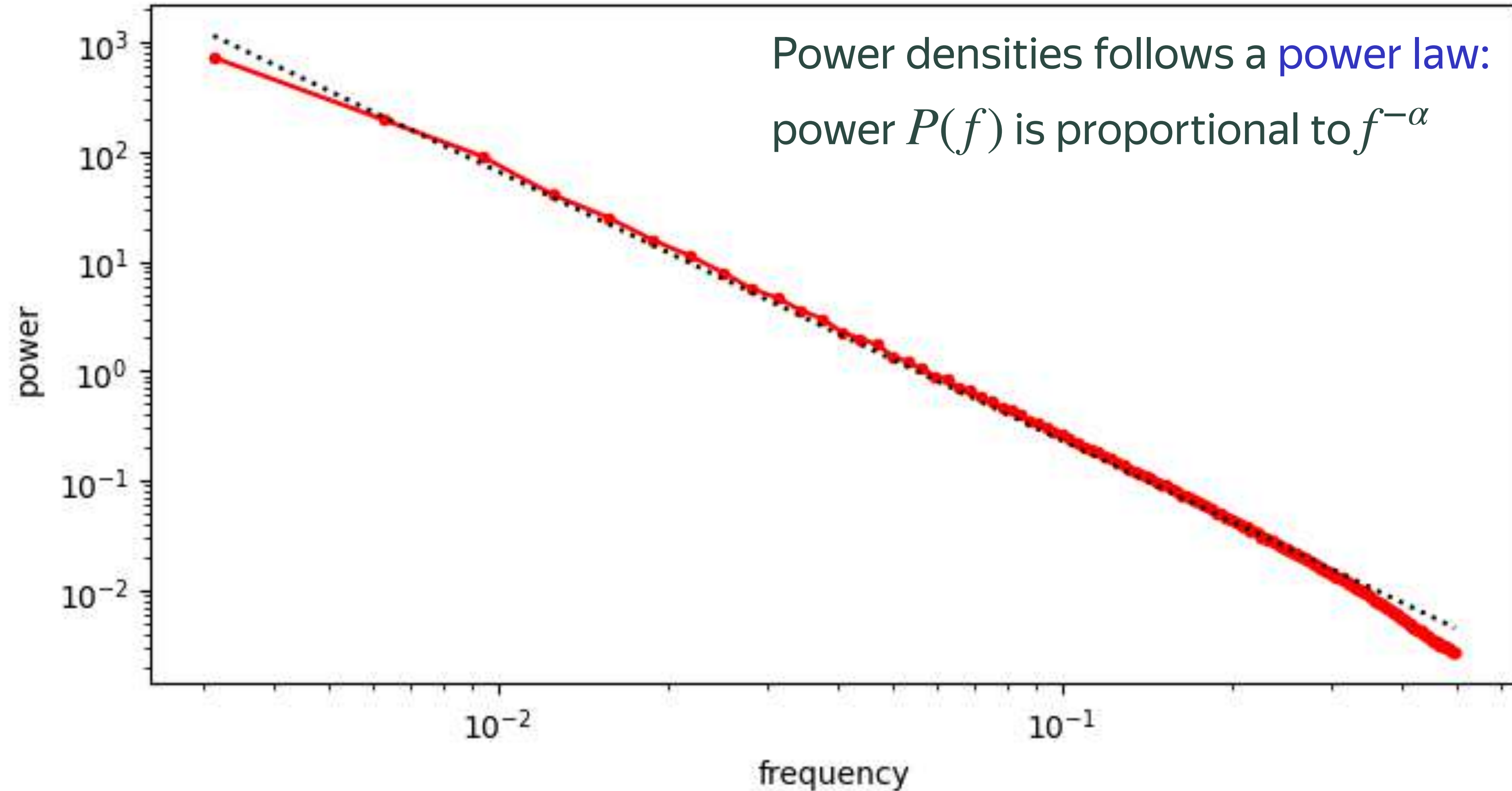
Radial spectral power densities



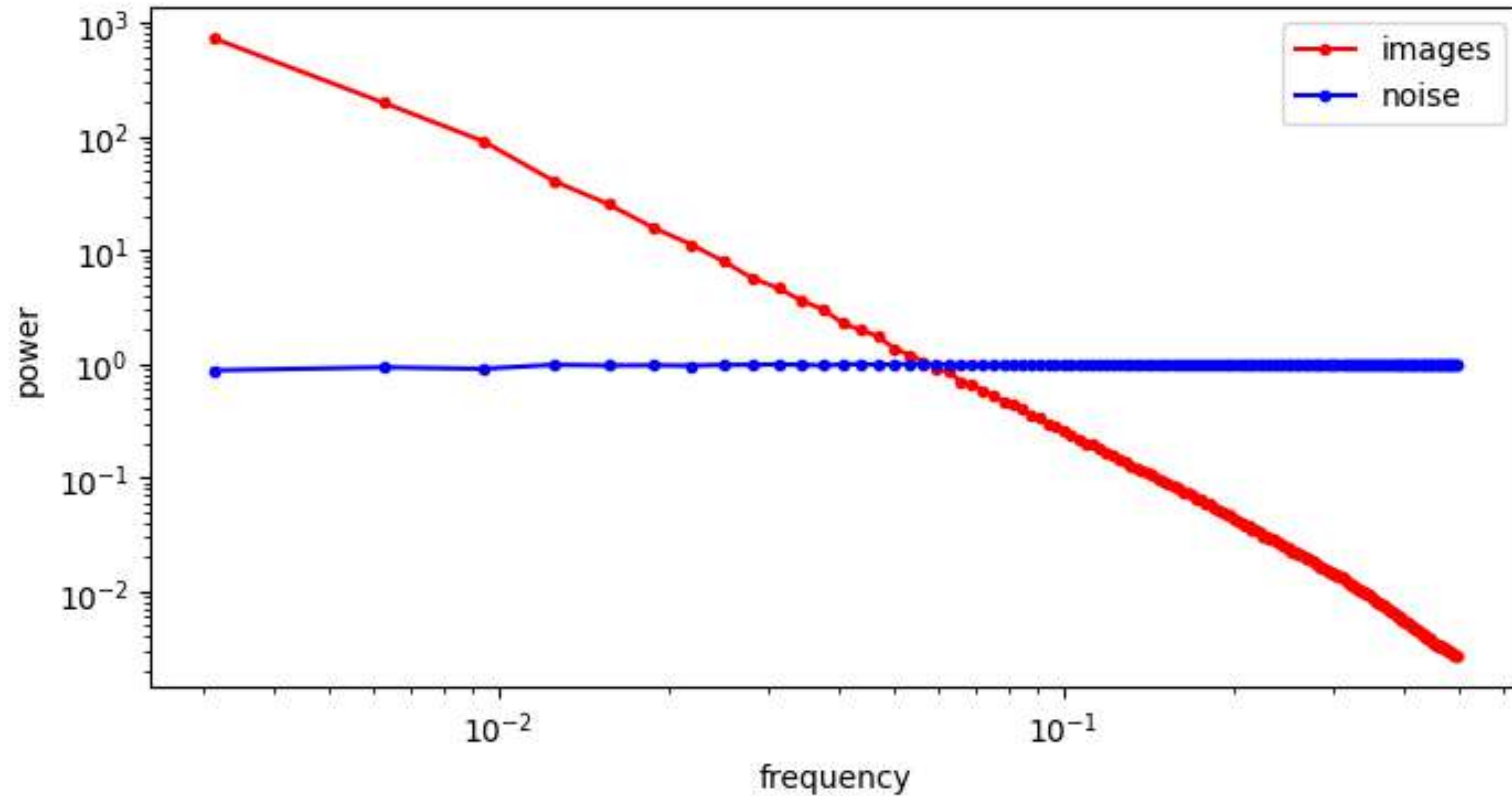
Radially averaged spectral power densities



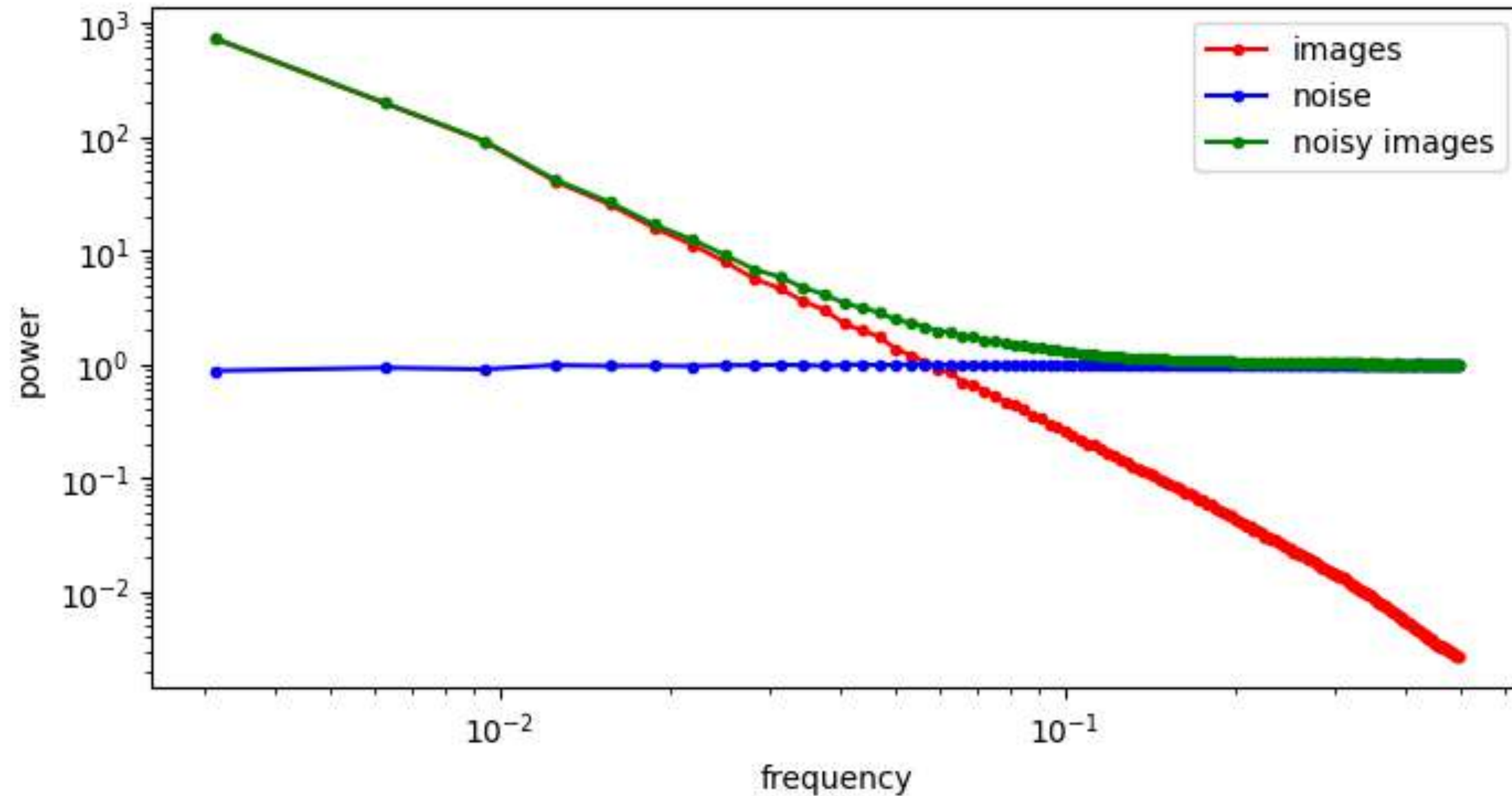
Radially averaged spectral power densities



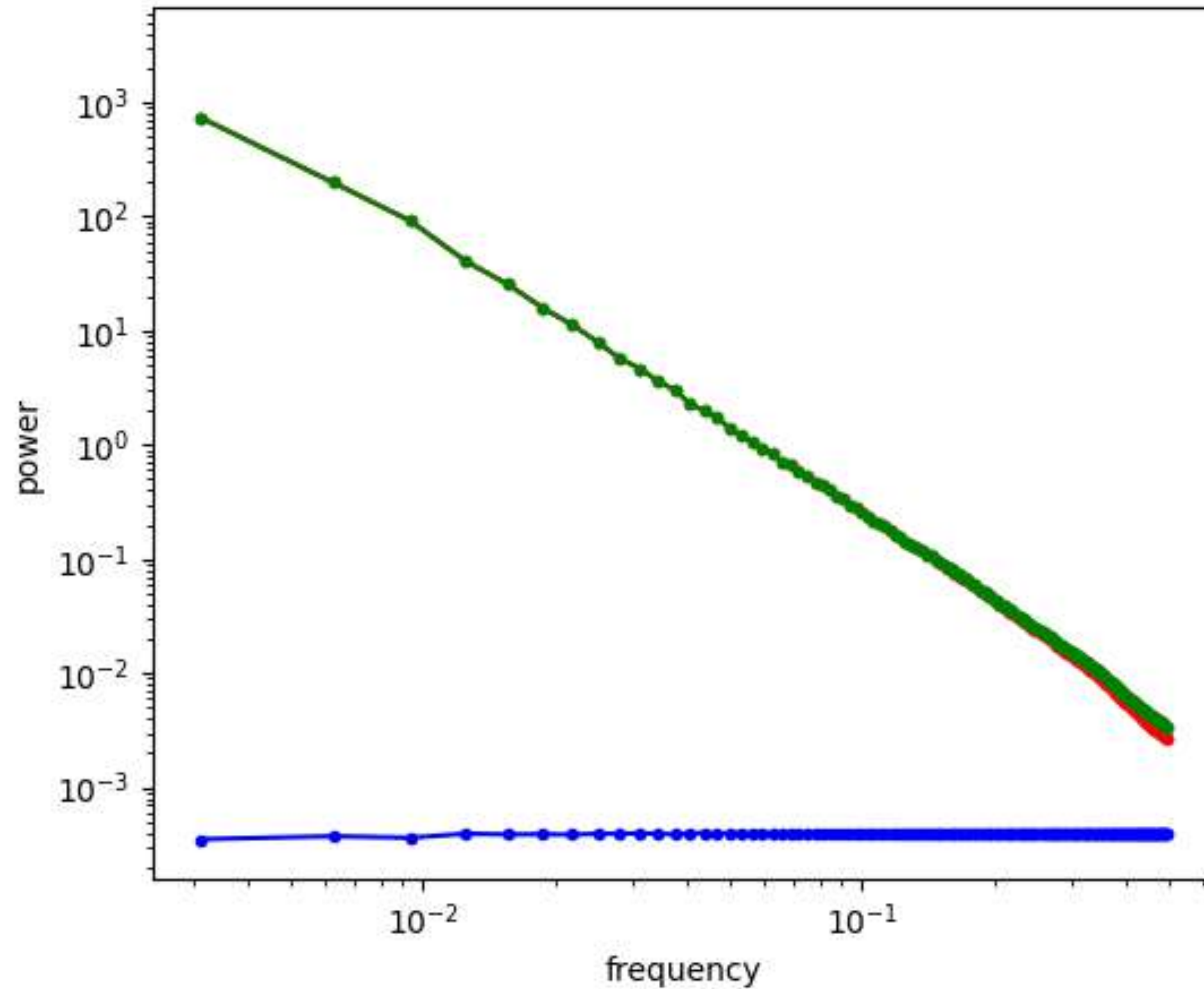
Radially averaged spectral power densities



Noise masks high frequencies

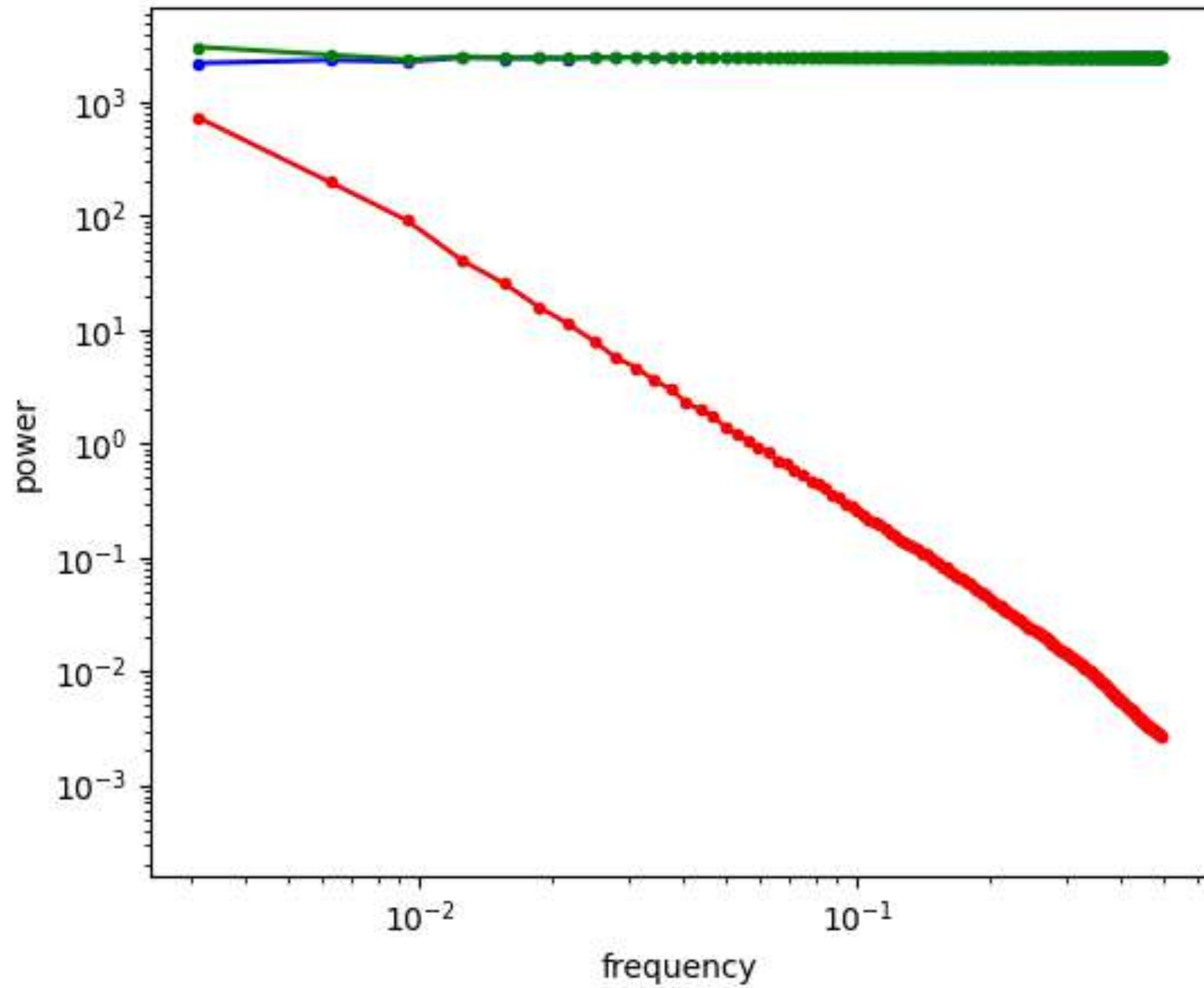


Diffusion process in RASPD

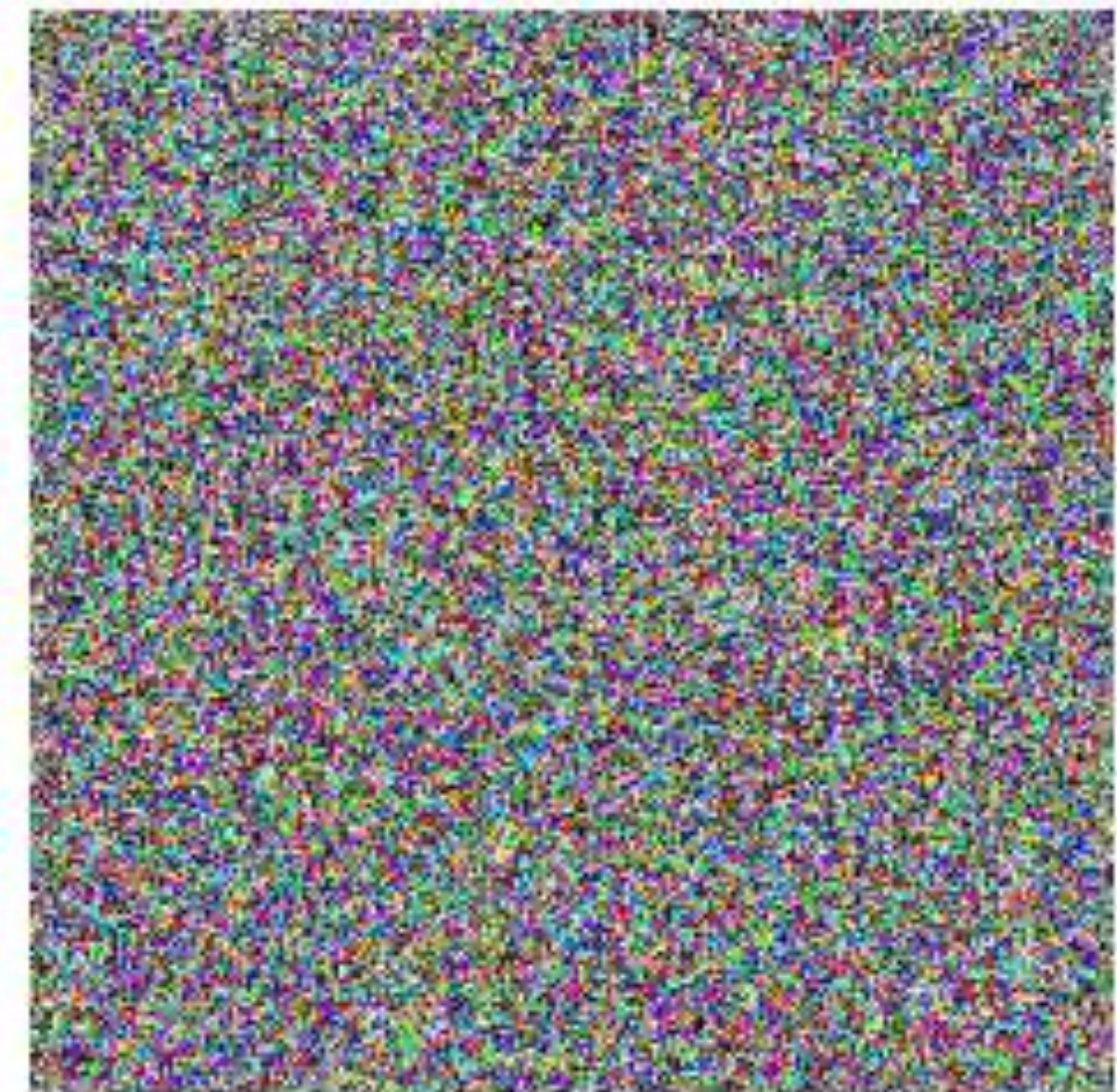


VE process

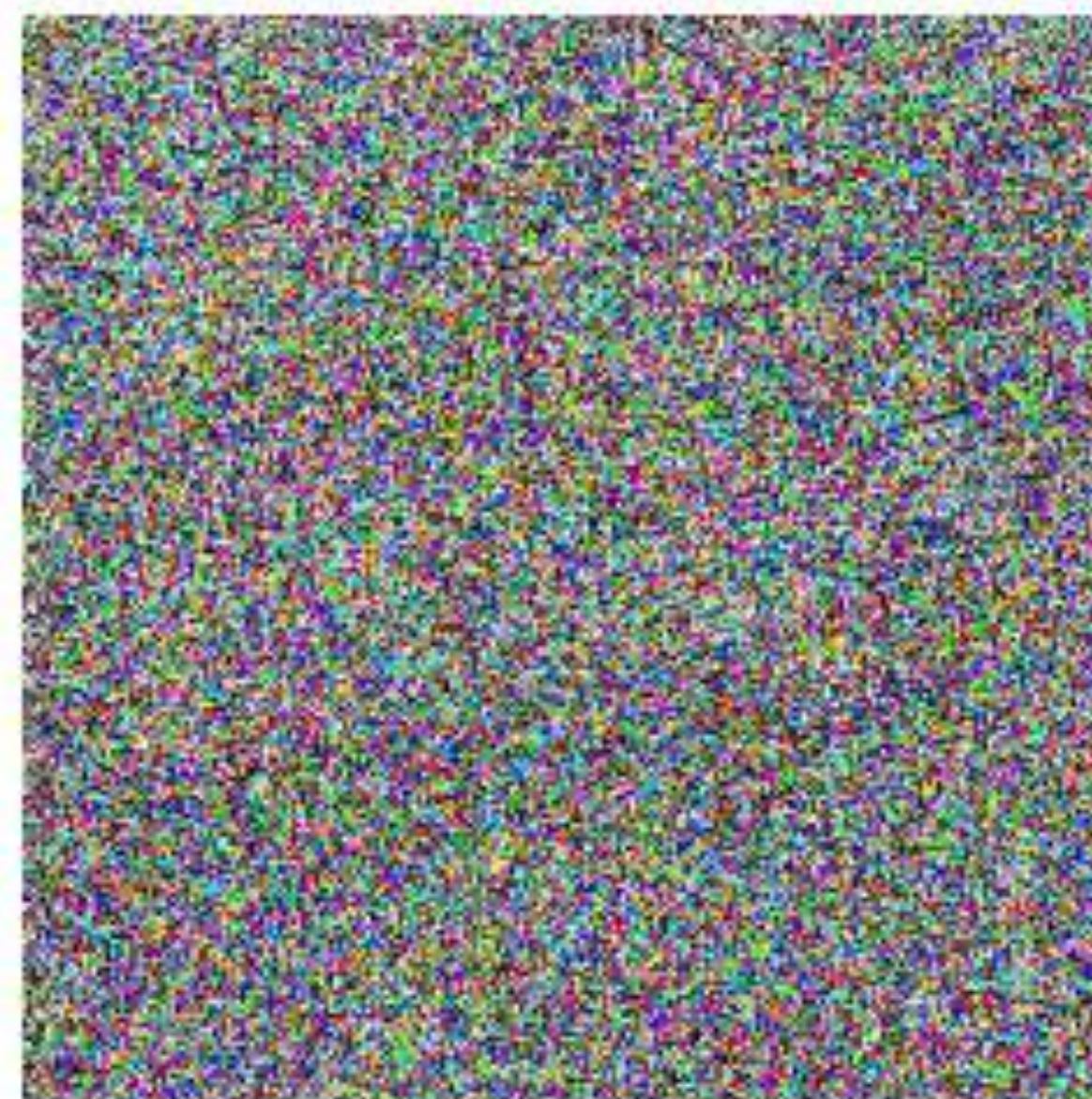
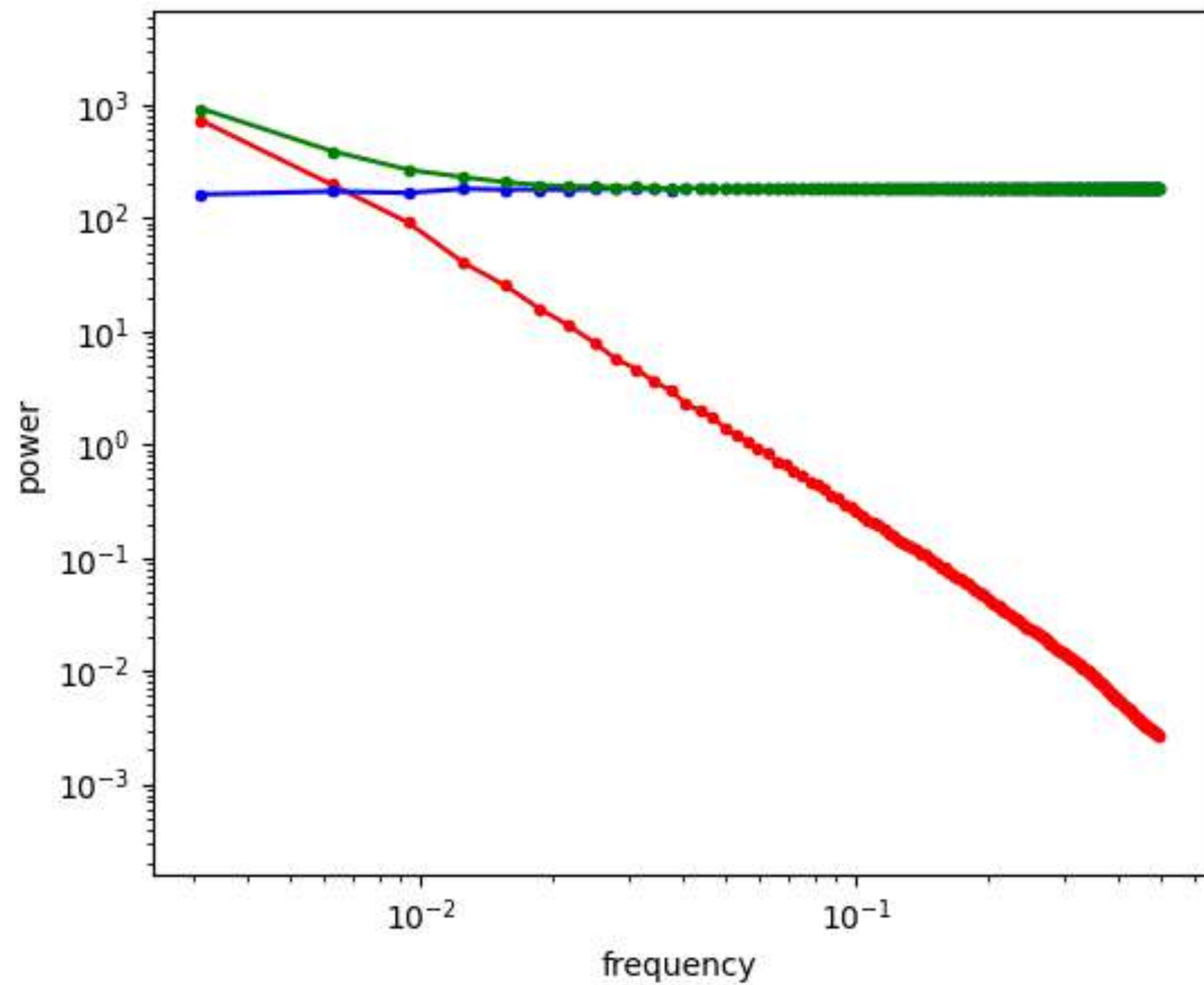




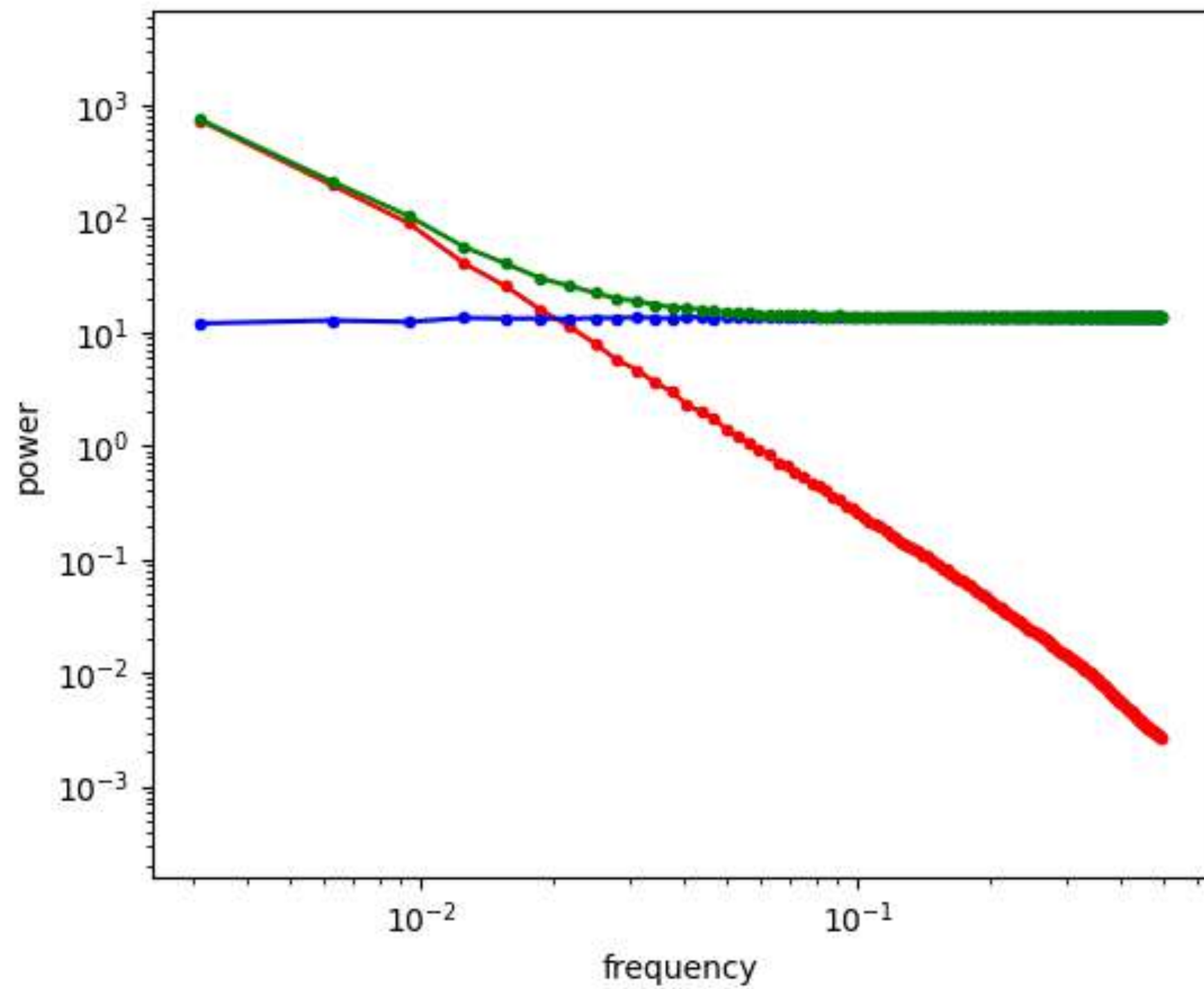
VE process



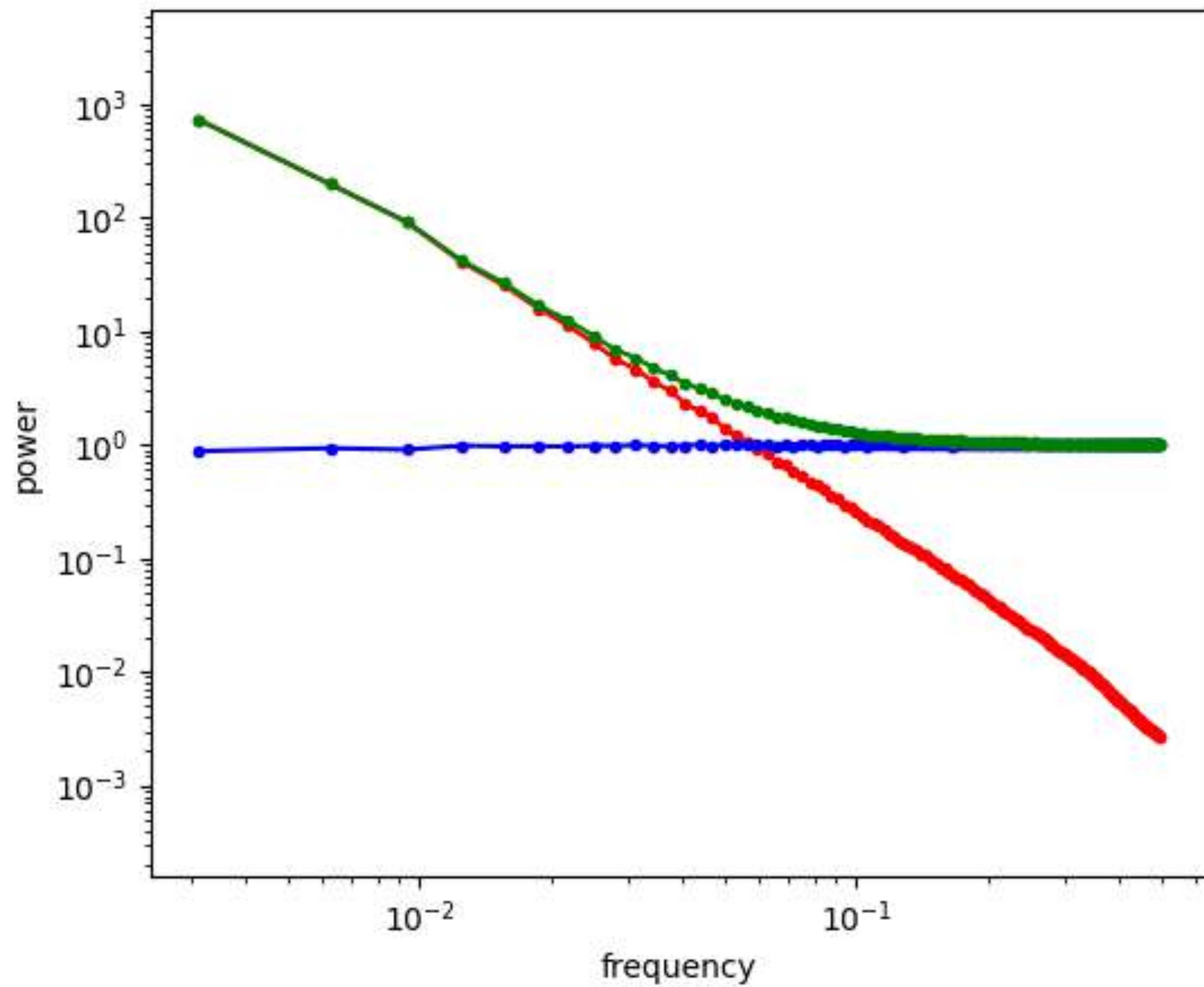
VE process



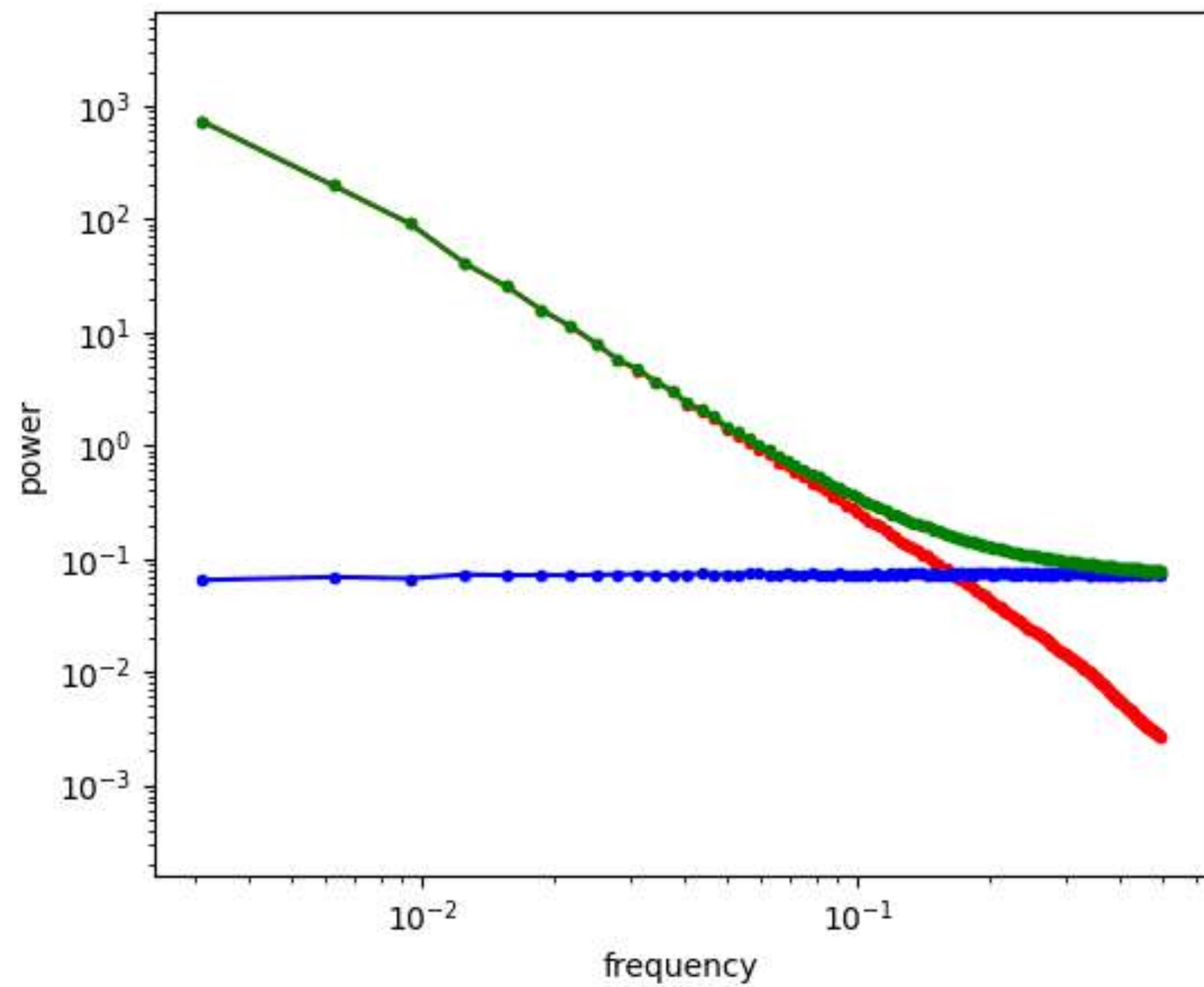
VE process



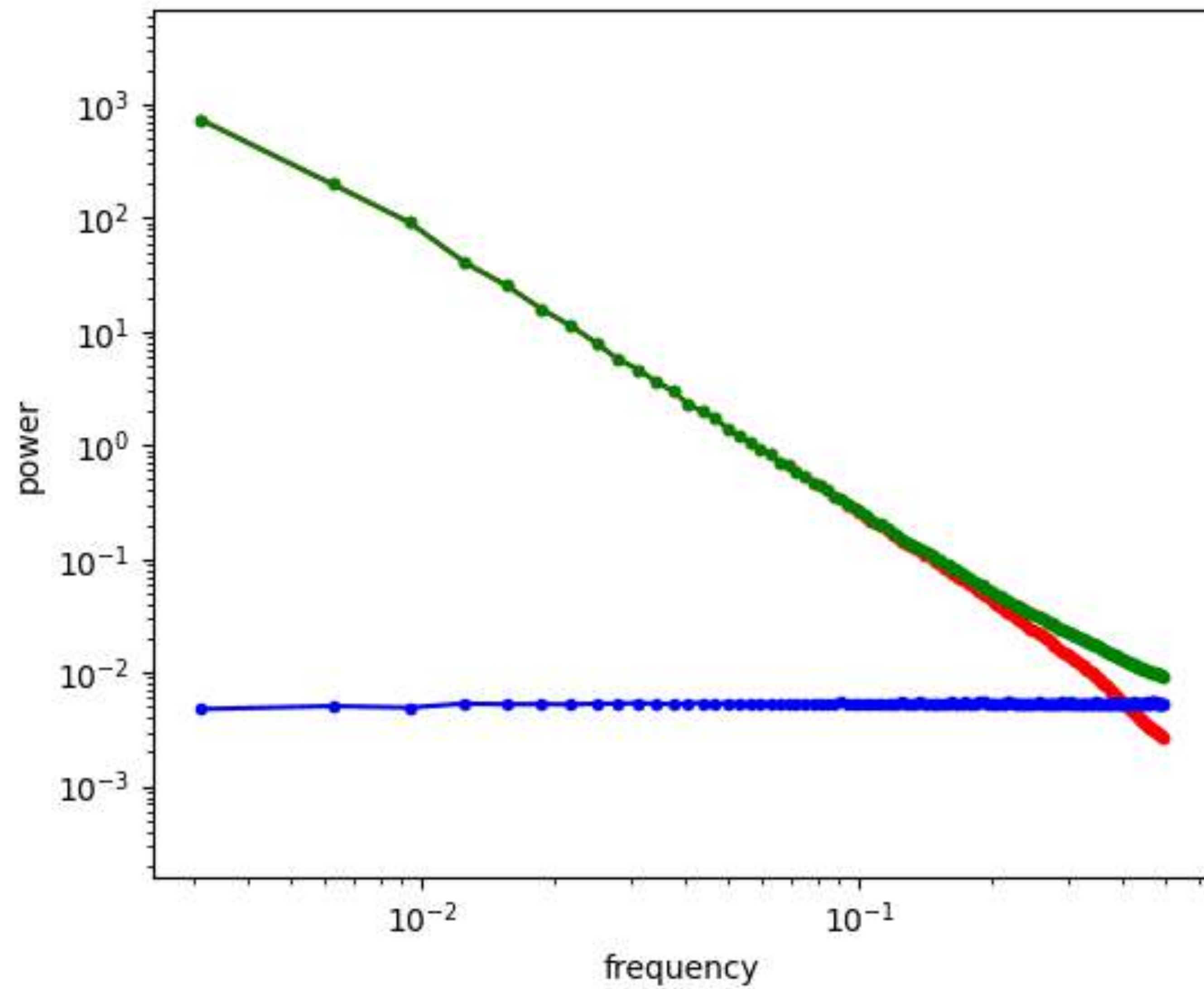
VE process



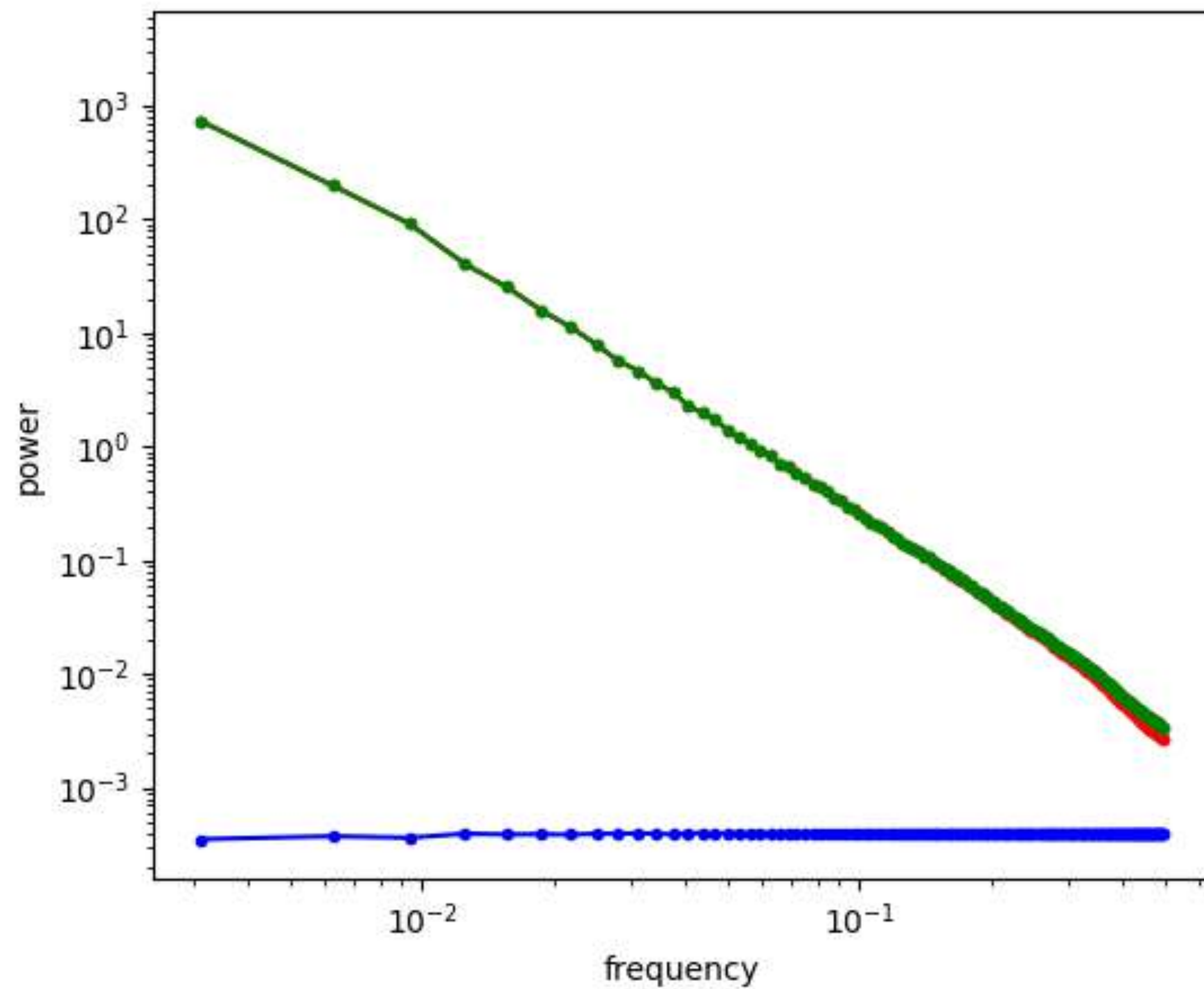
VE process



VE process



VE process

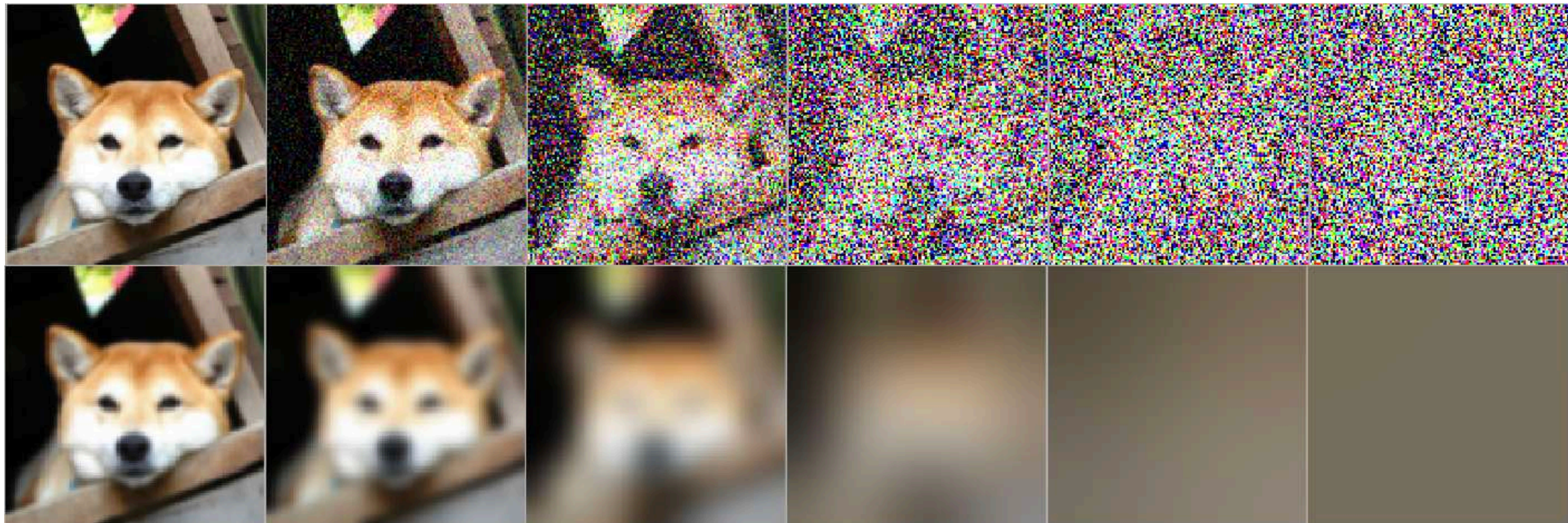


Diffusion as spectral autoregression

Diffusion **gradually** predicts **higher frequencies** given **all** previous **lower frequencies** as **input**



Diffusion is **implicit spectral** AR model



Summary of AR model types

GPT-like AR	Next-scale prediction	Continuous-token prediction	Diffusion
Unnatural inductive bias	Natural inductive bias	Unnatural inductive bias	Natural inductive bias
Discrete	Discrete	Continuous	Continuous
Slow inference	Fast inference	Slow inference	Slow inference

What does this perspective give us?

Connects *diffusion models* with *next-scale prediction* models (e.g., Switti)

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Connects *diffusion models* with *next-scale prediction* models (e.g., Switti)



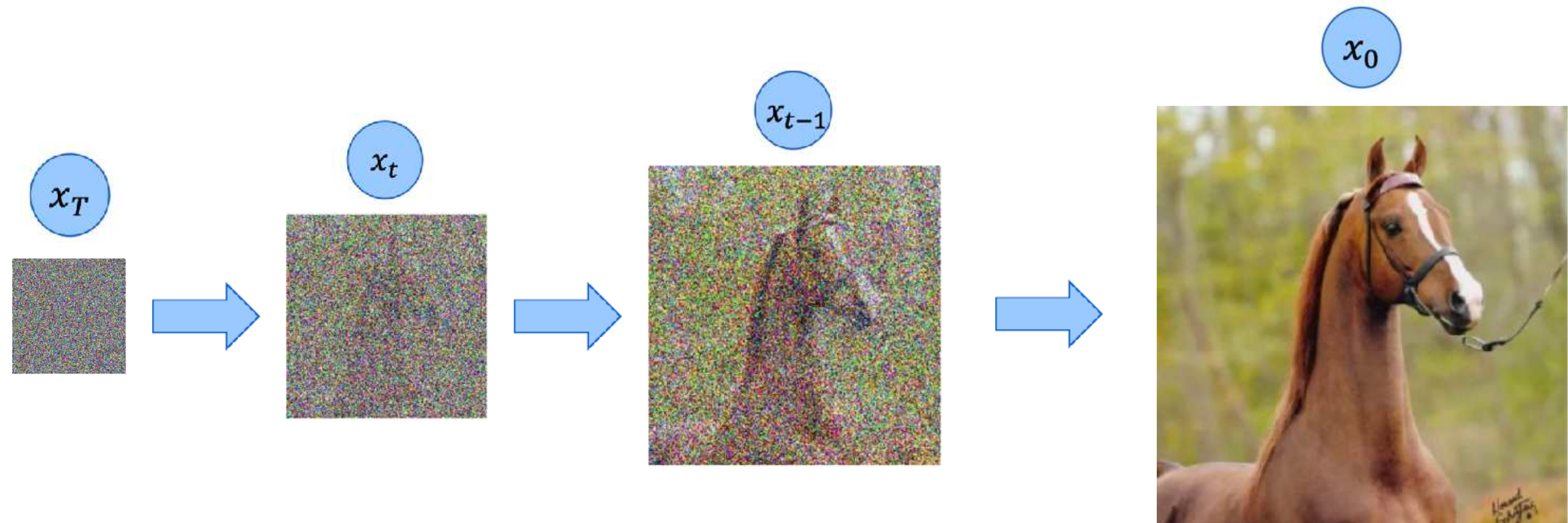
Does this mean that diffusion models can be similarly fast as VAR/Switti?



Yes! Scale-wise diffusion models

Scale-wise diffusion models

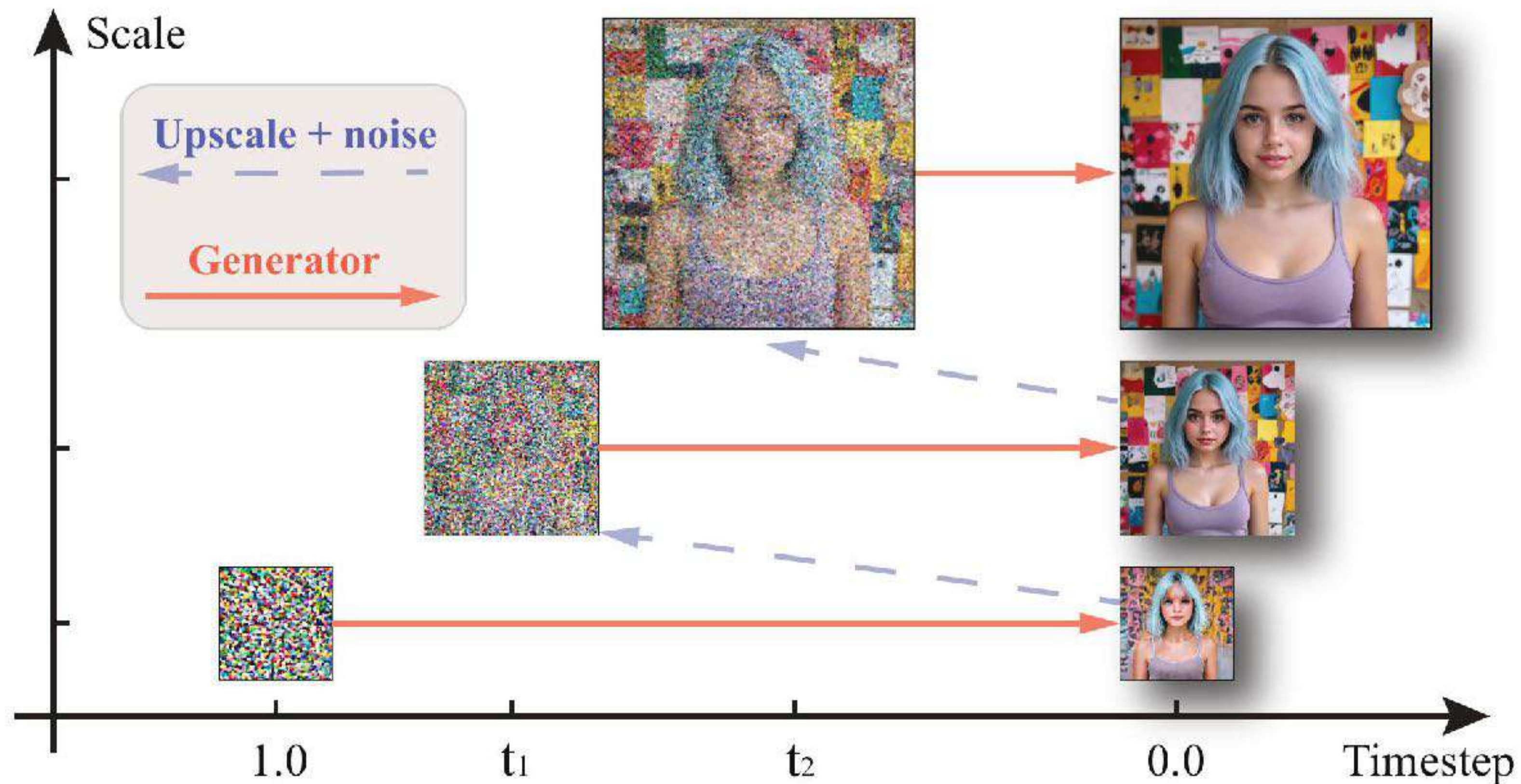
Spectral AR perspective suggests that noisy steps can be made in lower resolutions



Let's integrate scale-wise idea into the few-step models

Scale-wise Distillation | SwD

SwD: a **few-step** generator that progressively increases resolution **over a DM process**



Summary

Diffusion

Natural inductive bias

Continuous

Slow inference

Scale-wise diffusion

Natural inductive bias

Continuous

Mid-speed inference

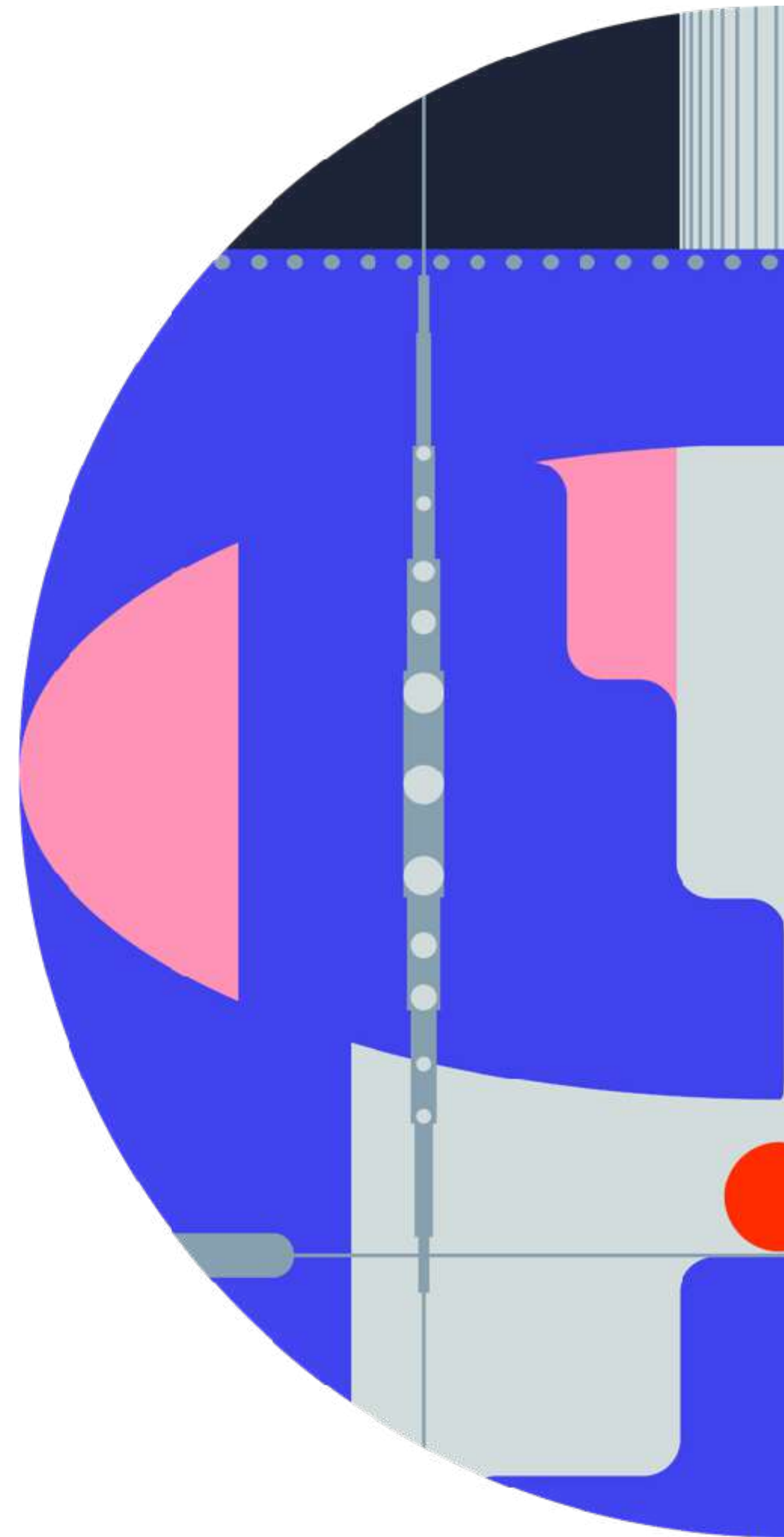
Scale-wise few-step diffusion

Natural inductive bias

Continuous

Fast inference

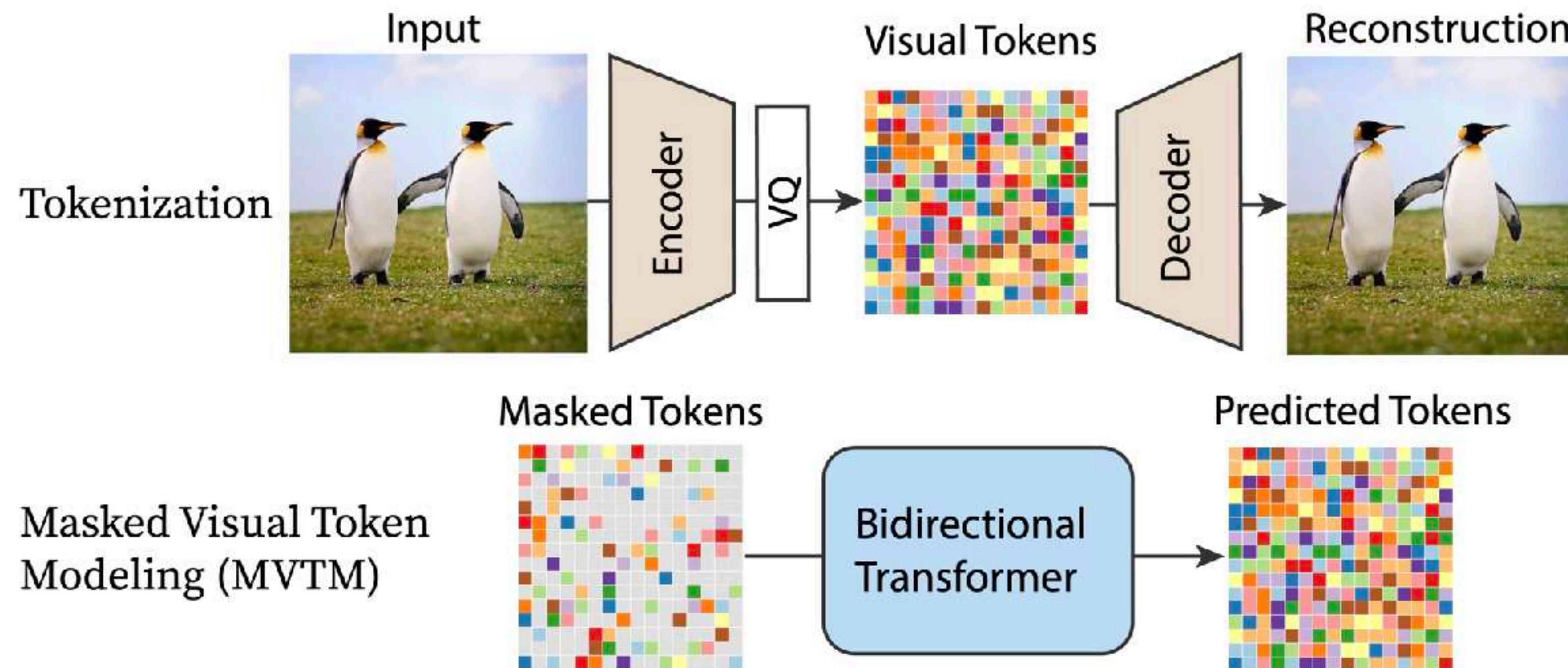
Bonus | Masked Diffusion Models



Masked Diffusion

MaskGIT: Masked Generative Image Transformer

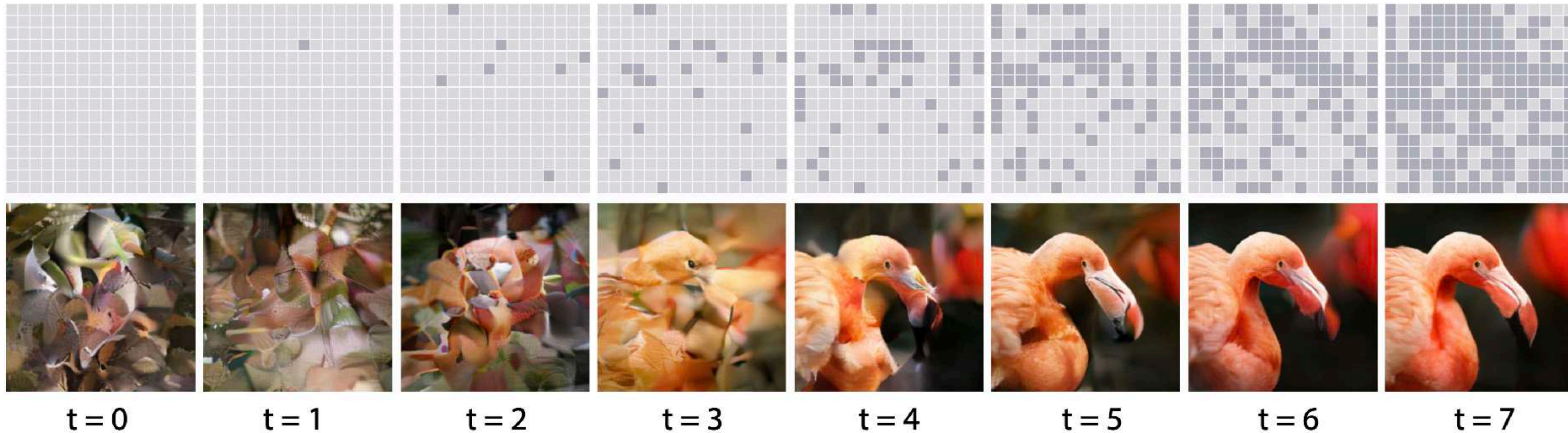
Among diverse discrete diffusion types, we typically consider **masked diffusion** in practice



Masked Diffusion

MaskGIT: Masked Generative Image Transformer

Scheduled Parallel Decoding



Masked diffusion == any-order AR model

Masked diffusion can be modeled with **any-order next-token prediction AR model**, e.g., RandAR.

Both have **identical objectives** but **different architectures**

Masked diffusion == any-order AR model

How to sample with AO-AR model in the diffusion way?

Given observed $\mathbf{x}_o$, use the AO-AR model to predict all masked tokens one by one

$$\{p(\mathbf{x}_m^1 | \mathbf{x}_o), \dots, p(\mathbf{x}_m^M | \mathbf{x}_o)\}$$

Sample τ % tokens with high confidence to unmask

Takeaways

Discrete diffusion is not popular in visual generation because images are inherently continuous

Masked diffusion can be modeled with **any-order next-token prediction AR model**